Human Decisions and Machine Predictions in Multistage Systems*

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Abstract

We study how algorithms can improve decision-making in multistage systems, where cases pass through successive stages of human review before a final decision is made. We develop a framework that compares humans and algorithms at each stage, recovers the system-wide effect of deploying an algorithm, and identifies how humans can add value to the final decision. We find three main results when applying our framework to child protective services (CPS) in Michigan, where hotline screeners decide whether an allegation warrants an investigation before investigators visit the family and decide whether to open a case. First, conventional single-stage comparisons show that hotline screeners significantly underperform an equally selective algorithm but investigators perform roughly as well as the algorithm on average. Second, these conventional comparisons overstate the gains of deploying an algorithm in the multistage system, because they do not account for how decision-making interacts across stages. Replacing hotline screeners with an algorithm changes which calls reach investigators, and their response largely offsets the gain from algorithmic screening. Third, humans can add value at the investigation stage. The decisions of the highest-performing investigators carry information about maltreatment beyond the algorithm's prediction, which we trace through their case notes to evidence gathered during the investigation itself. Motivated by these findings, we design a screening algorithm that accounts for the value of the downstream investigators, finding that it significantly improves on a conventional algorithm trained only to predict outcomes.

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1 Introduction

Many high-stakes decisions are made in multistage systems, where cases pass through successive stages of human review before a final decision is made. In hiring, job recruiters decide whether to advance an applicant for an interview, where interviewers meet with the candidate and collect additional information before extending an offer. In medicine, primary-care physicians decide whether to refer a patient to a specialist, where specialists meet with the patient and collect additional information before determining treatment options. In child protective services (CPS), hotline screeners decide whether an allegation warrants an investigation, where an investigator visits the family, gathers information, and decides whether to open a case. The final decision in each of these settings emerges from the sequence of choices made by decision-makers across multiple stages, with decision-makers at later stages collecting new information and interacting with cases in ways that may shape the outcome.

Predictive algorithms increasingly assist or replace human decision-makers in each of these systems (Kleinberg et al., 2015; Mullainathan, 2025). The conventional algorithms deployed in these settings, however, are designed to predict outcomes at just one stage of the decision-making process. In hiring, algorithms are designed to help job recruiters screen applicants at the resume stage, leaving the interview process unchanged (Hoffman, Kahn and Li, 2018; Li, Raymond and Bergman, 2025). In medicine, algorithms are designed to help physicians screen patients for additional testing at the examination stage, leaving the referral or triage process that brought the patient to the physician unchanged (Mullainathan and Obermeyer, 2022; Agarwal et al., 2023). In CPS, predictive risk assessments are designed to help hotline screeners screen allegations for investigation, leaving the investigation process unchanged (Fitzpatrick, Sadowski and Wildeman, 2025; Grimon and Mills, 2025).

Yet conventional comparisons of humans and algorithms at a single stage may fail to capture the effects of deploying an algorithm in a multistage system for at least two reasons. First, decisions across stages interact. An algorithm introduced early in the decision-making process changes the pool of cases that reaches the next stage, and its effects therefore depend on how the humans at later tages respond to cases they would not otherwise have seen. Second, humans at later stages do more than decide which cases to treat. Their job is to gather information that did not exist when the case entered the system, and their work on a case may shape its outcome. Fully replacing these humans with an algorithm removes both their predictions and these other contributions. Conventional tests of whether humans or algorithms predict outcomes better at a particular stage miss both features, and are therefore unable to recover the system-wide effect of deploying an algorithm in multistage systems.

In this paper, we develop a framework for evaluating humans and algorithms in mul-

tistage systems, allowing us to compare humans and algorithms at each stage, recover the system-wide effect of deploying an algorithm, and identify how humans add value to the decision-making process. We apply this framework to study CPS, a high-stakes setting where human caseworkers screen and investigate allegations of child maltreatment. Contact with CPS is surprisingly common in the United States, with 37 percent of children subject to investigation by age 18 (Kim et al., 2017). The goal of these investigations is to address any immediate safety concerns and prevent future maltreatment among the most at-risk children, a canonical prediction-policy problem in which costly interventions must be targeted at only the highest-risk cases (Kleinberg et al., 2015; Mullainathan, 2025). Predictive algorithms have accordingly attracted growing interest in child protection, with child welfare agencies in at least 26 states reporting interest in algorithmic tools (Samant et al., 2021) and the federal government providing funding to support their deployment (Administration for Children, Youth and Families, 2026).

CPS cases pass through two stages of human review before a decision is made, making it a natural setting for our study. First, incoming allegations of child abuse and neglect are routed to a centralized state hotline, where a hotline screener takes a brief call, consults the family's prior CPS records, and decides whether the allegation warrants an investigation. Second, screened-in calls then go to a regional investigator, who visits the family, interviews caregivers and children, arranges services if needed, and decides whether to open a case. The hotline screener's job is to determine whether an allegation warrants an investigation using only the information available at intake, while the investigator's job is to gather the necessary information to determine whether a case needs to be opened among the screened-in calls. Predictive algorithms can be deployed at the screening stage, the investigation stage, or at both the screening and investigation stages.

The first part of the paper develops our framework. We start by considering the conventional evaluation of an algorithm at just one stage of the decision-making process. Each stage-wise comparison asks whether an algorithm can identify high-risk cases more accurately than the human at a given decision threshold, following Kleinberg et al. (2018), Rambachan (2024), and Angelova, Dobbie and Yang (2025). At the screening stage, we compare the subsequent maltreatment rate among the calls human screeners screen out with the rate among the calls an equally selective algorithm would screen out. At the investigation stage, we compare the subsequent maltreatment among the calls human investigators leave unopened with the rate among the calls an equally selective algorithm would leave unopened. The screening-stage comparison captures how well screeners rank calls by risk compared to the algorithm, while the investigation-stage comparison captures both how well investigators rank calls by risk compared to the algorithm and whether the investigation itself impacts

outcomes directly.

We then show how to recover the system-wide effect of deploying an algorithm using the conventional stage-wise comparisons and terms that capture how decisions interact across stages. At the screening stage, the effect of deploying an algorithm depends on the screening-stage comparison and a term capturing the downstream response of the investigators. The screening algorithm will screen out the lowest-risk calls and only send the highest-risk calls to the human investigators. However, the potential performance gain from the screening algorithm will be attenuated if investigators do not open cases for these newly screened-in calls. The conventional screening-stage comparison holds the downstream investigator response fixed, and so it does not recover the effect of deploying the algorithm at the screening stage.

At the system level, the effect of deploying the algorithm in place of both stages equals the stage-wise comparisons plus a residual that we call the aggregation correction. The residual collects three ways the stage-wise comparisons miscount the gains from correcting a human mistake. Consider a low-risk call that the screener advanced but the algorithm would screen out. The screening-stage comparison credits the algorithm for screening it out. If the investigator also opened it, the investigation-stage comparison credits the algorithm a second time, which we call double counting. If the investigator left it unopened, the credited screen-out changes nothing, which we call a redundant screen-out. The third term, threshold disagreement, arises because the stage-wise algorithms act at the human decision rate of their own stage, whereas the system-wide algorithm opens cases at the status-quo case-opening rate over all calls and so uses a higher threshold, leaving unchanged some decisions that the stage-wise comparisons count as flipped.

The second part of our paper applies our framework to Michigan CPS, using administrative records that cover every call to the state's hotline over six years, link each call to the screener and, if the call is screened in, the investigator who handled it. We build an algorithm that predicts subsequent maltreatment from the information available at intake alone and use it as the benchmark for human decisions throughout the paper.

Comparing humans to this algorithm is complicated by a selective labels problem: subsequent maltreatment is observed only for calls left unopened, and the case-opening decision only for calls screened in (Lakkaraju et al., 2017; Kleinberg et al., 2018). Calls in Michigan CPS are quasi-randomly assigned to screeners who differ in their propensity to screen them in, so we can trace how subsequent maltreatment varies with the probability of intervention and extrapolate to the counterfactuals the framework requires (Arnold, Dobbie and Hull, 2022; Baron et al., 2024; Angelova, Dobbie and Yang, 2025). We show that this screener variation alone recovers each stage-wise comparison and each deployment effect, while allowing

investigators to affect outcomes through the investigation itself.

Our first set of results shows that hotline screeners underperform an equally selective algorithm by a wide margin, while investigators perform about as well as the algorithm on average. At the screening stage, the algorithm reduces subsequent maltreatment among screened-out calls by 11.5 percentage points (s.e. 0.5), nearly one-third of the status-quo maltreatment rate. The gains are driven by the algorithm reversing about half of the current screening decisions: the low-risk calls that screeners advance and the algorithm would not have a maltreatment rate of 24.1 percent, compared to 52.6 percent for the high-risk calls that the algorithm would advance and screeners do not. Estimating effects at the screener level, none of the 291 screeners in our sample are able to outperform the algorithm, and only 3 percent of the screeners perform better than random. Investigators fare better in the single-stage comparisons. The algorithm reduces subsequent maltreatment among unopened cases by a statistically insignificant 1.8 percentage points (s.e. 1.2). The algorithm only reverses 17 percent of case-opening decisions overall, and we find that 33.9 percent (s.e. 1.2) of investigators outperform the algorithm. Nonetheless, the single-stage comparisons suggest that the algorithm outperforms or equals human decision-makers at both stages and its deployment should meaningfully reduce maltreatment rates.

But our second set of results shows that these conventional single-stage comparisons overstate the gains of deploying an algorithm in the Michigan CPS system. Deploying the algorithm at the screening stage, where its advantage over the human decision-makers is largest, only reduces subsequent maltreatment by a statistically insignificant 1.1 percentage points (s.e. 2.3), about 3 percent of the status-quo maltreatment rate. Deploying the algorithm at the system level only reduces subsequent maltreatment by 1.3 percentage points (s.e. 0.5), about 4 percent of the status-quo maltreatment rate. Set beside the economically large single-stage comparisons, these deployment effects pose a puzzle. Judged stage by stage, the algorithm is able to reduce subsequent maltreatment among screened-out cases at the screening stage by 11.5 percentage points and unopened cases at the investigation stage by 1.8 percentage points. Yet deploying it in place of the screeners, or of the whole system, reduces subsequent maltreatment by about one point. How can so much human error, particularly at the screening stage, translate into such small deployment effects?

The multistage decompositions resolve this puzzle. Consider the screening stage deployment. The algorithm improves screening decisions by swapping the low-risk calls screeners advance for the high-risk calls they screen out, with the weighted single-stage comparison indicating that these changes will reduce subsequent maltreatment by 7.6 percentage points (s.e. 0.3). But realizing this gain depends on investigators opening cases on the high-risk calls the algorithm now advances. Those calls have far longer prior CPS histories than the

calls screeners advance, and investigators' case-opening decisions respond only weakly to prior history. Their case-opening rate therefore rises by only 1.4 percentage points (s.e. 2.2), and most of the new high-risk calls are left unopened. This limited downstream investigator response ultimately leads to a 6.7 percentage point (s.e. 2.3) increase in maltreatment, offsetting most of the gains from algorithmic screening.

Now consider the system-wide deployment. The algorithm improves both screening and case-opening decisions, with the weighted single-stage comparisons indicating that these changes will reduce subsequent maltreatment by 8.2 percentage points. But these single-stage comparisons do not account for an aggregation correction, which collects three ways the single-stage comparisons miscount the gains from deployment. The first is double counting: low-risk calls that screeners advance and investigators open are credited to the algorithm at both the screening and the investigation stage, even though maltreatment can be prevented only once. The second is redundant screen-outs: low-risk calls that screeners advance but investigators leave unopened are credited to the algorithm at the screening stage, even though the algorithm does not change the final decision. The third is threshold disagreement: many of the high-risk calls that screeners miss are credited to the algorithm at the screening stage but fall below the system-wide threshold and are unopened in the system-wide deployment. Together the three terms sum to 6.9 percentage points (s.e. 0.6), offsetting most of the gains implied by the single-stage comparisons.

Our third set of results turns to investigators themselves. A conventional algorithm forgoes most of the gain from better screening because of how investigators respond downstream, and understanding how to recover these gains requires understanding the value of the investigation stage. We therefore focus on the roughly one-third of investigators who outperform the algorithm and ask what allows them to do better. Caseload, demographics, and tenure explain almost none of the variation in performance across investigators, and the higher- and lower-performing investigators place similar weights on the intake characteristics, so decision rules distilled from each group's decisions are nearly identical. These results tell us that the higher-performing investigators add value to the decision-making process through the investigation process itself, either in what actions they take or in what they learn.

Investigators do a great deal on a case besides deciding whether to open it. They visit the home, interview the family, and may arrange services, put a safety plan in place, or offer guidance, any of which could change subsequent maltreatment whether or not a case is opened. Our framework allows for such direct effects, and it would be natural to expect higher-performing investigators to differ in them. We find no evidence that they do. Higher- and lower-performing investigators, who handle comparable calls by quasi-random

assignment, open cases, substantiate maltreatment, and place children at similar rates, and they are no more likely to take any of five informal actions that we label from their case notes. A direct statistical test of whether investigators affect maltreatment only through their case-opening decisions, which requires no channel to be named, fails to reject. The evidence thus points to a single channel: investigators affect subsequent maltreatment through which cases they open.

Having ruled out that investigators take actions beyond case opening that affect maltreatment, we turn to the other channel through which they can improve upon the algorithm: what they learn in the field. The algorithm predicts from the information recorded at intake, a brief phone call and the family's prior history. An investigator visits the home, sees the children and the caregivers, and speaks with relatives, teachers, and others who know the family, so by the time a case-opening decision is made, the investigator knows things about the family that no intake record contains. If investigators act on this information, then within a quintile of the algorithm's predicted risk the calls they leave unopened should be safer than the quintile as a whole. For investigators pooled together, they are not: within each quintile, the calls left unopened are about as risky as the quintile, so case openings on average carry little information beyond the algorithm. For higher-performing investigators, they are: the calls they leave unopened are safer than the quintile at every level of predicted risk. Their decisions carry information about maltreatment that the algorithm does not have.

The case notes provide us with a window into this information. At the close of every investigation, the investigator writes a narrative account of the case: the home visits and what was observed there, the interviews with children and caregivers, the collateral contacts, the actions taken, and the reasoning behind the disposition. The notes are long, unstructured, and written in the investigator's own words, and none of their content enters the administrative data or the algorithm. Rather than decide in advance which evidence matters, we use large language models to generate hypotheses from the notes (e.g., Ludwig and Mullainathan, 2024; Zhong et al., 2022; Zhou et al., 2024; Batista and Ross, 2025; Modarressi, Spiess and Venugopal, 2025): we show a model pairs of notes from calls that look alike at intake but whose text predicts opposite decisions, with the decision and the investigator masked, and ask it to name the evidence that distinguishes them. We then label every note for the candidate themes and keep the five that best predict case opening: corroborated harm, a caregiver involving the child in dangerous conduct, evidence of abuse that is absent or contradictory, corroboration that the child is adequately cared for, and a child left without a stable caregiver. These themes are the information an investigation produces, and none of it is available at intake, or to the algorithm.

Taken together, our results suggest a possible division of labor: the algorithm's advantage lies at the screening stage, and the investigators' lies in what they learn in the field. A conventional algorithm does not reap the benefits of this division of labor. The algorithm screens far better than human screeners, yet deploying it at the screening stage recovers little of that advantage, because investigators open no more of the riskier calls it sends them, and deploying it in place of the whole system forgoes the investigation altogether. The investigation, meanwhile, produces information that the algorithm cannot have, which higher-performing investigators turn into better case-opening decisions. Neither deployment makes use of what an investigation produces. Our final set of results takes up the design question this raises: how should a screening algorithm be built to reap the benefits of investigation?

We consider screening rules that, at each level of predicted risk, either screen a call out, open a case on it directly, or send it to an investigator and defer to the decision the investigator makes. Sending a call to investigation is worth its cost only when investigators' decisions use information beyond the prediction, opening the riskier calls among those with the same prediction and leaving the safer ones unopened, which is what we find higher-performing investigators do. We characterize the screening rule that maximizes a policymaker's welfare over subsequent maltreatment, case openings, and investigations, and we compare it with a conventional algorithm trained only to predict outcomes, which never investigates. We find that the optimal screening algorithm significantly improves on a conventional algorithm trained only to predict outcomes. When the calls it sends to investigation are handled by higher-performing investigators, it improves on the conventional algorithm by 18.5 percent.

Although we focus on CPS throughout, our framework and empirical tools apply more broadly. Multistage systems are pervasive: resumes are screened (Cowgill, 2018; Li, Raymond and Bergman, 2025) before applicants are interviewed (e.g., Jabarian and Henkel, 2025; Jabarian and Reshidi, 2025), tax returns are flagged before they are audited (e.g., Battaglini et al., 2025; Boning et al., 2025), and patients are referred before they are treated. In each, a decision is a bundle of tasks spread across stages and people (e.g., Autor, 2015; Acemoglu and Restrepo, 2019): someone must collect information, someone must predict, and someone must decide. The value of predictive algorithms in multistage systems like these depends not only on how well it predicts, but on which tasks, and at which stage, it would take over.

Our paper draws together several literatures on human and algorithmic decision-making. The closest, discussed above, compares humans and algorithms at a single decision. In these comparisons, the recorded choice is the decision-maker's only channel of influence on the outcome, and the rest of the system is held fixed; we relax both. A second literature studies algorithms that advise rather than replace human decision-makers: when an algorithm issues a recommendation, its effect on outcomes is often muted, because humans follow it

only selectively (Imai et al., 2023; Albright, 2024; Stevenson and Doleac, 2024), which has prompted work on how to design recommendations under explicit models of how humans respond (Agarwal, Moehring and Wolitzky, 2025; Dreyfuss and Hoong, 2026). A third asks how cases should be allocated between people and algorithms, through algorithmic triage and learning to defer (Madras, Pitassi and Zemel, 2018; Raghu et al., 2019; Mozannar and Sontag, 2020) and human-algorithm complementarity (Bansal et al., 2021; Agarwal et al., 2023). A fourth examines how selection and discrimination propagate through sequential decisions (Rambachan and Roth, 2020; Bohren, Hull and Imas, 2022; Baron et al., 2024). Finally, we draw on recent work that uses machine learning and language models to generate hypotheses from unstructured data (Ludwig and Mullainathan, 2024; Zhong et al., 2022; Zhou et al., 2024; Batista and Ross, 2025; Modarressi, Spiess and Venugopal, 2025), and on econometric frameworks for using language-model outputs in empirical work (Ludwig, Mullainathan and Rambachan, 2025; Carlson and Dell, 2026). We use these tools to study what investigators do and find, which the structured data do not record. We provide a unified framework for studying algorithms in multistage systems: by linking stage-wise human-algorithm comparisons to the effects of deployment, we show where in a system humans and algorithms are complementary, and how selection across stages and human influence beyond the recorded decision drive a wedge between comparing a decision-maker and replacing one.

2 Setting and Data

2.1 Decision-Making in Michigan CPS

CPS aims to prevent child maltreatment by identifying children at risk and directing investigative and service resources toward their families. Screening and investigation decisions seek to separate children who would benefit from continued intervention from those who can safely exit the system.

We study CPS in Michigan, whose process resembles that of many other U.S. states. The process begins when a mandated reporter (e.g., a teacher, doctor, or police officer) or anyone else (e.g., a neighbor or family member) contacts the state's central hotline about a child's safety.[1] Figure 1 summarizes the two decisions that follow. In the screening stage, a hotline screener at one of two intake centers (Detroit and Grand Rapids) gathers information about the child and the alleged maltreatment and decides whether to "screen in" the call for investigation. In the investigation stage, a screened-in call is transferred to a local child welfare office, where an investigator gathers additional information and decides whether to

[1] In Michigan, 97 percent of reports arrive by phone and the rest through an online system. We refer to all reports as "calls."

open a case.

Screeners learn the allegation, the reporter's identity, and information about the child and household from the reporter, and they see prior CPS contact, past investigations, and foster care involvement in administrative records. An allegation may concern abuse (e.g., bruises or burns) or neglect (e.g., improper supervision due to parental substance abuse).

Agency policy gives screeners substantial discretion, subject to state law and departmental guidelines, and instructs them to screen in calls whose children are most likely to suffer subsequent maltreatment if screened out (MDHHS, 2020).[2] Calls last about 15 minutes on average, and approximately 36 percent of calls in our sample are screened in. A screen-out ends the agency's involvement, and screeners play no other role in the process. Calls are assigned to screeners by a queue that routes each call to the longest-waiting available screener, so assignment is quasi-random conditional on day and shift.

If a call is screened in, investigators have 24 hours to begin the investigation, 72 hours to make face-to-face contact with the child, and 30 days to complete it. They visit the home, assess the child's immediate safety, and interview caregivers and children. They may also contact teachers, medical providers, or others outside the household and review records of prior CPS contact. Investigators may further connect a caregiver to services, put a safety plan in place, or offer advice, and such contact may change a child's subsequent maltreatment even if no case is opened. At its conclusion, investigators write detailed narrative case notes that summarize their activities and findings. Screened-in calls rotate among the investigators in the local office for the child's county, so assignment is quasi-random conditional on month, ZIP code, and shift.

The decision we study is whether to open a case, the final decision the system makes about a call. To open a case, the investigator substantiates the allegation, finding by a preponderance of the evidence that abuse or neglect occurred.[3] When a case is opened, the department lists the responsible adult on the state's central registry, monitors the family on an ongoing basis, and refers it to services such as parenting classes or substance abuse treatment. Approximately 11 percent of screened-in calls result in a case opening. We do not study the foster care placement decision, which is the system's most intensive intervention and has been the focus of prior work in economics (e.g., Doyle, 2007; Baron et al., 2024).

[2]In 2023, after our sample period, Michigan introduced Structured Decision Making (SDM) at the screening stage, which placed greater emphasis on whether the focal allegation meets the statutory definition of maltreatment rather than on broader assessments of subsequent maltreatment risk.

[3]Each investigation ends in one of five disposition categories defined by Michigan's Child Protection Law, ordered by the intensity of the department's response. In Categories I and II, the investigator substantiates the allegation and the department opens a case. We refer to these two categories together as a "case opening." In Category III, the allegation is substantiated but no case is opened. In Categories IV and V, it is not substantiated. In Categories III and IV, the investigator may still refer the family to services.

Placement follows only about 2 percent of screened-in calls and is increasingly reserved for the most severe safety concerns. The Family First Prevention Services Act of 2018 reinforced this trend by letting states spend federal child welfare funds on services that keep children at home. As a consequence, case opening is the central margin on which CPS allocates intervention to prevent maltreatment.

2.2 Data

We use administrative records from the Michigan Department of Health and Human Services (MDHHS) on all calls to the state's CPS hotline from 2017 to 2022. For each call, we observe the assigned screener, the screening decision, the date and time, and characteristics of the children and allegations involved. For screened-in calls, we further observe the investigator assigned to the intake, the child's ZIP code, the child's relationship to the alleged perpetrator, and the investigation's outcomes, including substantiation, case opening, and foster care placement. We also observe subsequent CPS contact, from which we construct our primary outcome. From these records we construct a screener sample of all calls, which we use to study the screening stage, and an investigator sample of screened-in calls, which we use to study the investigation stage.

Our primary maltreatment potential outcome is whether any child on the call is the subject of a new hotline report 31 to 180 days later. We exclude the first 30 days, the investigation window, to define the outcome consistently for screened-in and screened-out calls. Because our goal is to identify which children are most at risk of subsequent maltreatment absent intervention, we focus on the outcome that would occur if no case were opened. While subsequent hotline reports are observed for all calls, opening a case may itself affect whether a new report occurs, giving rise to a selective labels problem (Lakkaraju et al., 2017; Kleinberg et al., 2018). We use a new hotline report rather than more intensive forms of subsequent CPS contact because reports originate from third parties outside the CPS system, whereas subsequent investigations and substantiations also depend on the decisions of CPS personnel. Moreover, repeat calls are typically reassigned to the same investigator, who again decides whether to substantiate the allegation and open a case. We report robustness to alternative outcomes and time horizons in Section 4.4.

Our analysis is at the call level, since screen-in and case-opening decisions are consistent across all children involved in a call. The screener sample consists of 514,392 calls assigned to 291 screeners, and the investigator sample of the 184,884 screened-in calls assigned to 4,579 investigators. Appendix B.1 describes the sample construction, and Appendix Table A.1 reports summary statistics. Most calls involve children already known to the system: 78 percent include a child with a prior CPS investigation. Improper supervision is the most

common allegation, appearing in 72 percent of calls, and the youngest child on a call is 8.6 years old on average. Among screened-in calls, 23 percent are substantiated and 11 percent lead to a case opening. Opened cases involve younger children and more often a child with a prior investigation (89 versus 77 percent). Conditional on no case being opened, 34 percent of calls are followed by a new hotline report within 180 days, 36 percent among screened-out calls and 31 percent among screened-in calls.

3 Humans and Algorithms in a Multistage System

Using CPS as our running example, we develop a framework for evaluating predictive algorithms in multistage systems. We define the conventional stage-wise human-algorithm comparisons and the effects of deploying an algorithm at a single stage or in place of the entire system. We then decompose each deployment effect into the stage-wise comparisons and the terms they omit, which capture how decisions interact across stages. These decompositions illustrate that the stage-wise comparisons can overstate the effects of deploying an algorithm in a multistage system.

3.1 The Multistage System

We consider a population of calls to a child-welfare hotline, each with characteristics X_i recorded at intake. In the screening stage, a screener observes X_i and decides whether to advance the call for investigation, $S_i \in \{0,1\}$. In the investigation stage, an investigator then decides whether to open a case, $D_i \in \{0,1\}$, a decision defined for every call but observed only for those screened in ($S_i = 1$). The multistage system opens a case only when both stages act, $C_i = S_i D_i$. Each call has two potential outcomes for maltreatment, $Y_i^0, Y_i^1 \in \{0,1\}$: Y_i^0 is the call's maltreatment if it is immediately screened out, and Y_i^1 is its maltreatment if it is screened in and investigated but no case is opened. The two may differ because the investigator's contact with the family, the advice they give, and the intermediate actions they take may change maltreatment even when no case is opened. We define maltreatment as $Y_i^* := (1 - S_i)Y_i^0 + S_i Y_i^1$, and it is observed only when no case is opened, so $Y_i = (1 - C_i)Y_i^*$. A screened-out call reveals Y_i^0, a screened-in call with no case opened reveals Y_i^1, and a call with an opened case reveals neither. We take as given a fixed algorithm $m(X_i) \in [0,1]$ that predicts subsequent maltreatment risk from the intake characteristics, and consider algorithmic decision rules $\mathbf{1}\{m(X_i) \geq \tau\}$ that act whenever predicted risk exceeds a threshold τ.

Throughout this section, we treat each stage as having a single representative decision-maker. Heterogeneity across decision-makers will enter our empirical approach. Section 5 and Appendix D extend the framework to individual investigators.

3.2 Stage-Wise Human-Algorithm Comparisons

We first compare each human stage to the algorithm in isolation, holding the rest of the system fixed. For each stage, define an algorithmic decision rule that acts at the same rate as the human it replaces: $S_i^m := \mathbf{1}\{m(X_i) \geq \tau^S\}$, with τ^S chosen so that $\Pr(S_i^m = 1) = \Pr(S_i = 1)$, and $D_i^m := \mathbf{1}\{m(X_i) \geq \tau^I\}$, with τ^I chosen so that $\Pr(D_i^m = 1 \mid S_i = 1) = \Pr(D_i = 1 \mid S_i = 1)$. The *stage-wise human-algorithm comparisons* are

$$\Delta^S := \mathbb{E}[Y_i^0 \mid S_i = 0] - \mathbb{E}[Y_i^0 \mid S_i^m = 0],$$
$$\Delta^I := \mathbb{E}[Y_i^1 \mid S_i = 1, D_i = 0] - \mathbb{E}[Y_i^0 \mid S_i = 1, D_i^m = 0].$$

Each comparison contrasts maltreatment among the calls a human stage screens out or leaves unopened with maltreatment among the calls an equally selective algorithm would, so a positive value means the algorithm outperforms the human at that stage. These are the comparisons used to evaluate judges against pretrial risk tools (Kleinberg et al., 2018; Rambachan, 2024; Angelova, Dobbie and Yang, 2025), physicians against predicted diagnostic yield (Mullainathan and Obermeyer, 2022), and hiring managers against algorithmic screens (Li, Raymond and Bergman, 2025).

At the investigation stage, the algorithm acts on the intake characteristics alone, so the calls it leaves unopened produce Y_i^0, whereas the calls investigators leave unopened produce Y_i^1. We decompose Δ^I as

$$\Delta^I = \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 \mid S_i = 1, D_i = 0]}_{\text{Direct effect of investigation.}} + \underbrace{\mathbb{E}[Y_i^0 \mid S_i = 1, D_i = 0] - \mathbb{E}[Y_i^0 \mid S_i = 1, D_i^m = 0]}_{\text{Investigator selection.}}.$$

The second term measures whether investigators or the algorithm leave riskier calls unopened, and differences may reflect private information gathered in the field (Agarwal et al., 2023; Alur et al., 2023; Alur, Raghavan and Shah, 2024), systematic prediction errors (Mullainathan and Obermeyer, 2022; Rambachan, 2024), noisy decisions (Kahneman, Sibony and Sunstein, 2021), or omitted payoffs (Kleinberg et al., 2018). The first term is any direct effect of the investigation on maltreatment among the calls investigators leave unopened. The stage-wise comparison Δ^I therefore bundles two possible differences between human investigators and the algorithm—in which calls they leave unopened and in what the investigation itself does to those calls. Separating the two is part of our empirical analysis.

3.3 Incorporating Algorithms in a Multistage System

We take the perspective of a policymaker who cares about three outcomes of the multistage system: maltreatment, investigations, and case openings. The system seeks to reduce mal-

treatment through case openings, but investigations and case openings consume resources and intrude on families. A decision rule that lowers all three unambiguously improves on the status quo; otherwise, the ranking depends on the weights placed on each.[4] We describe the effects on these outcomes of deploying the algorithm at the screening stage only, at the investigation stage only, or in place of the entire system. We then decompose each deployment's effect on maltreatment into the stage-wise comparisons and additional terms that summarize how the stages interact.

3.3.1 Deploying the Algorithm

Under the status quo, subsequent maltreatment among unopened calls is

$$V^0 := \mathbb{E}[Y_i^* \mid C_i = 0] = \omega^S \mathbb{E}[Y_i^0 \mid S_i = 0] + \omega^I \mathbb{E}[Y_i^1 \mid S_i = 1, D_i = 0],$$

where $\omega^S := \Pr(S_i = 0 \mid C_i = 0)$ and $\omega^I := \Pr(S_i = 1, D_i = 0 \mid C_i = 0)$ are the shares of unopened calls that exit at the screening and investigation stages ($\omega^S + \omega^I = 1$).

First consider deploying the algorithm at only one stage, holding the rest of the multistage system fixed. Deploying the algorithm at the investigation stage alone leads to the case-opening rule $C_i^I := S_i D_i^m$ and the system-wide maltreatment rate $V^I := \mathbb{E}[Y_i^0 \mid C_i^I = 0]$. Because D_i^m acts at the status-quo case-opening rate among screened-in calls, this deployment holds case openings fixed. Because the algorithm decides from the intake characteristics alone, it eliminates investigations, so the calls left unopened produce Y_i^0.

Deploying the algorithm at the screening stage alone leads to the case-opening rule $C_i^S := S_i^m D_i$. The calls the algorithm screens out produce Y_i^0, and those it screens in that investigators leave unopened produce Y_i^1, so the system-wide maltreatment rate is $V^S := \mathbb{E}[(1 - S_i^m)Y_i^0 + S_i^m Y_i^1 \mid C_i^S = 0]$. Because S_i^m screens in calls at the status-quo rate, this deployment holds investigations fixed. But because the algorithm may screen in a different pool than human screeners, the human case-opening rate on that pool generally differs, so the number of case openings may change.

Finally, deploying the algorithm in place of the entire system dispenses with human screening, investigation, and case opening. The algorithm opens cases directly from the intake characteristics, leading to the decision rule $C_i^m := \mathbf{1}\{m(X_i) \geq \tau^*\}$, with τ^* chosen so that $\Pr(C_i^m = 1) = \Pr(C_i = 1)$, and the system-wide maltreatment rate $V^{System} := \mathbb{E}[Y_i^0 \mid C_i^m = 0]$. Like the investigation-stage deployment, it holds case openings fixed and eliminates investigations.

[4] Any objective of the form $-\kappa_S \mathbb{E}[S_i] - \kappa_C \mathbb{E}[C_i] - \kappa_Y \mathbb{E}[(1 - C_i)Y_i^*]$, with weights $\kappa_S, \kappa_C, \kappa_Y > 0$, ranks two decision rules consistently whenever one reduces all three quantities relative to the other.

3.3.2 From Stage-Wise Comparisons to Deployment Effects

How do the stage-wise human-algorithm comparisons Δ^S and Δ^I relate to the effects of these deployments on system-wide maltreatment? For a single-stage deployment, the answer depends on whether the stage is terminal or upstream in the multistage system. Proofs are in Appendix A.

Proposition 3.1 (Stage-specific deployment)**.** *The single-stage deployments satisfy*

$$V^0 - V^I = \omega^I \Delta^I, \tag{1}$$

$$V^0 - V^S = \omega^S \Delta^S + \omega^I \mathcal{R}^I + (\omega^S - \lambda^S)\Gamma^S, \tag{2}$$

where $\mathcal{R}^I := \mathbb{E}[Y_i^1 \mid S_i = 1, D_i = 0] - \mathbb{E}[Y_i^1 \mid S_i^m = 1, D_i = 0]$, $\lambda^S := \Pr(S_i^m = 0 \mid C_i^S = 0)$, *and* $\Gamma^S := \mathbb{E}[Y_i^0 \mid S_i^m = 0] - \mathbb{E}[Y_i^1 \mid S_i^m = 1, D_i = 0]$.

Proposition 3.1 shows that the comparison and the deployment effect coincide for the terminal investigation stage: deploying the algorithm there changes system-wide maltreatment by exactly $\omega^I \Delta^I$. Nothing downstream depends on the case-opening decision, and Δ^I already reflects the removal of the investigation, since the calls the algorithm leaves unopened produce Y_i^0.

For the upstream screening stage, however, the comparison and the deployment effect do not coincide. Deploying the algorithm at the screening stage contributes the weighted comparison $\omega^S \Delta^S$, but it also changes the pool of calls that investigators face, and their case-opening decisions on that pool may either reinforce or undo the gains from the algorithm. We call this the "downstream investigator response" $\omega^I \mathcal{R}^I$, which compares maltreatment among the calls investigators leave unopened under the human-screened and the algorithm-screened pools. The term $\mathcal{R}^I$ decomposes as

$$\mathcal{R}^I = \underbrace{\mathbb{E}[Y_i^1 \mid S_i = 1] - \mathbb{E}[Y_i^1 \mid S_i^m = 1]}_{\text{Change in maltreatment of screened-in calls}}$$
$$+ \underbrace{\big(\mathbb{E}[Y_i^1 \mid S_i^m = 1] - \mathbb{E}[Y_i^1 \mid S_i^m = 1, D_i = 0]\big) - \big(\mathbb{E}[Y_i^1 \mid S_i = 1] - \mathbb{E}[Y_i^1 \mid S_i = 1, D_i = 0]\big)}_{\text{Change in investigator selection}}.$$

The first term is the change in the risk of the screened-in pool when the algorithm rather than screeners selects it. The second is the change in how much risk investigators remove from the unopened calls by opening cases, which on each pool equals their case-opening rate times the maltreatment gap between the calls they open and the calls they leave unopened. The last term in equation (2) reweights the two exit routes: $\omega^S - \lambda^S$ is the change in the share of unopened calls that exit at the screening stage, and Γ^S is the difference in maltreatment

between the two routes. The screening comparison Δ^S holds the downstream investigator response fixed, and so it does not recover the effect of deploying the algorithm at the screening stage.

Our second result decomposes the effect of deploying the algorithm in place of the entire multistage system.

Proposition 3.2 (System-wide decomposition)**.** *The effect of system-wide deployment decomposes as*

$$V^0 - V^{System} = \omega^S \Delta^S + \omega^I \Delta^I + \omega^I \mathcal{C}^{\text{Sys}}, \tag{3}$$

for $\mathcal{C}^{\text{Sys}} := \mathbb{E}[Y_i^0 \mid S_i = 1, D_i^m = 0] - \mathbb{E}[Y_i^0 \mid S_i^m = 1, C_i^m = 0]$.

Proposition 3.2 shows that the weighted stage-wise comparisons $\omega^S \Delta^S$ and $\omega^I \Delta^I$ do not sum to the effect of deploying the algorithm in place of the system; the difference is the "aggregation correction" $\omega^I \mathcal{C}^{\text{Sys}}$. Appendix A shows that the aggregation correction is proportional to

$$\begin{aligned} &\underbrace{\mathbb{E}\big[Y_i^0 \mathbf{1}\{S_i = 1, D_i = 1, S_i^m = 0\}\big]}_{\text{Double counting}} + \underbrace{\mathbb{E}\big[Y_i^0 \mathbf{1}\{S_i = 1, D_i = 0, S_i^m = 0\}\big]}_{\text{Redundant screen-outs}} \\ &\quad - \underbrace{\mathbb{E}\big[Y_i^0 \mathbf{1}\{S_i = 1, D_i^m = 1, C_i^m = 0\}\big]}_{\text{Threshold disagreement (investigation)}} - \underbrace{\mathbb{E}\big[Y_i^0 \mathbf{1}\{S_i = 0, S_i^m = 1, C_i^m = 0\}\big]}_{\text{Threshold disagreement (screening)}}. \end{aligned} \tag{4}$$

The first two terms arise because the stage-wise comparisons double count. Consider the calls that screeners advance but the algorithm would screen out, $S_i = 1$ and $S_i^m = 0$, all of which enter Δ^S. Those that investigators open, $D_i = 1$, also enter Δ^I, and so are "double counted." Those that investigators leave unopened, $D_i = 0$, are unopened under the status quo and the system-wide deployment alike, so Δ^S credits screen-outs that do not change whether a case is opened; we refer to these as "redundant screen-outs." The last two terms arise because the stage-wise algorithms act at each stage's human rate, whereas the system-wide algorithm opens cases at the status-quo case-opening rate over all calls and so uses a higher threshold, $\tau^* \geq \tau^S$ and $\tau^* \geq \tau^I$. Among screened-out calls, those with $\tau^S \leq m(X_i) < \tau^*$ are screened in by the screening-stage algorithm, and among screened-in calls, those with $\tau^I \leq m(X_i) < \tau^*$ are opened by the investigation-stage algorithm. The comparisons Δ^S and Δ^I count these decisions, but the system-wide algorithm opens none of these calls; we refer to them as "threshold disagreement."

4 From Comparisons to Deployment Effects in Michigan CPS

Propositions 3.1 and 3.2 show that, at the screening stage and for the system as a whole, the stage-wise comparisons do not recover the deployment effects. In this section, we identify and estimate in Michigan CPS the stage-wise comparisons, the deployment effects, and the terms that capture how the stages interact.

4.1 Empirical Approach

The central challenge is two selective labels problems: subsequent maltreatment is observed only when no case is opened, and the case-opening decision is observed only when a call is screened in. Calls are quasi-randomly assigned to screeners who differ in their propensity to screen them in, so we can trace how subsequent maltreatment varies with case-opening probability and extrapolate to the counterfactuals the framework requires (Arnold, Dobbie and Hull, 2022; Baron et al., 2024; Angelova, Dobbie and Yang, 2025). This screener variation alone recovers each stage-wise comparison and each deployment effect.

4.1.1 The Predictive Algorithm

We construct an algorithm $m(X_i)$ that predicts subsequent maltreatment from the structured information available at the time of the hotline call: complaint timing, allegation type, reporter and contact-method characteristics, child age, and prior CPS history. The algorithm uses intake information only, so it never sees what investigators learn in the field. It is a gradient-boosted decision tree, trained on the calls in a training sample whose outcome is observed (screened-out calls and screened-in calls with no case opened). All estimates in the main text use a separate evaluation sample, which consists of a random half of screened-out calls and the screened-in calls of the 806 investigators with at least 75 screened-in calls, who handle 126 calls each on average (Appendix B.2). Appendix Figure A.1 summarizes its performance in the evaluation sample. The area under the receiver operating characteristic curve is 0.693, and realized maltreatment among unopened calls rises from 11 percent in the lowest predicted-risk bin to 70 percent in the highest, close to the 45-degree line throughout.

4.1.2 Identification from Screener Variation

Each stage-wise comparison and deployment effect in Section 3 contrasts a status-quo term with an algorithmic term. The status-quo terms are observed. The maltreatment rate V^0 and its two components, $\mathbb{E}[Y_i^0 \mid S_i = 0]$ and $\mathbb{E}[Y_i^1 \mid S_i = 1, D_i = 0]$, are the observed

maltreatment rates among screened-out calls and among screened-in but unopened calls, and the exit shares ω^S and ω^I are the observed shares of unopened calls that exit at each stage. On each risk set $\{m(X_i) \leq t\}$, we also observe two quantities used below, the screen-in rate $\sigma(t) := \Pr(S_i = 1 \mid m(X_i) \leq t)$ and the maltreatment rate among screened-out calls $\mu_0(t) := \mathbb{E}[Y_i^0 \mid S_i = 0, m(X_i) \leq t]$. Panel A of Appendix Figure A.2 plots both across thresholds.

The algorithmic terms require maltreatment among the calls the algorithm leaves unopened. Define $\mu(t) := \mathbb{E}[Y_i^0 \mid m(X_i) \leq t]$, the maltreatment rate among all calls with predicted risk at or below t, and $\mu_I(t) := \mathbb{E}[Y_i^0 \mid m(X_i) \leq t, S_i = 1]$, the same rate among the screened-in calls in that set. These functions deliver the algorithmic terms: $\mu(\tau^S)$ enters Δ^S, $\mu_I(\tau^I)$ enters Δ^I and V^I, and $\mu(\tau^*) = V^{System}$. The first selective labels problem keeps both from being observed, because humans opened cases on some calls in every risk set, which censors their outcomes. The screening-stage deployment V^S requires two further moments on the algorithm's screened-in set: the case-opening rate $\Pr(D_i = 1 \mid m(X_i) \geq \tau^S)$ and the maltreatment rate $\mathbb{E}[Y_i^1 \mid m(X_i) \geq \tau^S, D_i = 0]$ among the calls investigators leave unopened.[5] The second selective labels problem keeps these from being observed, because humans screened out some of the calls the algorithm would screen in, so no investigator ever decided whether to open them.

To build intuition for how our empirical approach addresses both selective labels problems, consider two hypothetical screeners. A "supremely lenient" screener, whose calls never result in a case opening, leaves the outcome observed across their entire caseload, so their calls in the risk set $\{m(X_i) \leq t\}$ identify $\mu(t)$. A "supremely strict" screener, who screens in every call, leaves the case-opening decision observed on the algorithm's screened-in set $\{m(X_i) \geq \tau^S\}$ and the outcome observed on the calls left unopened there, identifying both moments that V^S requires. In the absence of such screeners, we extrapolate the observed screening, case-opening, and maltreatment rates across screeners toward these two limits. Section 4.4 uses this screener variation in other ways, replacing extrapolations with worst-case bounds that rely only on the quasi-random assignment of screeners (Manski, 1990; Rambachan, 2024).

We recover $\mu(t)$ by extrapolating across screeners to a case-opening rate of zero, the limit that the supremely lenient screener attains. Let K_i denote the screener assigned to call i and W_i the indicators for the day-by-shift strata within which screeners are assigned. For

[5]The share λ^S in equation (2) follows from the first of these by Bayes' rule, $\lambda^S = \Pr(S_i^m = 0)/\big(1 - \Pr(D_i = 1 \mid S_i^m = 1)\Pr(S_i^m = 1)\big)$.

each risk threshold t, we estimate the regressions

$$C_i = \sum_k \left[c_{1,k}(t)\, \mathbf{1}\{K_i = k\} + c_{2,k}(t)\, \mathbf{1}\{K_i = k, m(X_i) \leq t\} \right] + W_i^\top \gamma + \varepsilon_i^C, \tag{5}$$

$$Y_i = \sum_k \left[r_{1,k}(t)\, \mathbf{1}\{K_i = k\} + r_{2,k}(t)\, \mathbf{1}\{K_i = k, m(X_i) \leq t\} \right] + W_i^\top \zeta + \varepsilon_i^Y \text{ on } \{C_i = 0\}. \tag{6}$$

The first regression, estimated on all calls, recovers screener k's strata-adjusted case-opening rate on the risk set, $c_k(t) := c_{1,k}(t) + c_{2,k}(t)$; the second, estimated on calls with no case opened, recovers their strata-adjusted maltreatment rate among unopened calls on the risk set, $r_k(t) := r_{1,k}(t) + r_{2,k}(t)$. We regress $\{r_k(t)\}_k$ on $\{c_k(t)\}_k$ across screeners, weighting each screener by the inverse variance of their estimated maltreatment rate, and evaluate the fitted line at $c_k(t) = 0$.

Panel B of Appendix Figure A.2 presents these extrapolations for all calls and for the risk sets at τ^* and τ^S. The extrapolations are short: only 3.8 percent of calls result in an opened case, so the limit of no case openings lies just beyond the observed case-opening rates, and the resulting estimates are correspondingly precise, with standard errors of 0.5 percentage points. The intercept recovers $\mu(t)$ under the assumptions stated below, and we evaluate it at $t = \tau^*$ and $t = \tau^S$. We next recover $\mu_I(t)$ from $\mu(t)$ by iterating expectations over the screening decision on the risk set: $\mu(t) = (1 - \sigma(t))\mu_0(t) + \sigma(t)\mu_I(t)$. Rearranging gives $\mu_I(t) = \mu_0(t) + \big(\mu(t) - \mu_0(t)\big)/\sigma(t)$.

We recover the two moments that V^S requires by extrapolating across screeners to a screen-in rate of one, the limit that the supremely strict screener attains. On the algorithm's screened-in set, we estimate each screener's strata-adjusted screen-in rate, case-opening rate, and maltreatment rate among unopened calls from regressions of the same form, and extrapolate the latter two to a screen-in rate of one. Appendix Figure A.3 presents these extrapolations. They are long, since observed screen-in rates range from roughly 20 to 45 percent, and the resulting estimates are less precise. To assess robustness, Section 4.4 also re-estimates the screening-stage deployment effect under alternative identification strategies for these two moments. Appendix B.3 describes these extrapolations, how we assemble the estimates of the stage-wise comparisons and the deployment effects, and the bootstrap procedure used to calculate standard errors in more detail.

4.1.3 Identifying Assumptions

Our identification strategy rests on three assumptions. First, conditional on the day-by-shift strata, screeners are quasi-randomly assigned to calls. Second, a screener's identity affects a call's subsequent maltreatment and its case-opening decision only through the

screening decision. Third, within a risk set, screening a call in does not change its subsequent maltreatment on average, so there is no average screening-in effect.

First, as mentioned in Section 2.1, a centralized queue routes each call to the longest-waiting available screener within a day and shift, so screener assignment is quasi-random within these strata. Prior work has examined the same assignment process in this setting (Baron et al., 2024; Baron, Jacob and Lombardo, 2026). Appendix Figure A.4 reports a balance test. Call characteristics strongly predict the screening decision (joint F-statistic of 279.1) but are much less predictive of the assigned screener's leave-out screen-in propensity (joint F-statistic of 1.4).

Second, the exclusion restriction on screeners is plausible given their narrow role (Section 2.1). A call lasts about 15 minutes on average, and the screener's only task is to decide whether it meets the criteria for investigation. A screen-out ends the agency's involvement, and once a call is screened in, screeners play no part in the investigation, case management, or service provision. Appendix D.4 develops a test of this exclusion restriction, which fails to reject when applied to screeners.

Third, the extrapolation in equation (5) pools screened-out calls, whose outcome is Y_i^0, with screened-in but unopened calls, whose outcome is Y_i^1, so its intercept recovers $\mu(t)$ only if the two agree on average on the risk set. The assumption restricts the average effect of investigation, not the effect of any individual investigator. Because investigators are assigned quasi-randomly and independently of the screener (Appendix Figure A.5), every screener's screened-in calls receive the same average effect of investigation, so the extrapolation requires only that this common average be zero within the risk set. Prior work finds the average effect of CPS investigations on subsequent maltreatment to be small (Baron et al., 2024; Baron, Jacob and Lombardo, 2026), and Appendix C.4 bounds it in our sample and reports how our estimates move as it varies.

Our strategy does not require the conventional IV monotonicity assumption that screeners share a common ranking of calls. Monotonicity is suspect in settings like ours, where decision-makers may differ in skill as well as in leniency (Chan, Gentzkow and Yu, 2022; Frandsen, Lefgren and Leslie, 2023). Instead, the extrapolations rely on the three assumptions above and on their functional form, which Section 4.4 varies (see Arnold, Dobbie and Hull, 2022; Baron et al., 2024; Angelova, Dobbie and Yang, 2025, for further discussion).

4.2 Stage-Wise Human-Algorithm Comparisons

Figure 2 traces maltreatment among the calls the algorithm leaves unopened, $\widehat{\mu}(t)$, and among the screened-in calls in that set, $\widehat{\mu}_I(t)$, across thresholds expressed as the case-opening rate each algorithmic rule would induce. It also marks the thresholds τ^*, τ^I, and τ^S and

the observed status-quo maltreatment rates. The screened-in curve lies slightly below the all-calls curve at every threshold, by one to three percentage points. Screeners therefore appear to add no selection on maltreatment risk beyond the algorithm's predictions. Panel A of Table 1 reports the stage-wise comparisons $\widehat{\Delta}^S$ and $\widehat{\Delta}^I$. Figure 2 shows the same comparisons as the vertical gaps between the status-quo rates and the counterfactual curves at the matched thresholds.

At the screening stage, the gap between human screeners and the algorithm is large: $\widehat{\Delta}^S = 11.5$ percentage points (s.e. 0.5). Among the calls human screeners screen out, 35.6 percent are followed by subsequent maltreatment, against 24.2 percent among the calls an equally selective algorithm would screen out, a reduction of nearly one third. The gap is of the size that comparisons of humans and algorithms find in other, well-known settings (e.g., Mullainathan, 2025).[6] It holds for every screener in our sample: all 291 screeners underperform an equally selective algorithm, and 97 percent underperform random screening (Appendix D.3).

The gap reflects pervasive disagreement between screeners and the algorithm. Panel A of Appendix Table A.2 cross-tabulates their decisions at the same screen-in rate. The algorithm would reverse 51 percent of screening decisions, and the calls on which the two disagree differ sharply in risk. Among the calls screeners advance and the algorithm would not, 24.1 percent (s.e. 1.1) are followed by subsequent maltreatment, against 52.6 percent (s.e. 0.2) among the calls the algorithm would advance and screeners do not. Panel A of Figure 3 reports standardized differences in intake characteristics between the two groups. The calls screeners advance and the algorithm would not have far less prior CPS history, with about 1.24 standard deviations fewer prior investigations and 0.98 fewer prior substantiations, and more often involve a priority allegation, a physical-abuse allegation, or a law-enforcement reporter.

At the investigation stage, by contrast, human decisions roughly match the algorithm: $\widehat{\Delta}^I = 1.8$ percentage points (s.e. 1.2), indistinguishable from zero. Because $\widehat{\Delta}^I$ bundles investigator selection with any effect of the investigation itself and averages across investigators (Section 3.2), a value near zero has three possible explanations. Investigators may select about as well as the algorithm; they may select worse while their investigations reduce maltreatment among the calls they leave unopened; or the average may mask investigators who outperform the algorithm alongside others who underperform. Section 5 tests whether investigations affect outcomes beyond the case-opening decision and studies this heterogene-

[6] At the judges' release rate, the algorithm of Kleinberg et al. (2018) reduces pretrial crime by roughly 25 percent; Angelova, Dobbie and Yang (2025) and Mullainathan and Obermeyer (2022) report similarly large gaps between judges and physicians and the algorithms they are compared against.

ity.

Investigators and the algorithm disagree far less often than screeners and the algorithm do. Panel B of Appendix Table A.2 cross-tabulates case-opening decisions among screened-in calls. The algorithm would reverse about 17 percent of case-opening decisions, against 51 percent of screening decisions. Among the calls investigators open and the algorithm would not, 37.0 percent (s.e. 11.7) would be followed by subsequent maltreatment if left unopened, against 56.1 percent (s.e. 0.6) among the calls the algorithm would open and investigators do not. Panel B of Figure 3 shows the same pattern as at the screening stage. The calls investigators open and the algorithm would not have about 1.30 standard deviations fewer prior investigations and more often involve threatened harm or a law-enforcement reporter. At both stages, the algorithm's disagreements with humans are concentrated on prior history.

4.3 Deployment Effects and Multistage Decompositions

Panel B of Table 1 reports the effects on subsequent maltreatment of the three deployments of Section 3.3, and Appendix Table A.3 adds their effects on investigations and case openings. Deploying the algorithm at the investigation stage alone reduces subsequent maltreatment by 0.6 percentage points (s.e. 0.4), holding case openings fixed and eliminating investigations. As Proposition 3.1 implies for a terminal stage, this effect equals the weighted investigation-stage comparison $\widehat{\omega}^I \widehat{\Delta}^I$. Deploying the algorithm in place of the multistage system, opening cases directly on its predictions at the status-quo rate, reduces subsequent maltreatment by 1.3 percentage points (s.e. 0.5), about 4 percent of the status-quo rate of 33.8 percent. It holds the case-opening rate fixed by construction and eliminates investigations, which fall by 36.2 percentage points, so a policymaker who values all three quantities would prefer it to the status quo.

Deploying the algorithm at the screening stage alone reduces subsequent maltreatment by an estimated 1.1 percentage points (s.e. 2.3), with a 95 percent confidence interval from -3.3 to 5.6. The interval is wide because of the long extrapolation to a screen-in rate of one, and the point estimate is consistent with other, more informative identification strategies discussed in Section 4.4. This deployment holds investigations fixed and raises the case-opening rate by 1.4 percentage points (s.e. 2.2), a change we cannot distinguish from zero.

Set beside the stage-wise comparisons, these deployment effects pose a puzzle. Judged stage by stage, human screeners leave 11.5 percentage points more subsequent maltreatment than the algorithm, yet deploying the algorithm at the screening stage reduces subsequent maltreatment by an estimated 1.1 points, and deploying it in place of the entire system by 1.3 points. Propositions 3.1 and 3.2 show why the two need not agree. Figure 4 shows that the terms responsible, the downstream investigator response and the aggregation correction,

are both large.

Panel A of Figure 4 decomposes the screening-stage deployment effect following Proposition 3.1. Better screening decisions contribute the weighted screening-stage comparison, $\widehat{\omega}^S\widehat{\Delta}^S = 7.6$ percentage points (s.e. 0.3), and the exit-margin reweighting term is negligible, $(\widehat{\omega}^S - \widehat{\lambda}^S)\widehat{\Gamma}^S = 0.2$ (s.e. 0.4). But deploying the algorithm at the screening stage also changes the pool of calls that human investigators face, and the downstream investigator response offsets most of the gain: $\widehat{\omega}^I\widehat{\mathcal{R}}^I = -6.7$ percentage points (s.e. 2.3), with a 95 percent confidence interval from -11.2 to -2.2. On net, the deployment reduces subsequent maltreatment by 1.1 percentage points (s.e. 2.3), and its confidence interval includes zero but excludes a gain as large as the 7.6 point weighted comparison.

The downstream investigator response combines a change in the risk of the screened-in pool with a change in investigator selection (Section 3.3), and the first is large. The algorithm's screened-in pool replaces the calls that screeners advance and the algorithm would not, 24.1 percent of which are followed by subsequent maltreatment, with the calls it would advance and screeners do not, 52.6 percent of which are. Yet investigators' case-opening rate on this pool barely moves, rising by 1.4 percentage points (s.e. 2.2) to about 12 percent, so maltreatment among the calls they leave unopened rises from 30.0 percent (s.e. 0.4) to 49.8 percent (s.e. 7.2).

Why does the case-opening rate change so little when the pool becomes so much riskier? Figure 5 answers this question with a Kitagawa-Oaxaca-Blinder decomposition of investigators' case-opening rate across the two pools. Each point is an intake characteristic, plotted by how much its mean shifts between the screeners' pool and the algorithm's, in standard deviations, against how much investigators' case-opening rate responds to it, in percentage points per standard deviation; its contribution to the change in the opening rate is the product of the two. The pool shifts most along prior history, by half to nine tenths of a standard deviation, but openings respond to prior history by only one to two points per standard deviation, and the characteristic openings respond to most, a law-enforcement reporter, is no more common in the algorithm's pool. Investigators are relatively insensitive to the characteristics that change most, so they keep opening the calls they always opened and leave most of the higher-risk calls the algorithm screens in unopened.

Panel B of Figure 4 decomposes the system-wide deployment effect following Proposition 3.2. The weighted stage-wise comparisons, $\widehat{\omega}^S\widehat{\Delta}^S = 7.6$ percentage points (s.e. 0.3) and $\widehat{\omega}^I\widehat{\Delta}^I = 0.6$ (s.e. 0.4), together credit the algorithm with 8.2 points, about six times the system-wide effect of 1.3. The gap is the aggregation correction, $\widehat{\omega}^I\widehat{\mathcal{C}}^{\text{Sys}} = -6.9$ percentage points (s.e. 0.6), with a 95 percent confidence interval from -8.1 to -5.6.

Equation (4) says which calls produce the correction. The first two terms concern the

calls that screeners advance and the algorithm would screen out, and investigators open only 8 percent of them. In the status quo, investigators therefore correct most of screeners' low-risk screen-ins, yet the screening comparison credits the algorithm's screen-out of each of these calls, including the redundant screen-outs that leave the case-opening decision unchanged. The few calls investigators open are double counted. Screeners and investigators alike underweight prior history relative to the algorithm (Figure 3), so the low-history calls screeners advance that the algorithm would screen out are also the calls investigators open that the algorithm would not. Threshold disagreement matters too. The screening comparison credits the algorithm with screening in the high-risk calls that screeners screen out, but the system-wide algorithm, with its higher threshold, opens only the riskiest of them.

4.4 Robustness and Extensions

Our estimates rest on the extrapolation across screeners, on our measure of subsequent maltreatment, and on the assumption that screening a call in has no average effect on its subsequent maltreatment. We examine each in turn and then attribute the system-wide deployment effect to the two stages; Appendix C describes these exercises in more detail.

Appendix Table A.5 re-estimates Table 1 and Figure 4 under four alternative implementations of the extrapolation: a local-linear fit, a quadratic fit, a linear fit without the inverse-variance weights, and a linear fit with screener-level rates estimated without the day-by-shift fixed effects. In each, the screening-stage comparison remains between 10.4 and 12.5 percentage points, the weighted stage-wise comparisons sum to well above the system-wide deployment effect, and the screening-stage deployment effect stays between 0.5 and 1.8 percentage points. Extrapolating instead from each screener's screened-out calls alone, without case-opening variation, yields much less precise estimates.

We next replace the extrapolation with worst-case bounds that use only the quasi-random assignment of screeners and the binary support of the outcome (e.g., Manski, 1990). Panels A and B of Appendix Figure A.6 plot the resulting bounds on the two counterfactual curves, and columns (6) and (7) of Appendix Table A.5 report the objects they imply. The bounds are wide, and those on the investigation-stage objects contain zero. Even so, the screening-stage comparison remains between 9.2 and 11.7 percentage points, the weighted comparisons sum to well above the system-wide effect, and the screening-stage deployment effect lies between -0.4 and 1.3 percentage points.

The screening-stage deployment effect also rests on the long extrapolation to a screen-in rate of one, so Appendix Table A.6 re-estimates it under two alternative identification strategies for the two moments on the algorithm's screened-in pool. The first bounds these moments under a monotone selection assumption: the calls screeners screen out would be

opened at weakly lower rates and followed by weakly less maltreatment than the calls they screen in (e.g., Manski and Pepper, 2000). Only the upper bound on maltreatment is informative, since the lower bound imputes no maltreatment to every unobserved call. The second assumes selection on observables: conditional on the intake characteristics the algorithm uses, the screening decision is as good as random with respect to a call's case opening and subsequent maltreatment. Under the informative bound and under selection on observables, the downstream investigator response is -5.4 and -6.3 percentage points, and the deployment effect is 2.5 and 1.6 percentage points, a fraction of the screening-stage comparison.

Appendix Table A.7 re-estimates every object under eight alternative outcome definitions: re-referral horizons of two to five months, and four outcomes that require an escalating system response within six months (a new investigation, a substantiated investigation, a subsequent case opening, and a subsequent removal). Under every definition, the screening-stage comparison is positive and precisely estimated, and the weighted comparisons sum to more than the system-wide effect. For outcomes that require both a later screener and investigator to act, the downstream investigator response is small, and the screening-stage deployment effect is comparable to or larger than its weighted stage-wise comparison. We prefer the re-referral measure because these alternative outcomes register maltreatment only when later screeners and investigators act on it, so they inherit any errors in those decisions.

Appendix Table A.8 bounds the average screening-in effect on the risk sets $\{m(X_i) \leq \tau^*\}$ and $\{m(X_i) \leq \tau^S\}$ using variation across screeners (Appendix C.4). With case opening as the censoring margin the bounds are wide, but with the rarer placement decision as the margin they lie within a few percentage points of zero, in line with the small screening-in effect that Baron et al. (2024) estimate in Michigan over an earlier period. Because each estimate is a known linear function of the assumed effect, Appendix Table A.9 also reports them for effects between -4 and 4 percentage points. Across this range, the screening-stage comparison stays between 9.8 and 13.1 percentage points, while the screening-stage deployment effect stays between 0.1 and 2.2.

The system-wide deployment effect can also be attributed to the two stages. The incremental effect of deploying the algorithm at one stage depends on whether the other stage has already been automated, so the attribution depends on the order of deployment. Following Shapley (1953), Appendix C.5 averages each stage's incremental effect over the two orderings. The screening and investigation stages contribute 0.9 (s.e. 1.2) and 0.4 (s.e. 1.2) percentage points of the 1.3 point system-wide effect, respectively.

5 Understanding Investigator Heterogeneity and Case-Opening Decisions

A conventional screening algorithm forgoes most of its gain because of how investigators respond to the calls it sends them (Section 4.3), so recovering that gain requires understanding what the investigation stage contributes. In this section, we therefore study investigators themselves. Although investigators roughly match the algorithm on average, about a third of them outperform it, and we ask what allows them to do better. Screeners, by contrast, vary little, and every one of them performs substantially worse than the algorithm (Appendix D.3).

5.1 Variation in Performance across Investigators

We summarize each investigator g's performance by the investigator-level analog of the investigation-stage comparison,

$$\Delta_g^I := \mathbb{E}\big[Y_i^{1,g} \mid G_i = g,\ D_i = 0\big] - \mathbb{E}\big[Y_i^0 \mid G_i = g,\ D_i^{m,g} = 0\big]\,,$$

where G_i denotes the investigator assigned to call i, $Y_i^{1,g}$ is call i's subsequent maltreatment if investigated by g and left unopened, and $D_i^{m,g}$ is the algorithmic rule matched to g's case-opening rate (Appendix D.1). Because the investigation stage is terminal, deploying the algorithm in place of g changes system-wide maltreatment by $\Pr(G_i = g, S_i = 1 \mid C_i = 0)\,\Delta_g^I$ (Proposition D.1), so a negative Δ_g^I means that g outperforms the algorithm. As at the stage level, the comparison splits into the effect of g's investigation on the calls left unopened and g's selection of which calls to open,

$$\Delta_g^I = \underbrace{\mathbb{E}[Y_i^{1,g} - Y_i^0 \mid G_i = g, D_i = 0]}_{\text{Direct effect of investigator } g.} + \underbrace{\mathbb{E}[Y_i^0 \mid G_i = g, D_i = 0] - \mathbb{E}[Y_i^0 \mid G_i = g, D_i^{m,g} = 0]}_{\text{Investigator } g\text{'s selection.}}, \tag{7}$$

so an investigator may outperform the algorithm by acting on calls, by ranking them better, or both.

We estimate each Δ_g^I with the same empirical strategy as the stage-level comparison, comparing investigator g's strata-adjusted maltreatment rate among unopened calls with the curve $\mu_I(\cdot)$ evaluated at the threshold matched to g's case-opening rate. To summarize the variation across investigators, we shrink each estimate toward the population mean with a parametric empirical Bayes model and report posterior average effects (Bonhomme and Weidner, 2022), as described in Appendix D.2.

Figure 6 plots each investigator's estimated maltreatment rate among unopened calls

against their estimated case-opening rate, together with the algorithm's curve $\mu_I(t)$, so an investigator's vertical distance above the curve at their own case-opening rate is Δ_g^I. Case-opening rates range from near zero to about 40 percent, which is why we compare each investigator to the algorithm at their own case-opening rate. At a given case-opening rate, investigators also differ in the maltreatment they leave unopened: the caseload-weighted standard deviation of Δ_g^I, reported in the figure, is 3.2 percentage points (s.e. 0.2). This is variation in performance rather than in leniency, and it is evidence against the strict monotonicity assumption that investigators with the same case-opening rate leave the same subsequent maltreatment (Chan, Gentzkow and Yu, 2022; Frandsen, Lefgren and Leslie, 2023). Investigators' maltreatment rates are only weakly related to their case-opening rates: an OLS regression of one on the other yields a coefficient of -0.07 (s.e. 0.04), indistinguishable from zero.

Accounting for sampling error with posterior average effects, 33.9 percent (s.e. 1.2) of investigators outperform the algorithm at their own case-opening rate. We refer to investigators with a negative posterior mean of Δ_g^I as higher-performing and to the rest as lower-performing, a descriptive split that the rest of this section uses to ask what allows the higher performers to do better.

As a first step to understand what differentiates the higher-performing investigators, Appendix Table A.10 regresses each investigator's Δ_g^I, and the posterior probability that the investigator outperforms the algorithm, on caseload, gender, race, and ethnicity predicted from names, and, for the roughly half of investigators we can link to MDHHS personnel records, age and tenure (Appendix B.1). Investigators predicted to be Black perform somewhat better and longer-tenured investigators somewhat worse, but no other characteristic predicts performance, and together they explain almost none of its variation ($R^2 = 0.03$). As a result, who investigators are does not account for the heterogeneity in performance, which leaves the two channels in the decomposition of Δ_g^I: what investigators do during an investigation and what they learn from it.

5.2 Do Investigators Affect Outcomes Beyond Case Opening?

The first way investigators could come to differ in performance is through what an investigation does to a family apart from the case-opening decision. Over the course of a home visit, an investigator can connect a caregiver to substance-use or mental health treatment, put a safety plan in place, arrange material assistance, offer advice, or simply signal that the family is being watched, and any of these could change subsequent maltreatment. If higher-performing investigators do more of this, the effect of their investigations on the calls they leave unopened could account for their advantage.

We begin with the dispositions recorded in the structured data. Besides opening a case, an investigator can substantiate maltreatment without opening one, a lighter response that still puts a finding on the record, or remove a child to foster care, the most intensive response, which is rare and typically accompanies an opening. An investigator who opens fewer cases could act more along these other margins, substantiating more allegations without opening them or removing children at a different rate. Panel A of Figure 7 reports, for each action, the coefficient from regressing an indicator for the action on an indicator that the case was handled by a higher-performing investigator, with fixed effects for the month-by-ZIP-code-by-shift strata within which investigators are assigned. Higher-performing investigators are slightly less likely to open a case and to substantiate maltreatment, by 1.4 and 1.5 percentage points, and slightly less likely to remove a child, by 0.6 points, and they substantiate without opening at the same rate. On the formal margins MDHHS records, higher-performing investigators do not act more.

Formal dispositions are not the only actions investigators take, and the treatment referrals, safety plans, assistance, and advice described above go unrecorded in the department's structured data. Investigators do describe them in narrative case notes, which record each investigation as it unfolds and are available for calls from 2018 on (Appendix E). The notes are detailed accounts of the whole investigation, documenting home visits, interviews with children and caregivers, contacts with teachers, doctors, and other collateral sources, the investigator's assessment of the child's safety and the household, the actions taken, and the rationale for the final disposition. They therefore offer a far richer record of each investigation than the structured data.

We use a large language model to label every note for five actions that the notes often mention and the structured data omit. Substance-use services include a test, assessment, referral, or treatment requirement that the investigator arranged, and mental health services include a referral to, or help accessing, counseling, psychological evaluation, or psychiatric treatment. A safety plan is a safety agreement or a concrete restriction, such as an alternate-caregiver arrangement or a rule against physical discipline; material assistance is help with housing, food, utilities, transportation, childcare, or other essentials; and parenting guidance is advice, education, or a referral about discipline, supervision, safe sleep, or child development. For each action, GPT-5-mini reads the de-identified note with a prompt that defines the action, gives coding rules separating an action the investigator took from an incidental mention, and supplies worked examples, so that a caregiver's history of substance use, for instance, is not coded as an action. Appendix E.1 describes the labeling, and Appendix Table A.11 gives example excerpts for each action.

Even on these informal margins, higher- and lower-performing investigators act alike.

None of the five differences in Panel B of Figure 7 is distinguishable from zero, and within quintiles of the algorithm's predicted risk, partitioning cases into opened, closed with a labeled action, and closed without one shows no difference between the two groups (Appendix Figure A.7).

The actions we can measure may still miss what some investigators do, so we also statistically test for direct effects across investigators without specifying the channel. Suppose that investigator identity has no direct effect on outcomes, $Y_i^{1,g} = Y_i^1$ for all g, so that a call investigated and left unopened has the same subsequent maltreatment whoever investigates it. Under quasi-random assignment within strata, every investigator's caseload then has the same average outcome $\bar{\mu} = \mathbb{E}[Y_i^1 \mid S_i = 1]$, and each investigator's case-opening rate d_g and share of calls left unopened and followed by maltreatment L_g bound it, $L_g \leq \bar{\mu} \leq L_g + d_g$, an interval as wide as the investigator's case-opening rate. If no value of $\bar{\mu}$ lies in every investigator's interval, some investigators must affect subsequent maltreatment beyond the case-opening decision, an implication in the spirit of instrument validity tests (Kitagawa, 2015; Frandsen, Lefgren and Leslie, 2023). We group investigators into quintiles of their strata-adjusted case-opening rate and compare the largest lower bound with the smallest upper bound using a least-favorable critical value (Appendix D.4). Appendix Figure A.8 plots each quintile's bracket at its case-opening rate. The lowest quintile opens 2.4 percent of its cases, so its bracket is narrow, from 29.7 to 32.1 percent, and every other quintile's bracket contains it. The brackets therefore intersect, with the largest lower bound 2.4 percentage points below the smallest upper bound, and we fail to reject the hypothesis that investigators affect subsequent maltreatment only through their case-opening decisions (p-value of 1.00; Appendix Table A.12).

We therefore find no evidence that investigators affect subsequent maltreatment beyond their case-opening decisions, whether in the formal actions the administrative data record, the informal actions the case notes describe, or a statistical test that specifies no specific channel. This leads us to take the case-opening decision as the only channel through which an investigator affects outcomes, $Y_i^0 = Y_i^1$, and to write Y_i^* for the common value. The comparison Δ_g^I is then a comparison of selection alone, so the heterogeneity in Section 5.1 must come from how investigators decide which cases to open.

5.3 Uses of Intake Characteristics and Private Information

The second way investigators could come to differ in performance is through what they learn over the course of an investigation. An interview, a look inside the home, or a call to a teacher or doctor can reveal risk, or its absence, that the intake characteristics do not capture, so that an investigator could rank calls better than the algorithm. If investigators' case-opening

decisions carry information about maltreatment beyond the algorithm's prediction (Alur et al., 2023; Alur, Raghavan and Shah, 2024), then within a bin of predicted risk the calls they open are riskier than the calls they leave unopened. If they do not, so that $D_i \perp Y_i^* \mid m(X_i)$, the calls left unopened are no safer than the bin as a whole: within a quintile q of predicted risk among screened-in calls, $\mathbb{E}[Y_i^* \mid S_i = 1, q] = \mathbb{E}[Y_i \mid S_i = 1, D_i = 0, q]$.

The maltreatment rate among unopened calls is observed, but, because investigators opened cases on some calls, the overall rate within a quintile is not. Because investigators show no evidence of direct effects, their quasi-random assignment can now play the role that screener assignment played in Section 4.1. Within each quintile, we extrapolate investigators' strata-adjusted maltreatment rates among unopened calls to a case-opening rate of zero, which identifies $\mathbb{E}[Y_i^* \mid S_i = 1, q]$. We call the difference between the overall maltreatment rate in a quintile and the rate among its unopened calls the "outcome gap." It equals the quintile's case-opening rate times the difference in maltreatment between the calls investigators open and the calls they leave unopened, so it is positive when investigators open the riskier calls. We compute the outcome gap for all investigators and for higher- and lower-performing investigators separately.

Figure 8 reports the outcome gaps. Pooled across investigators, the gap is small in every quintile and slightly negative in the lower ones, so within a quintile of predicted risk, case openings on average carry no information beyond the algorithm. The two groups, however, differ sharply. For higher-performing investigators, the gap is positive in every quintile, with intervals that exclude zero, and largest in the highest quintile; for lower-performing investigators, it is negative in every quintile. Within a quintile of predicted risk, higher performers open the riskier calls and lower performers open the safer ones. Where does the higher performers' information come from? Their decisions may covary with the intake characteristics X_i within a quintile, or they may draw on what the investigation reveals beyond those characteristics ("private information").

Consider first whether higher performers use the intake characteristics differently. Appendix Table A.13 regresses the case-opening decision on the intake characteristics and the algorithm's predicted risk, separately for the two groups, so that each coefficient is the weight a group places on a characteristic beyond its contribution to predicted risk. The two groups weight the intake characteristics similarly. Lower performers put more weight on improper-supervision and other allegations, by 1.8 and 1.9 percentage points, and the remaining differences are smaller and indistinguishable from zero. Nor do these differences matter for outcomes. We distill each group's case-opening decisions into a rule that uses only the intake characteristics and trace the subsequent maltreatment each rule would leave across case-opening rates, as $\mu_I(t)$ does for the algorithm. The two curves in Appendix Figure A.9

are nearly indistinguishable, so differences in how investigators use the intake information do not explain the gap. If higher performers rank cases better, it must be because of what they learn in the field.

The case notes of Section 5.2 record what investigators learn as well as what they do, describing in their own words the evidence they assemble during home visits, interviews, and contacts with teachers, doctors, and others. A note reflects both what the investigator observed and what they chose to write down, and we cannot separate the two. We can therefore use the notes to ask whether higher- and lower-performing investigators differ in the evidence they collect and record.

Rather than decide in advance which evidence matters, we use large language models to surface the evidence in the notes that predicts case opening, following recent work on hypothesis generation from text (e.g., Ludwig and Mullainathan, 2024; Zhong et al., 2022; Zhou et al., 2024; Batista and Ross, 2025; Modarressi, Spiess and Venugopal, 2025). Within each decile of an algorithm that predicts case opening from the intake characteristics alone, we contrast the notes whose text most predicts an opening with the notes whose text most predicts none, so that the contrast isolates evidence the intake characteristics do not capture. A large language model reads these contrasting sets, with the decision and any information about the investigator masked, and names recurring, concrete evidence that distinguishes them, and a second model aggregates the answers into a compact set of candidate themes. We then label every note for each candidate theme and keep the five themes most predictive of case opening on a held-out sample. Throughout, the models are instructed to flag and discard language that explicitly states the disposition rather than describes the case. Appendix E.2 provides additional detail on each step.

The pipeline yields five themes, which Appendix Table A.15 defines and illustrates with excerpts. Two are potentially incriminating evidence of maltreatment: "corroborated harm or injury," in which a physical injury, a child's disclosure, or external documentation substantiates the allegation, and a "caregiver actively involving the child in dangerous conduct." Two are potentially exculpatory: an "absence, ambiguity, or contradiction of abuse evidence," in which the allegation is unsupported, recanted, or medically indeterminate, and "exculpatory corroboration of adequate care," in which reassuring observations and collateral contacts indicate that the home is adequate. The fifth, "caregiver absence or unstable placement," marks children left without a secure caregiving arrangement.

How much of the notes' content do these five themes capture? Appendix Table A.16 benchmarks them against embeddings of the full text, which predict case opening almost perfectly, with an AUC of 0.995. Because a note often restates the decision it describes, this is an upper bound on the evidence the notes carry about the decision. The five themes

alone reach an AUC of 0.914, which in the completeness measure of Fudenberg et al. (2022) recovers about 84 percent of the full text's predictive content relative to an uninformative prediction.

With the five themes in hand, we ask whether higher- and lower-performing investigators record them at different rates. Figure 9 reports, for each theme, the coefficient from regressing an indicator that the note records the theme on an indicator that the case was handled by a higher-performing investigator, with fixed effects for the strata within which investigators are assigned. Because assignment within strata is quasi-random, a nonzero coefficient reflects a difference in what investigators observe and record, not in the cases they encounter. Higher-performing investigators are 1.5 percentage points (s.e. 0.7) more likely to record exculpatory corroboration of adequate care and 2.0 points (s.e. 0.6) more likely to record an absence, ambiguity, or contradiction of abuse evidence. They record the three remaining themes at rates indistinguishable from those of lower performers.

Higher performers thus record slightly more of the evidence that cuts against the allegation and about as much of the evidence that supports it. If the notes only justified the decision after the fact, an investigator who opened a case would record the harm and one who closed it would record the reassurance. The themes do not follow that pattern (Appendix Table A.17). Corroborated harm is recorded in 41 percent of the cases investigators leave unopened, ambiguous or contradictory abuse evidence in 40 percent of the cases they open, corroboration of adequate care in 17 percent of opened cases, and four in ten notes record evidence on both sides of the decision. Because a call reaches an investigator only after a screener has found the allegation credible, evidence of adequate care or of ambiguity is the evidence that would overturn the working hypothesis. Recording more of it is what a search for disconfirming evidence, the classic corrective for confirmation bias, would produce (Nickerson, 1998; Heller et al., 2017; Dube, MacArthur and Shah, 2025), although it is equally consistent with a higher evidentiary standard for substantiating an allegation.

Do the two groups also act differently on the evidence they record? Appendix Table A.18 regresses the case-opening decision on the five themes separately for each group, controlling for the intake characteristics and the algorithm's predicted risk. Both groups respond in the expected directions, and for three of the themes their responses coincide. Lower performers respond more strongly to the other two, with a note recording adequate care lowering their probability of opening by 4.6 percentage points more and one recording corroborated harm raising it by 1.3 points more.

To summarize, higher-performing investigators record more evidence that challenges the initial allegation, but respond similarly to lower-performing investigators when such evidence is recorded. Importantly, this does not necessarily mean that they are less likely to open cases.

Instead, searching for exculpatory evidence may help them better distinguish cases that warrant intervention from those that do not. These patterns suggest that their advantage may lie in the information gathered during the investigation, rather than in how they use that information to make case-opening decisions

The themes are hypotheses generated from the notes, and the differences between the groups are descriptive. Whether a theme itself carries information about maltreatment is harder to assess. Unopened calls are selected on the very decisions the themes predict, and because investigators write the notes, conditioning on what they record breaks the quasi-random assignment that our extrapolations require.

6 Designing a Screening Algorithm for the Multistage System

Judged stage by stage, the algorithm has the most to add at the screening stage, where screeners leave far more subsequent maltreatment than an equally selective algorithm and appear to add no selection on maltreatment risk beyond its prediction. At the investigation stage, by contrast, the case-opening decisions of higher-performing investigators are informative about maltreatment beyond the algorithm's prediction. Yet neither deployment of Section 4.3 combines the two. Deploying the algorithm at the screening stage recovers little of its stage-wise gain, because investigators open few of the riskier calls it sends them, and deploying it in place of the entire system forgoes the investigation altogether. We therefore ask whether a screening algorithm designed for the multistage system, one that accounts for how investigators decide downstream, can do better.

6.1 The Design Problem

Motivated by Section 5.2, we assume for simplicity that investigations have no direct effects, so that a call's subsequent maltreatment absent a case opening is Y_i^* whether or not it is investigated (Appendix F relaxes this assumption). We also hold investigators' decisions fixed, so that call i receives the case-opening decision D_i whichever screening rule sends it to investigation. A screening algorithm acts on the prediction $m(X_i)$, which we treat as taking discrete values $\tilde{m}$, and for each value either screens the call out, opens a case directly, or sends the call to investigation, where the investigator decides. As in Section 3.3, the policymaker cares about subsequent maltreatment among unopened calls, case openings, and investigations, and we weight maltreatment by one, a case opening by κ_C, and an investigation by κ_S.

A conventional algorithm that fully automates the multistage system opens a case whenever a call's predicted risk, $\mathbb{E}[Y_i^* \mid m(X_i) = \tilde{m}]$, exceeds the cost of a case opening κ_C, and

screens the call out otherwise. It never investigates, and it attains the highest welfare among rules that decide directly from the prediction (Appendix F). When κ_C is set so that it opens cases at the status-quo rate, it is the system-wide deployment of Section 3.3. The design question is whether, and on which calls, a screening algorithm should send calls to investigators instead.

Sending calls with prediction $\tilde{m}$ to investigators can do better than deciding directly if investigators, using what they learn, open the riskier of these calls and leave the safer ones unopened. We measure this by the "value of an investigation," the welfare of sending the calls to investigators minus that of the better immediate decision,

$$\mathrm{VOI}(\tilde{m}) := \min\left\{\mathbb{E}[Y_i^* \mid m(X_i)=\tilde{m}],\, \kappa_C\right\} - \mathbb{E}\left[(1-D_i)\,Y_i^* \mid m(X_i)=\tilde{m}\right] - \kappa_C \Pr(D_i=1 \mid m(X_i)=\tilde{m}). \tag{8}$$

The first term is the cost of deciding directly, the smaller of the predicted risk and the cost of a case opening, and the last two are the maltreatment investigators leave unopened and the cost of the cases they open. Appendix F shows that $\mathrm{VOI}(\tilde{m}) > 0$ if and only if investigators open some but not all of these calls, and the calls they open are riskier than κ_C while the calls they leave unopened are safer. An investigation therefore has value only if investigators' decisions use information beyond the algorithm's prediction, as Section 5.3 found for higher-performing investigators.

The following result, proved in Appendix F, characterizes the optimal screening algorithm.

Proposition 6.1 (Optimal screening algorithm)**.** *Fix $\kappa_C > 0$ and $\kappa_S \geq 0$. The screening algorithm that sends calls with prediction $\tilde{m}$ to investigation if $\mathrm{VOI}(\tilde{m}) \geq \kappa_S$, and otherwise decides as the conventional algorithm does, attains the highest welfare among all screening algorithms. Its welfare gain over the conventional algorithm is $\mathbb{E}\left[(\mathrm{VOI}(m(X_i)) - \kappa_S)^+\right]$, which is zero if and only if $\mathrm{VOI}(\tilde{m}) \leq \kappa_S$ at every $\tilde{m}$.*

Proposition 6.1 shows that the optimal algorithm thresholds the value of an investigation at its cost, rather than the predicted risk at the cost of a case opening, and falls back on the conventional rule only where an investigation is not worth its cost. Because the value of an investigation describes the decisions of a particular set of investigators, both the optimal algorithm and its gain depend on who investigates. We next compute them for all investigators and for higher-performing investigators.

6.2 The Welfare Effects of Alternative Screening Algorithms

How much would the optimal screening algorithm add in Michigan CPS, and how does its gain depend on the cost of an investigation and on who investigates? We bin the predic-

tion into quintiles of predicted risk, which leave enough screened-in calls in each to estimate investigators' decisions precisely, so that a screening algorithm assigns each quintile one of the three actions. We compute the value of an investigation, the optimal algorithm, and its welfare gain for all investigators and for higher-performing investigators alone, in which case D_i is the decision of an investigator drawn from that group. We identify the averages in equation (8) under selection on observables at the screening stage, so that screeners' decisions are as good as random given the intake characteristics. This assumption is reasonable here, because screeners appear to add no selection beyond the prediction and our earlier estimates are nearly unchanged under it (Section 4.4), and it yields far more precise estimates than extrapolating across screeners within each quintile. Finally, we set κ_C so that the conventional algorithm opens as many cases as the multistage system does, and we report results as κ_S varies (Appendix F.3).

Panel A of Figure 10 plots the value of an investigation within each quintile of predicted risk. For all investigators, it is negative in the two lowest quintiles, at about -1.9 and -1.5 percentage points with intervals that exclude zero, indistinguishable from zero in the third and fourth, and about 4 points in the highest, so on average an investigation is worth having only for the riskiest calls. For higher-performing investigators, it is positive in every quintile, with intervals that exclude zero, and largely rises with predicted risk, from about 3 percentage points in the lowest quintile to about 15 in the highest. If investigations were costless, the optimal algorithm would therefore screen out the three lowest quintiles and send the two highest to investigators when all investigators are downstream, and send every quintile to investigators when higher-performing investigators are.

Panel B plots the optimal algorithm's welfare gain over the conventional algorithm, as a share of the conventional algorithm's loss, as the cost of an investigation rises from zero to three tenths of the cost of a case opening. With higher-performing investigators downstream, the gain is large. When investigations are free, the optimal algorithm improves welfare by 18.5 percent over the conventional algorithm, with a 95 percent confidence interval from about 12 to 26 percent, about seven times the 2.6 percent gain with all investigators downstream, whose interval runs from about 1.5 to 5 percent. As investigations become more costly, the optimal algorithm sends fewer quintiles to investigators, starting with those where the value of an investigation is smallest, and the gain falls. With higher-performing investigators, it remains positive until an investigation costs about a quarter of a case opening, and even at a tenth it exceeds the gain with all investigators when investigations are free. With all investigators downstream, the gain vanishes once an investigation costs less than a tenth of a case opening.

A screening algorithm that accounts for how investigators decide downstream improves

on one trained only to predict outcomes, and by far the most when the investigators downstream are those whose decisions use information beyond the prediction. The improvement comes from the structure of the multistage system, in which the screening stage decides which calls to pass on and the investigation stage is tasked with collecting the information the screening stage lacks before it decides.[7] The value of an algorithm at one stage of a multistage system depends on how the other stages decide.

7 Conclusion

Many consequential decisions are made in multistage systems, where cases pass through successive stages of human review and the people at later stages gather information that did not exist when a case entered. We developed a framework that compares humans and algorithms at each stage, recovers the system-wide effect of deploying an algorithm, and identifies how humans add value to the final decision. We applied it to child protective services in Michigan, where hotline screeners decide which allegations to investigate and investigators decide which cases to open.

Judged stage by stage, hotline screeners underperform the algorithm by a wide margin and none outperforms it, while investigators perform about as well as the algorithm on average and a substantial minority outperform it. Yet deploying the algorithm, at the screening stage or in place of the entire system, reduces subsequent maltreatment by far less than these comparisons suggest, because they do not account for how decisions interact across stages. Investigators respond only weakly to the riskier calls a screening algorithm sends them, and this response offsets most of the gain from algorithmic screening; at the system level, the stage-wise comparisons miscount how screening and case-opening decisions combine into the final decision. We find that higher-performing investigators open cases using information they gather in the field and record in their case notes, which is unavailable to an algorithm at intake. A screening algorithm that sends calls to investigation where this information is worth its cost improves on one trained only to predict outcomes, and by far the most when higher-performing investigators handle the calls.

Most work on algorithms in decision-making compares a human and an algorithm at a single decision, holding the rest of the system fixed. Our results show that in a multistage system the effects of an algorithm depend on which stage it takes over, on how the humans at the other stages respond, and on what those humans learn that the algorithm cannot. The best way to use an algorithm may therefore not be to simply replace human decisions, but

[7] Recent machine learning methods optimize directly whether a decision should be made by an algorithm or deferred to an existing decision-maker (e.g., Madras, Pitassi and Zemel, 2018; Raghu et al., 2019; Mozannar and Sontag, 2020). In a multistage system, the deferral is built into the institution, and the decision-maker deferred to is tasked with collecting information the algorithm does not have.

to allocate tasks between humans and algorithms according to their respective advantages. Algorithms can process information already available, while humans can gather new information that changes the appropriate decision. Understanding these respective advantages, and how decisions interact across stages, opens new avenues for algorithm evaluation and design in multistage systems such as hiring, medicine, and tax enforcement.

References

Acemoglu, Daron, and Pascual Restrepo. 2019. "Automation and New Tasks: How Technology Displaces and Reinstates Labor." *Journal of Economic Perspectives*, 33(2): 3–30.

Administration for Children, Youth and Families. 2026. "Modernizing Child Welfare Technologies and Tools: Opportunities for Predictive Risk Modeling to Improve Child Safety and Outcomes." Administration for Children, Youth and Families, U.S. Department of Health and Human Services Issue Brief.

Agarwal, Nikhil, Alex Moehring, and Alexander Wolitzky. 2025. "Designing Human-AI Collaboration: A Sufficient-Statistic Approach." National Bureau of Economic Research Working Paper 33949.

Agarwal, Nikhil, Alex Moehring, Pranav Rajpurkar, and Tobias Salz. 2023. "Combining Human Expertise with Artificial Intelligence: Experimental Evidence from Radiology." National Bureau of Economic Research Working Paper 31422.

Albright, Alex. 2024. "The Hidden Effects of Algorithmic Recommendations." Federal Reserve Bank of Minneapolis Opportunity and Inclusive Growth Institute Working Paper 104.

Alur, Rohan, Loren Laine, Darrick Li, Manish Raghavan, Devavrat Shah, and Dennis Shung. 2023. "Auditing for Human Expertise." Vol. 36, 79439–79468. Curran Associates, Inc.

Alur, Rohan, Manish Raghavan, and Devavrat Shah. 2024. "Human Expertise in Algorithmic Prediction." Vol. 37, 138088–138129. Curran Associates, Inc.

Angelova, Victoria, Will S. Dobbie, and Crystal Yang. 2025. "Algorithmic Recommendations and Human Discretion." *Review of Economic Studies*.

Arnold, David, Will S. Dobbie, and Peter Hull. 2022. "Measuring Racial Discrimination in Bail Decisions." *American Economic Review*, 112(9): 2992–3038.

Autor, David H. 2015. "Why Are There Still So Many Jobs? The History and Future of Workplace Automation." *Journal of Economic Perspectives*, 29(3): 3–30.

Bansal, Gagan, Besmira Nushi, Ece Kamar, Eric Horvitz, and Daniel S. Weld. 2021. "Is the Most Accurate AI the Best Teammate? Optimizing AI for Teamwork." Vol. 35, 11405–11414.

Baron, E. Jason, Joseph J. Doyle, Natalia Emanuel, Peter Hull, and Joseph Ryan. 2024. "Discrimination in Multiphase Systems: Evidence from Child Protection." *Quarterly Journal of Economics*, 139(3): 1611–1664.

Baron, Jason, Brian A. Jacob, and Richard Lombardo. 2026. "The Impact of Maltreatment Investigations on Child Wellbeing." Working Paper.

Batista, Rafael M., and James Ross. 2025. "Words that Work: Using Large Language Models to Generate and Refine Hypotheses from Text." *SSRN Working Paper No. 4926398*.

Battaglini, Marco, Luigi Guiso, Chiara Lacava, Douglas L. Miller, and Eleonora Patacchini. 2025. "Refining Public Policies with Machine Learning: The Case of Tax Auditing." *Journal of Econometrics*, 249: 105847.

Bohren, J Aislinn, Peter Hull, and Alex Imas. 2022. "Systemic Discrimination: Theory and Measurement." NBER Working Paper 29820.

Bonhomme, Stéphane, and Martin Weidner. 2022. "Minimizing Sensitivity to Model Misspecification." *Quantitative Economics*, 13(3): 907–954.

Boning, William C, Nathaniel Hendren, Ben Sprung-Keyser, and Ellen Stuart. 2025. "A Welfare Analysis of Tax Audits Across the Income Distribution." *Quarterly Journal of Economics*, 140(1): 63–112.

Carlson, Jacob, and Melissa Dell. 2026. "A Unifying Framework for Robust and Efficient Inference with Unstructured Data."

Chan, David C, Matthew Gentzkow, and Chuan Yu. 2022. "Selection with Variation in Diagnostic Skill: Evidence from Radiologists." *Quarterly Journal of Economics*, 137(2): 729–783.

Cowgill, Bo. 2018. "Bias and Productivity in Humans and Algorithms: Theory and Evidence from Resume Screening." Columbia Business School Working Paper.

Doyle, Joseph J. 2007. "Child Protection and Child Outcomes: Measuring the Effects of Foster Care." *American Economic Review*, 97(5): 1583–1610.

Dreyfuss, Bnaya, and Ruru Hoong. 2026. "Calibrated Coarsening: Designing Information for AI-Assisted Decisions." Working Paper.

Dube, Oeindrila, Sandy Jo MacArthur, and Anuj K. Shah. 2025. "A cognitive view of policing." *Quarterly Journal of Economics*, 140(2).

Fitzpatrick, Maria D, Katharine Sadowski, and Christopher Wildeman. 2025. "Algorithms and Decision-making: Evidence from Child Maltreatment Reports." *Journal of Human Resources*.

Frandsen, Brigham, Lars Lefgren, and Emily Leslie. 2023. "Judging Judge Fixed Effects." *American Economic Review*, 113(1): 253–277.

Fudenberg, Drew, Jon Kleinberg, Annie Liang, and Sendhil Mullainathan. 2022. "Measuring the Completeness of Economic Models." *Journal of Political Economy*, 130(4): 956–990.

Grimon, Marie-Pascale, and Christopher Mills. 2025. "Better Together? A Field Experiment on Human-Algorithm Interaction in Child Protection." *Working Paper*.

Heller, Sara, Anuj Shah, Jonathan Guryan, Jens Ludwig, Sendhil Mullainathan, and Harold Pollack. 2017. "Thinking fast and slow? Some field experiments to reduce crime and dropout in Chicago." *Quarterly Journal of Economics*, 132(1): 1–54.

Hoffman, Mitchell, Lisa B Kahn, and Danielle Li. 2018. "Discretion in hiring." *Quarterly Journal of Economics*, 133(2): 765–800.

Imai, Kosuke, Zhichao Jiang, D. James Greiner, Ryan Halen, and Sooahn Shin. 2023. "Experimental Evaluation of Algorithm-Assisted Human Decision-Making: Application to Pretrial Public Safety Assessment." *Journal of the Royal Statistical Society Series A: Statistics in Society*, 186(2): 167–189.

Jabarian, Brian, and Luca Henkel. 2025. "Voice AI in Firms: A Natural Field Experiment on Automated Job Interviews." Working paper, SSRN 5395709.

Jabarian, Brian, and Pëllumb Reshidi. 2025. "Choice as Signal: Designing AI Adoption in Labor Market Screening." University of Chicago SSRN Working Paper 5856382. Available at SSRN: `https://ssrn.com/abstract=5856382`.

Kahneman, Daniel, Olivier Sibony, and Cass R. Sunstein. 2021. *Noise: A Flaw in Human Judgment.* New York:Little, Brown Spark.

Kim, Hyunil, Christopher Wildeman, Melissa Jonson-Reid, and Brett Drake. 2017. "Lifetime Prevalence of Investigating Child Maltreatment Among US Children." *American Journal of Public Health*, 107(2): 274–280.

Kitagawa, Toru. 2015. "A Test for Instrument Validity." *Econometrica*, 83(5): 2043–2063.

Kleinberg, Jon, Himabindu Lakkaraju, Jure Leskovec, Jens Ludwig, and Sendhil Mullainathan. 2018. "Human Decisions and Machine Predictions." *Quarterly Journal of Economics*, 133(1): 237–293.

Kleinberg, Jon, Jens Ludwig, Sendhil Mullainathan, and Ziad Obermeyer. 2015. "Prediction Policy Problems." *American Economic Review*, 105(5): 491–95.

Lakkaraju, Himabindu, Jon Kleinberg, Jure Leskovec, Jens Ludwig, and Sendhil Mullainathan. 2017. "The Selective Labels Problem: Evaluating Algorithmic Predictions in the Presence of Unobservables." *KDD '17*, 275–284. New York, NY, USA:Association for Computing Machinery.

Li, Danielle, Lindsey Raymond, and Peter Bergman. 2025. "Hiring as Exploration." *Review of Economic Studies*, 93(2): 1200–1240.

Ludwig, Jens, and Sendhil Mullainathan. 2024. "Machine Learning as a Tool for Hypothesis Generation." *Quarterly Journal of Economics*, 139(2): 751–827.

Ludwig, Jens, Sendhil Mullainathan, and Ashesh Rambachan. 2025. "Large Language Models: An Applied Econometric Framework."

Madras, David, Toniann Pitassi, and Richard Zemel. 2018. "Predict Responsibly: Improving Fairness and Accuracy by Learning to Defer." Vol. 31.

Manski, Charles F. 1990. "Nonparametric Bounds on Treatment Effects." *American Economic Review: Papers and Proceedings*, 80(2): 319–323.

Manski, Charles F., and John V. Pepper. 2000. "Monotone Instrumental Variables: With an Application to the Returns to Schooling." *Econometrica*, 68(4): 997–1010.

MDHHS. 2020. "Children's Protective Services Policy Manuals." Michigan Department of Health and Human Services, `https://mplp.org/sites/default/files/2020-08/child-protective-services-manual.pdf`.

Modarressi, Iman, Jann Spiess, and Amar Venugopal. 2025. "Causal Inference on Outcomes Learned from Text." *arXiv preprint arXiv:2503.00725*.

Mozannar, Hussein, and David Sontag. 2020. "Consistent Estimators for Learning to Defer to an Expert." Vol. 119 of *Proceedings of Machine Learning Research*, 7076–7087.

Mullainathan, Sendhil. 2025. "Economics in the Age of Algorithms." *AEA Papers and Proceedings*, 115: 1–23.

Mullainathan, Sendhil, and Ziad Obermeyer. 2022. "Diagnosing Physician Error: A Machine Learning Approach to Low-Value Health Care." *Quarterly Journal of Economics*, 137(2): 679–727.

Nickerson, Raymond S. 1998. "Confirmation bias: A ubiquitous phenomenon in many guises." *Review of General Psychology*, 2(2): 175–220.

Raghu, Maithra, Katy Blumer, Greg Corrado, Jon Kleinberg, Ziad Obermeyer, and Sendhil Mullainathan. 2019. "The Algorithmic Automation Problem: Prediction, Triage, and Human Effort." *arXiv preprint arXiv:1903.12220*.

Rambachan, Ashesh. 2024. "Identifying Prediction Mistakes in Observational Data." *Quarterly Journal of Economics*, 139(3): 1665–1711.

Rambachan, Ashesh, and Jonathan Roth. 2020. "Bias In, Bias Out? Evaluating the Folk Wisdom." Vol. 156 of *Leibniz International Proceedings in Informatics (LIPIcs)*, 6:1–6:15. Schloss Dagstuhl – Leibniz-Zentrum für Informatik.

Samant, Anjana, Aaron Horowitz, Sophie Beiers, and Kath Xu. 2021. "Family Surveillance by Algorithm: The Rapidly Spreading Tools Few Have Heard Of." *American Civil Liberties Union*, `https://www.aclu.org/news/womens-rights/family-surveillance-by-algorithm-the-rapidly-spreading-tools-few-have-heard-of`.

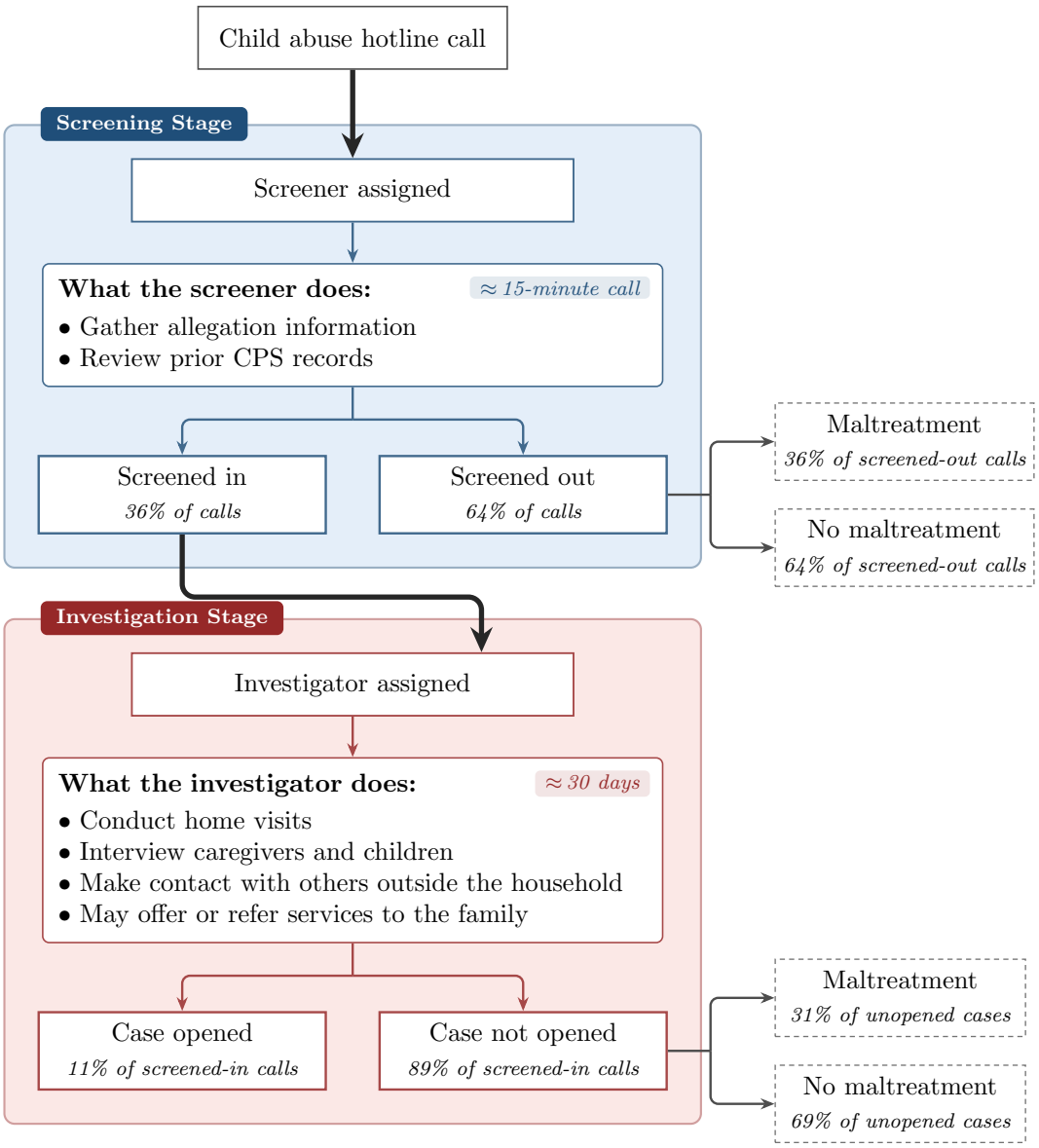
Shapley, L. S. 1953. "A Value for n-Person Games." In *Contributions to the Theory of Games, Volume II. Annals of Mathematics Studies*, , ed. H. W. Kuhn and A. W. Tucker, 307–317. Princeton, NJ:Princeton University Press.

Stevenson, Megan T., and Jennifer L. Doleac. 2024. "Algorithmic Risk Assessment in the Hands of Humans." *American Economic Journal: Economic Policy*, 16(4).

Zhong, Ruiqi, Charlie Snell, Dan Klein, and Jacob Steinhardt. 2022. "Describing Differences between Text Distributions with Natural Language."

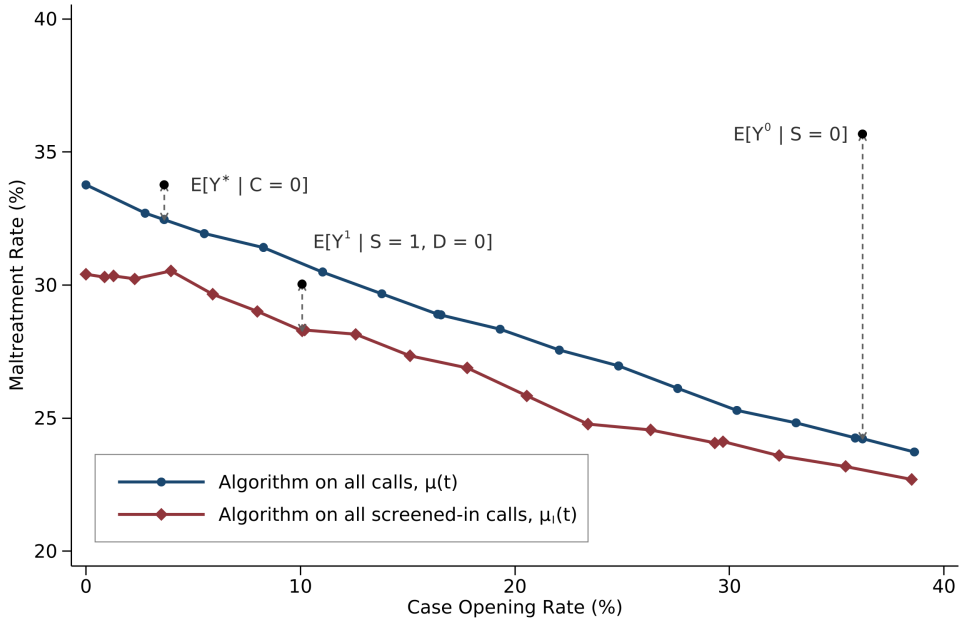
Zhou, Yangqiaoyu, Haokun Liu, Tejes Srivastava, Hongyuan Mei, and Chenhao Tan. 2024. "Hypothesis Generation with Large Language Models." 117–139. Association for Computational Linguistics.

Figure 1: Child Protective Services Process in Michigan



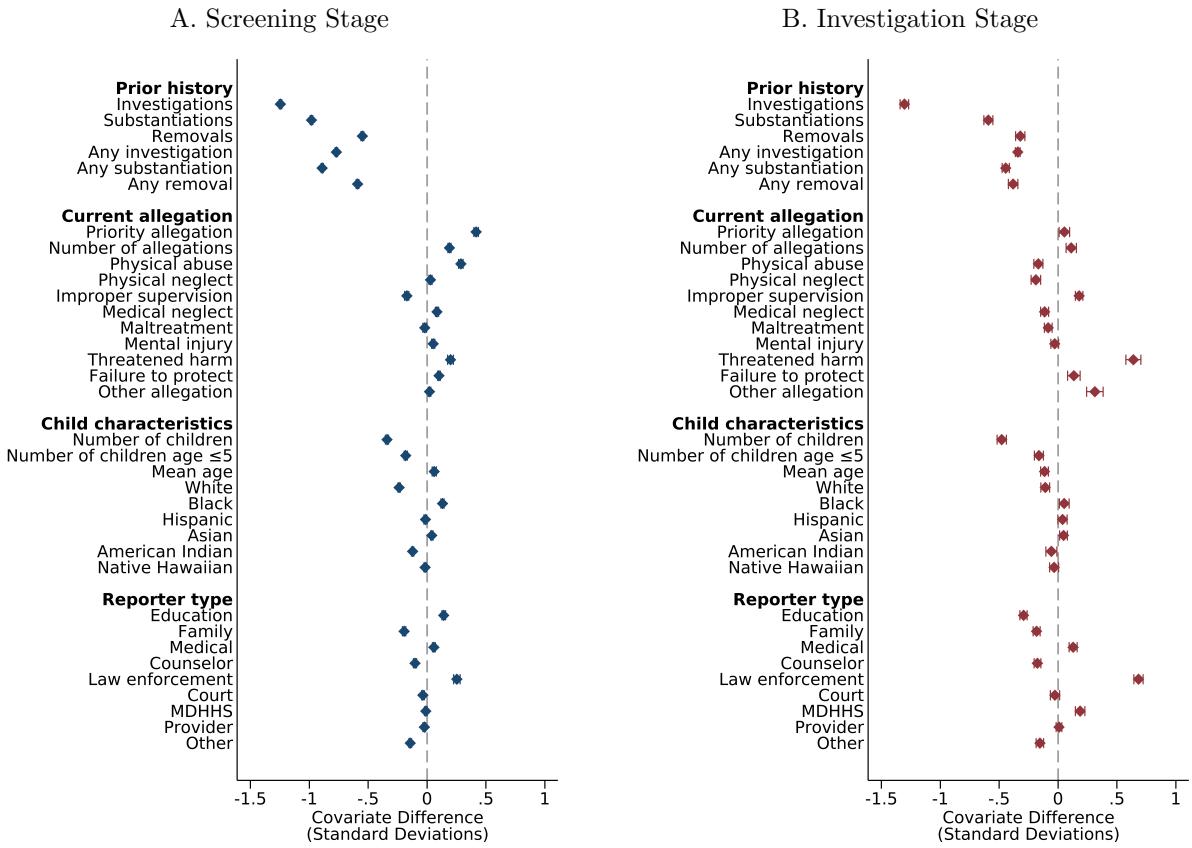
Notes. The figure illustrates the two-stage structure of the child protective services process in Michigan. In the screening stage, screeners gather the relevant information from administrative records and decide whether to screen in the call for investigation. In the investigation stage, investigators conduct one or more home visits, interview caregivers and children, make contact with other relevant parties, and decide whether to offer services and/or open a case.

Figure 2: Counterfactual Maltreatment Rates Under the Algorithm



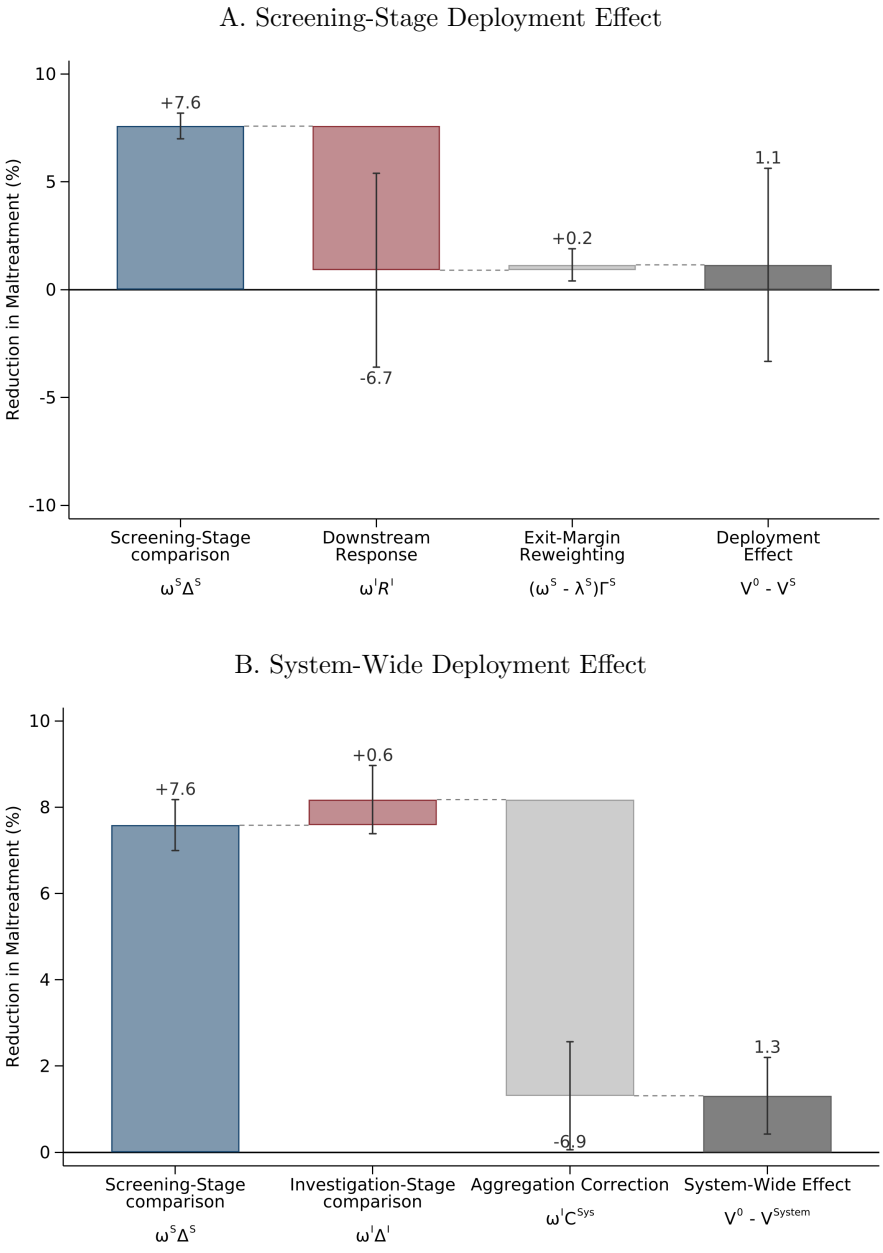
Notes. This figure plots the algorithm's counterfactual maltreatment rates against the case-opening rate each threshold t would induce. The blue curve plots the maltreatment rate among all calls the algorithm leaves unopened, $\widehat{\mu}(t)$, estimated from the extrapolations in Panel B of Appendix Figure A.2. The red curve plots the same rate among screened-in calls, $\widehat{\mu}_I(t)$, estimated from $\widehat{\mu}(t)$ and Panel A of Appendix Figure A.2. The black points are the observed status-quo maltreatment rates among all unopened calls, $\mathbb{E}[Y_i^* \mid C_i = 0]$, screened-in but unopened calls, $\mathbb{E}[Y_i^1 \mid S_i = 1, D_i = 0]$, and screened-out calls, $\mathbb{E}[Y_i^0 \mid S_i = 0]$, and the dashed arrows connect each to the counterfactual curve at the matched threshold. See Section 4.2 for discussion.

Figure 3: Intake Characteristics of Calls on Which Humans and the Algorithm Disagree



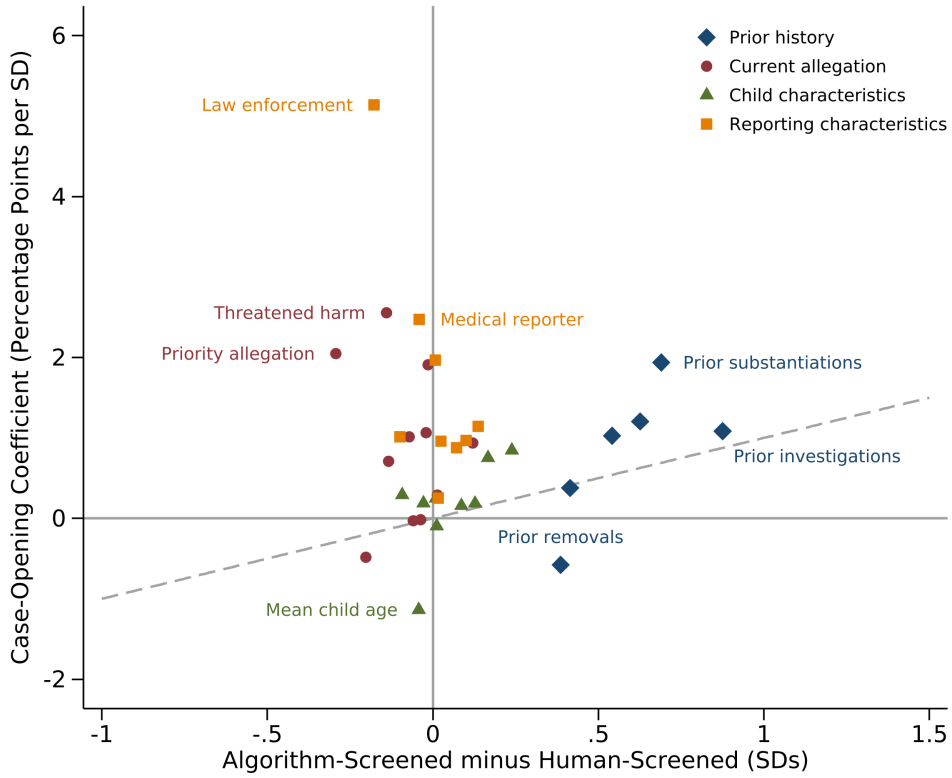
Notes. This figure plots standardized differences in intake characteristics between the calls on which humans and an equally selective algorithm disagree, in the evaluation sample. Panel A plots the mean among calls screeners advance and the algorithm would not ($S_i = 1, S_i^m = 0$) minus the mean among calls the algorithm would advance and screeners do not ($S_i = 0, S_i^m = 1$). Panel B plots the mean among calls investigators open and the algorithm would leave unopened ($S_i = 1, D_i = 1, D_i^m = 0$) minus the mean among calls the algorithm would open and investigators leave unopened ($S_i = 1, D_i = 0, D_i^m = 1$). Differences are divided by each characteristic's standard deviation across all calls in the evaluation sample, so negative values indicate higher means among the calls the algorithm selects. Horizontal bars are 95 percent confidence intervals, with standard errors clustered by screener in Panel A and by investigator in Panel B. See Section 4.2 for discussion.

Figure 4: Decomposing the Effects of Deploying the Algorithm



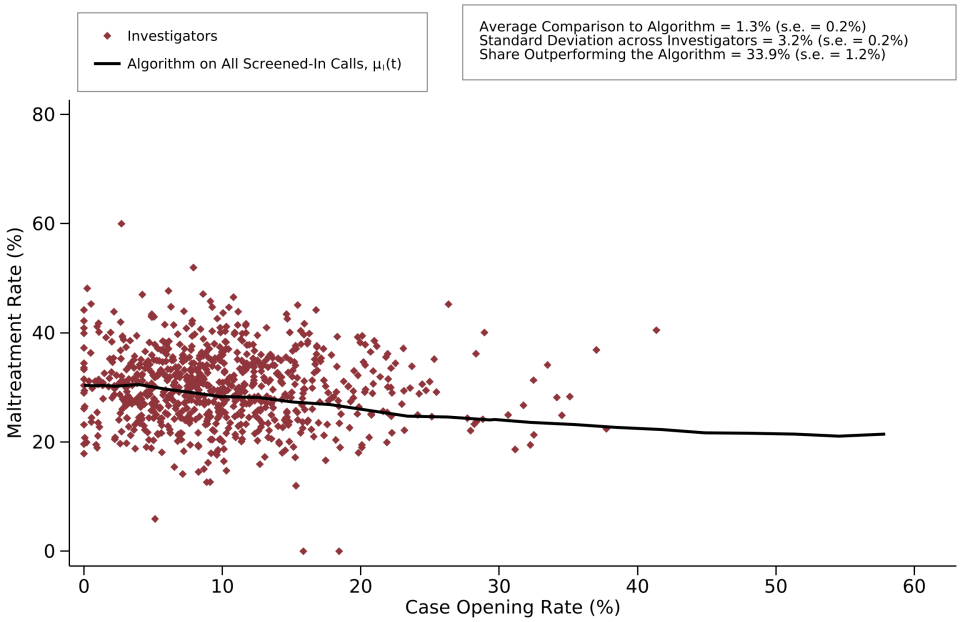
Notes. This figure decomposes the effects of deploying the algorithm at the screening stage and in place of the entire system, in percentage points of subsequent maltreatment. Panel A decomposes the screening-stage deployment effect $V^0 - V^S$ into the weighted screening-stage comparison $\omega^S \Delta^S$, the downstream investigator response $\omega^I \mathcal{R}^I$, and the exit-margin reweighting term $(\omega^S - \lambda^S)\Gamma^S$, following Proposition 3.1. Panel B decomposes the system-wide deployment effect $V^0 - V^{System}$ into the weighted screening-stage comparison $\omega^S \Delta^S$, the weighted investigation-stage comparison $\omega^I \Delta^I$, and the aggregation correction $\omega^I \mathcal{C}^{\text{Sys}}$, following Proposition 3.2. Each bar starts where the previous one ends, and whiskers are 95 percent confidence intervals around the running total, obtained via a bootstrapping procedure (with 500 replications). See Section 4.3 for discussion.

Figure 5: Shifts in the Screened-In Pool and the Response of Investigators' Case Openings



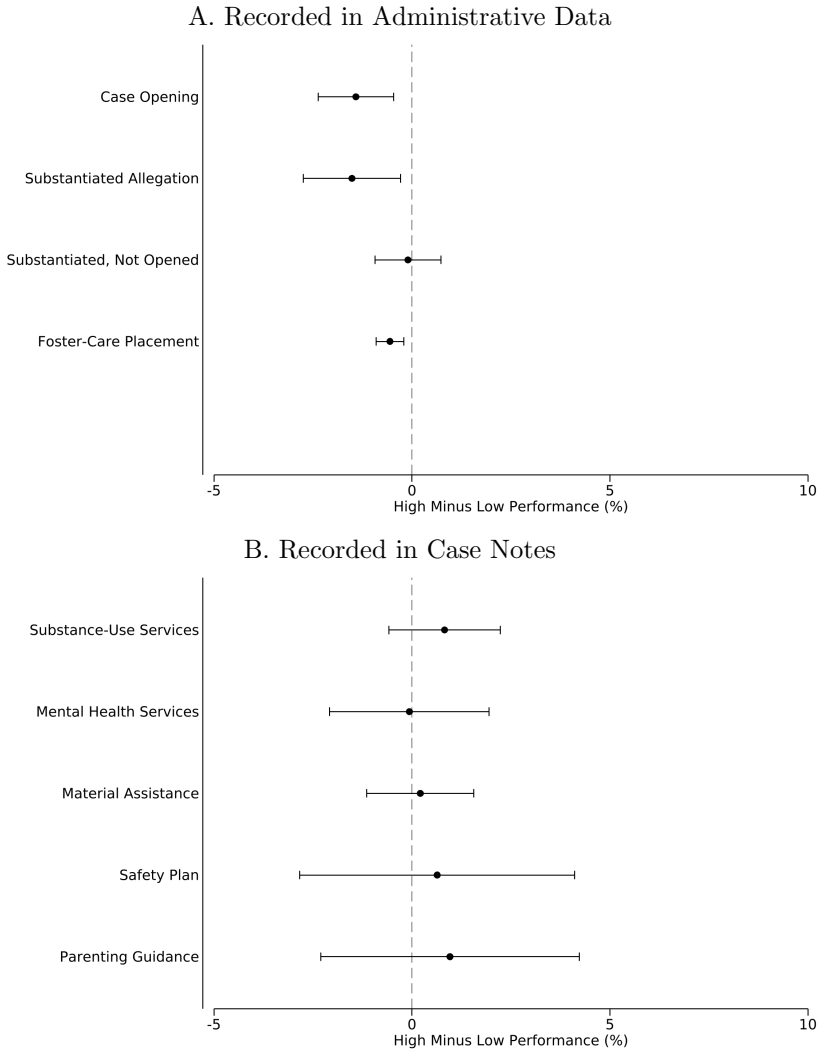
Notes. This figure plots, for each intake characteristic, the shift in its mean between the calls screeners advance and the calls the algorithm would advance, against the response of investigators' case openings to it. The horizontal coordinate is the mean among calls the algorithm would advance ($S_i^m = 1$) minus the mean among calls screeners advance ($S_i = 1$), divided by the characteristic's standard deviation across all calls in the evaluation sample. The vertical coordinate is the coefficient on the standardized characteristic, in percentage points per standard deviation, from a single regression of case opening on all displayed characteristics among calls screeners advance in the evaluation sample, with month-by-ZIP-code-by-shift fixed effects. The dashed line is the 45-degree line. See Section 4.3 for discussion.

Figure 6: Maltreatment Rates of Investigators Relative to the Algorithm



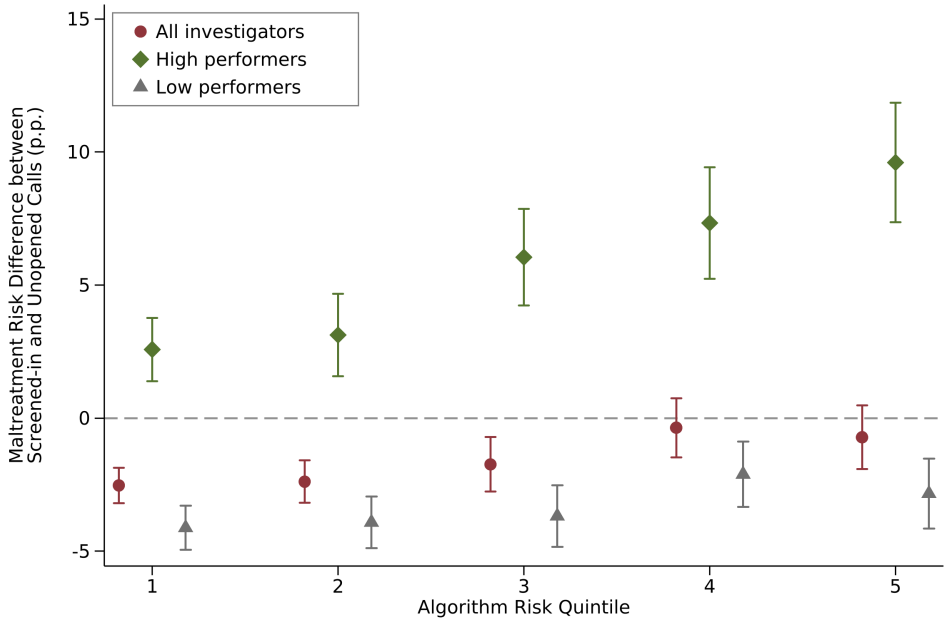
Notes. This figure plots each investigator's strata-adjusted maltreatment rate among unopened calls against their strata-adjusted case-opening rate, together with the algorithm's counterfactual curve $\widehat{\mu}_I(t)$ from Figure 2. Investigator-level rates are unshrunk estimates that adjust for month-by-ZIP-code-by-shift fixed effects. The box reports the caseload-weighted average and standard deviation of Δ_g^I across investigators and the caseload-weighted share of investigators who outperform the algorithm, estimated from posterior average effects as described in Appendix D.2. All standard errors are obtained via a bootstrapping procedure (with 500 replications) and appear in parentheses. See Section 5.1 for discussion.

Figure 7: Investigation Actions of Higher- versus Lower-Performing Investigators



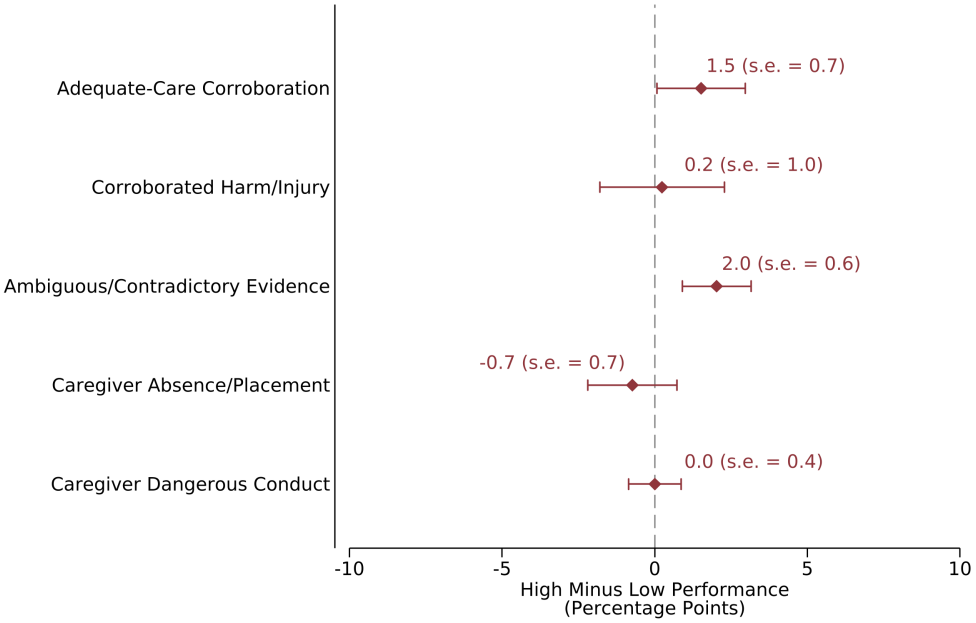
Notes. This figure plots, for each investigation action, the coefficient from a regression of an indicator for the action on an indicator that the case was handled by a higher-performing investigator, one whose posterior mean of Δ_g^I is negative, with month-by-ZIP-code-by-shift fixed effects. Panel A reports actions recorded in the structured administrative data, and Panel B reports actions labeled from investigators' narrative case notes, as described in Appendix E.1. Coefficients are in percentage points, and a negative coefficient means higher-performing investigators are less likely to take the action. Horizontal bars are 95 percent confidence intervals, with standard errors clustered by investigator. See Section 5.2 for discussion.

Figure 8: Outcome Gaps within Quintiles of the Algorithm's Predicted Risk



Notes. This figure plots the outcome gap within each quintile of the algorithm's predicted risk among screened-in calls, for all investigators and for higher- and lower-performing investigators separately. The outcome gap is the quintile's overall maltreatment rate, $\mathbb{E}[Y_i^* \mid S_i = 1, q]$, minus its observed maltreatment rate among unopened calls, $\mathbb{E}[Y_i \mid S_i = 1, D_i = 0, q]$, in percentage points. The overall rate is recovered by extrapolating investigators' strata-adjusted maltreatment rates among unopened calls to a case-opening rate of zero across quasi-randomly assigned investigators. Whiskers are 95 percent confidence intervals. See Section 5.3 for discussion.

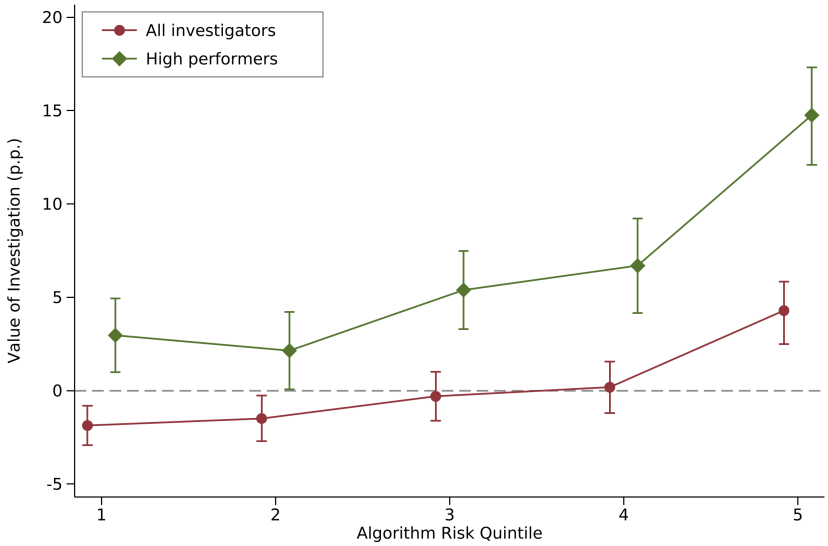
Figure 9: Narrative Evidence Themes Recorded by Higher- versus Lower-Performing Investigators



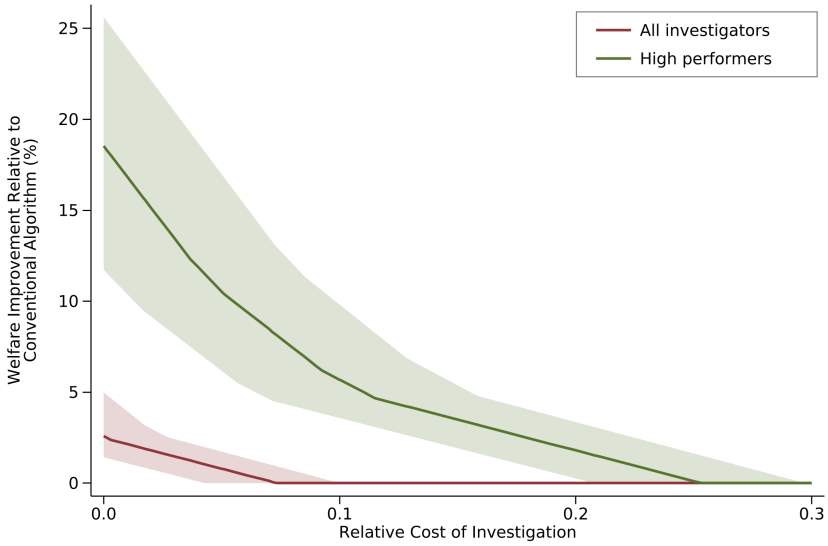
Notes. This figure plots, for each of the five evidence themes, the coefficient from a regression of an indicator that the case note records the theme on an indicator that the case was handled by a higher-performing investigator, one whose posterior mean of Δ_g^I is negative, with month-by-ZIP-code-by-shift fixed effects. The themes are extracted from investigators' narrative case notes with large language models, as described in Appendix E.2. Coefficients are in percentage points, and a negative coefficient means higher-performing investigators are less likely to record the theme. Horizontal bars are 95 percent confidence intervals, with standard errors clustered by investigator, and labels report each coefficient with its standard error in parentheses. See Section 5.3 for discussion.

Figure 10: The Value of an Investigation and the Welfare Gain of the Optimal Screening Algorithm

A. Value of an Investigation by Quintile of Predicted Risk



B. Welfare Improvement as the Cost of an Investigation Varies



Notes. This figure plots the value of an investigation and the welfare gain of the optimal screening algorithm for all investigators and for higher-performing investigators, estimated under selection on observables at the screening stage, with the cost of a case opening κ_C set so that the conventional algorithm opens cases at the status-quo rate. Panel A plots the value of an investigation in equation (8) within each quintile of the algorithm's predicted risk, in percentage points. Panel B plots the optimal algorithm's welfare gain over the conventional algorithm, as a share of the conventional algorithm's loss, against the cost of an investigation relative to the cost of a case opening. Whiskers in Panel A and shaded bands in Panel B are confidence intervals that hold jointly across quintiles and across costs of an investigation with probability at least 95 percent (Appendix F.3). See Section 6.2 for discussion.

Table 1: Stage-Wise Comparisons and Deployment Effects

Panel A: Excess Maltreatment Relative to an Equally Selective Algorithm		
Δ^S	11.5 (0.5) [10.6, 12.3]	Among screened-out calls at the screening stage
Δ^I	1.8 (1.2) [-0.6, 4.1]	Among unopened calls at the investigation stage
Panel B: Reduction in Maltreatment from Deploying the Algorithm		
$V^0 - V^I$	0.6 (0.4) [-0.2, 1.4]	Deployment at only the investigation stage
$V^0 - V^S$	1.1 (2.3) [-3.3, 5.6]	Deployment at only the screening stage
$V^0 - V^{System}$	1.3 (0.5) [0.4, 2.2]	Deployment in place of the multistage system

Notes. This table reports the stage-wise human-algorithm comparisons of Section 3.2 and the effects of the three algorithm deployments of Section 3.3 on subsequent maltreatment among unopened calls. Panel A reports excess maltreatment among the calls each human stage screens out or leaves unopened, relative to an equally selective algorithm, and Panel B reports the reduction in maltreatment from each deployment. Counterfactual maltreatment rates under the algorithm are estimated from the screener extrapolations described in Section 4.1 and Appendix B.3. All estimates are in percentage points. Standard errors are in parentheses and 95 percent confidence intervals are in brackets, both obtained via a bootstrapping procedure (with 500 replications). See Sections 4.2 and 4.3 for discussion.

Human Decisions and Machine Predictions in Multistage Systems

Online Appendix

Jason Baron, Will Dobbie, Richard Lombardo,
Ashesh Rambachan & Joseph Ryan

A Proofs of the Multistage Decompositions

This appendix proves Propositions 3.1 and 3.2 and derives equation (4). Each is an accounting identity. Throughout, we assume that $m(X_i)$ is continuously distributed, or that ties at a threshold are broken at random, so that each algorithmic rule acts at exactly its matched rate. Let $s := \Pr(S_i = 1)$ and $d := \Pr(D_i = 1 \mid S_i = 1)$, so that $\Pr(C_i = 1) = sd$. By construction, $\Pr(S_i^m = 1) = s$, $\Pr(D_i^m = 1 \mid S_i = 1) = d$, and $\Pr(C_i^m = 1) = sd$.

Since $sd \leq s$, the thresholds satisfy $\tau^* \geq \tau^S$, so $\{C_i^m = 1\} = \{m(X_i) \geq \tau^*\} \subseteq \{m(X_i) \geq \tau^S\} = \{S_i^m = 1\}$. Likewise, $\Pr(m(X_i) \geq \tau^I) \geq \Pr(S_i = 1, m(X_i) \geq \tau^I) = sd = \Pr(m(X_i) \geq \tau^*)$, so $\tau^* \geq \tau^I$. Hence $\{C_i^m = 0\}$ partitions into $\{S_i^m = 0\}$ and $\{S_i^m = 1, C_i^m = 0\}$, with

$$\Pr(S_i^m = 0 \mid C_i^m = 0) = \frac{1-s}{1-sd} = \omega^S, \qquad \Pr(S_i^m = 1, C_i^m = 0 \mid C_i^m = 0) = \frac{s(1-d)}{1-sd} = \omega^I,$$

which coincide with the status-quo exit shares $\Pr(S_i = 0 \mid C_i = 0) = (1-s)/(1-sd)$ and $\Pr(S_i = 1, D_i = 0 \mid C_i = 0) = s(1-d)/(1-sd)$.

A.1 Proof of Proposition 3.1

Since D_i^m acts at the status-quo case-opening rate, $\Pr(C_i^I = 0) = \Pr(C_i = 0)$, and the exit shares are again ω^S and ω^I. Under this deployment, screened-out calls stay screened out and the screened-in calls the algorithm leaves unopened are not investigated, so both pools produce Y_i^0:

$$V^I = \omega^S \, \mathbb{E}[Y_i^0 \mid S_i = 0] + \omega^I \, \mathbb{E}[Y_i^0 \mid S_i = 1, D_i^m = 0].$$

The screened-out terms coincide with those of V^0, so

$$V^0 - V^I = \omega^I \big(\mathbb{E}[Y_i^1 \mid S_i = 1, D_i = 0] - \mathbb{E}[Y_i^0 \mid S_i = 1, D_i^m = 0]\big) = \omega^I \Delta^I.$$

Under screening-stage deployment, the calls the algorithm screens out produce Y_i^0, while the calls it screens in are investigated by humans, so those left unopened produce Y_i^1. Since $C_i^S = S_i^m D_i$, the unopened calls $\{C_i^S = 0\}$ partition into $\{S_i^m = 0\}$ and $\{S_i^m = 1, D_i = 0\}$, so $V^S = \lambda^S \mathbb{E}[Y_i^0 \mid S_i^m = 0] + \lambda^I \mathbb{E}[Y_i^1 \mid S_i^m = 1, D_i = 0]$, with $\lambda^S := \Pr(S_i^m = 0 \mid C_i^S = 0)$ and $\lambda^I := \Pr(S_i^m = 1, D_i = 0 \mid C_i^S = 0)$. Subtracting V^S from V^0 and adding and subtracting $\omega^S \mathbb{E}[Y_i^0 \mid S_i^m = 0]$ and $\omega^I \mathbb{E}[Y_i^1 \mid S_i^m = 1, D_i = 0]$,

$$\begin{aligned} V^0 - V^S = {} & \omega^S \big(\mathbb{E}[Y_i^0 \mid S_i = 0] - \mathbb{E}[Y_i^0 \mid S_i^m = 0]\big) \\ & + \omega^I \big(\mathbb{E}[Y_i^1 \mid S_i = 1, D_i = 0] - \mathbb{E}[Y_i^1 \mid S_i^m = 1, D_i = 0]\big) \\ & + (\omega^S - \lambda^S) \, \mathbb{E}[Y_i^0 \mid S_i^m = 0] + (\omega^I - \lambda^I) \, \mathbb{E}[Y_i^1 \mid S_i^m = 1, D_i = 0]. \end{aligned}$$

The first term is $\omega^S \Delta^S$, and the second is $\omega^I \mathcal{R}^I$. Since $\omega^S + \omega^I = \lambda^S + \lambda^I = 1$, we have $\omega^I - \lambda^I = -(\omega^S - \lambda^S)$, so the last two terms combine into $(\omega^S - \lambda^S)\big(\mathbb{E}[Y_i^0 \mid S_i^m = 0] - \mathbb{E}[Y_i^1 \mid S_i^m = 1, D_i = 0]\big) = (\omega^S - \lambda^S)\,\Gamma^S$, giving equation (2).

A.2 Proof of Proposition 3.2

Since $V^{System} = \mathbb{E}[Y_i^0 \mid C_i^m = 0]$, the law of iterated expectations over the partition of $\{C_i^m = 0\}$ above gives

$$V^{System} = \omega^S\,\mathbb{E}[Y_i^0 \mid S_i^m = 0] + \omega^I\,\mathbb{E}[Y_i^0 \mid S_i^m = 1, C_i^m = 0].$$

Subtracting this from the decomposition of V^0 in Section 3.3,

$$\begin{aligned} V^0 - V^{System} = {} & \omega^S\big(\mathbb{E}[Y_i^0 \mid S_i = 0] - \mathbb{E}[Y_i^0 \mid S_i^m = 0]\big) \\ & + \omega^I\big(\mathbb{E}[Y_i^1 \mid S_i = 1, D_i = 0] - \mathbb{E}[Y_i^0 \mid S_i^m = 1, C_i^m = 0]\big). \end{aligned}$$

The first term is $\omega^S \Delta^S$. Adding and subtracting $\omega^I \mathbb{E}[Y_i^0 \mid S_i = 1, D_i^m = 0]$ in the second term gives

$$\begin{aligned} V^0 - V^{System} = {} & \omega^S \Delta^S + \omega^I \underbrace{\big(\mathbb{E}[Y_i^1 \mid S_i = 1, D_i = 0] - \mathbb{E}[Y_i^0 \mid S_i = 1, D_i^m = 0]\big)}_{\Delta^I} \\ & + \omega^I \underbrace{\big(\mathbb{E}[Y_i^0 \mid S_i = 1, D_i^m = 0] - \mathbb{E}[Y_i^0 \mid S_i^m = 1, C_i^m = 0]\big)}_{\mathcal{C}^{\mathrm{Sys}}}, \end{aligned}$$

which is equation (3).

A.3 Proof of Equation (4)

Start from the definition of $\mathcal{C}^{\mathrm{Sys}}$ in Proposition 3.2. Since D_i^m opens a share d of screened-in calls, $\Pr(S_i = 1, D_i^m = 0) = s(1-d)$, and since $\{C_i^m = 1\} \subseteq \{S_i^m = 1\}$, $\Pr(S_i^m = 1, C_i^m = 0) = \Pr(S_i^m = 1) - \Pr(C_i^m = 1) = s - sd$. Both equal $\Pr(S_i = 1, D_i = 0)$. Multiplying $\mathcal{C}^{\mathrm{Sys}}$ by $\Pr(S_i = 1, D_i = 0)$ turns each conditional expectation into an expectation over its pool,

$$\Pr(S_i = 1, D_i = 0)\,\mathcal{C}^{\mathrm{Sys}} = \mathbb{E}\big[Y_i^0 \mathbf{1}\{S_i = 1, D_i^m = 0\}\big] - \mathbb{E}\big[Y_i^0 \mathbf{1}\{S_i^m = 1, C_i^m = 0\}\big].$$

The first pool is the screened-in calls with $m(X_i) < \tau^I$, and the second is the calls with $\tau^S \le m(X_i) < \tau^*$. Substituting these thresholds and splitting the second pool by S_i,

$$\begin{aligned} \Pr(S_i = 1, D_i = 0)\,\mathcal{C}^{\mathrm{Sys}} = {} & \mathbb{E}\big[Y_i^0\,\mathbf{1}\{S_i = 1\}\big(\mathbf{1}\{m(X_i) < \tau^I\} - \mathbf{1}\{\tau^S \le m(X_i) < \tau^*\}\big)\big] \\ & - \mathbb{E}\big[Y_i^0\,\mathbf{1}\{S_i = 0,\, \tau^S \le m(X_i) < \tau^*\}\big]. \end{aligned}$$

Since $\mathbf{1}\{a \le m < b\} = \mathbf{1}\{m < b\} - \mathbf{1}\{m < a\}$ for $a \le b$, and $\tau^S \le \tau^*$ and $\tau^I \le \tau^*$,

$$\begin{aligned} & \mathbf{1}\{m(X_i) < \tau^I\} - \mathbf{1}\{\tau^S \le m(X_i) < \tau^*\} \\ & \quad = \mathbf{1}\{m(X_i) < \tau^I\} - \mathbf{1}\{m(X_i) < \tau^*\} + \mathbf{1}\{m(X_i) < \tau^S\} \\ & \quad = \mathbf{1}\{m(X_i) < \tau^S\} - \mathbf{1}\{\tau^I \le m(X_i) < \tau^*\}. \end{aligned}$$

Substituting the identity into the first term gives

$$\begin{aligned}
&\Pr(S_i = 1, D_i = 0)\,\mathcal{C}^{\text{Sys}} \\
&\quad = \mathbb{E}\big[Y_i^0 \mathbf{1}\{S_i = 1, m(X_i) < \tau^S\}\big] - \mathbb{E}\big[Y_i^0 \mathbf{1}\{S_i = 1, \tau^I \le m(X_i) < \tau^*\}\big] \\
&\qquad - \mathbb{E}\big[Y_i^0 \mathbf{1}\{S_i = 0, \tau^S \le m(X_i) < \tau^*\}\big].
\end{aligned}$$

Finally, we split the first term by investigators' decisions. The event $\{m(X_i) < \tau^S\}$ is $\{S_i^m = 0\}$, and partitioning it by D_i gives the double-counting and redundant-screen-out terms, $\{S_i = 1, D_i = 1, S_i^m = 0\}$ and $\{S_i = 1, D_i = 0, S_i^m = 0\}$. The event $\{S_i = 1, \tau^I \le m(X_i) < \tau^*\}$ is $\{S_i = 1, D_i^m = 1, C_i^m = 0\}$, the investigation-stage threshold-disagreement term, and the event $\{S_i = 0, \tau^S \le m(X_i) < \tau^*\}$ is $\{S_i = 0, S_i^m = 1, C_i^m = 0\}$, the screening-stage term. Substituting these events gives the four-term expression in equation (4). Since $\omega^I = \Pr(S_i = 1, D_i = 0)/\Pr(C_i = 0)$, dividing both sides by $\Pr(C_i = 0)$ shows that the aggregation correction $\omega^I \mathcal{C}^{\text{Sys}}$ equals that expression times $1/\Pr(C_i = 0)$.

B Additional Details on the Empirical Approach

B.1 Sample Definition and Summary Statistics

We begin with the 937,457 hotline calls received in Michigan from January 2017 to December 2022 and define the sample at the call level, since screening and case-opening decisions apply to every child on a call. We drop calls with a missing screener identifier and screened-in calls with a missing investigator identifier (143,110 calls). We drop calls involving a child younger than one year old, because state policy requires that calls about substance-exposed infants be screened in, leaving screeners little or no discretion (95,546). To preserve quasi-random assignment, we drop calls involving sexual abuse, which a non-random subset of investigators handles (43,814); calls with trafficking allegations reported by law enforcement, which a non-random subset of screeners handles (4); and screened-in calls on a child investigated in the past 60 days or with a case opened in the past 365 days, which are often assigned to the same investigator (40,450). We also drop screened-in calls with a missing child ZIP code (33,428), duplicate calls on the same child on the same day (22,687), calls assigned to screeners with fewer than 100 calls in the sample (1,095), and screened-in calls in month-by-ZIP-code-by-shift cells with a single investigator (42,931).

The resulting screener sample consists of 514,392 calls, or 822,832 call-by-child records, assigned to 291 screeners, and the investigator sample consists of the 184,884 screened-in calls assigned to 4,579 investigators. Appendix Table A.1 summarizes the screener sample in Columns (1)–(3) and the investigator sample in Columns (4)–(6), and Section 2.2 discusses both.

Investigators' age and tenure come from MDHHS personnel snapshots, which cover only some of the investigators in our sample. We predict investigator race and ethnicity from first and last names with the Python package *ethnicolr*, which draws on U.S. census data, Florida voter registration data, and the Wikipedia data of Skiena and Ward (2014), and investigator gender from first names with the R package *predictrace*, which draws on U.S. Social Security Administration data.

B.2 Building the Predictive Algorithm

We train the algorithm $m(X_i)$ of Section 4.1 on a training sample and evaluate it on a separate evaluation sample. The training sample consists of a random half of screened-out calls (255,137 call-by-child records) and the screened-in calls of investigators with fewer than 75 screened-in calls (153,480 records). The evaluation sample consists of the remaining screened-out calls and the screened-in calls of the 806 investigators with at least 75. Both steps below use only the records whose subsequent maltreatment is observed, the screened-out calls and the screened-in calls with no case opened.

In the first step, a gradient-boosted decision tree (XGBoost) fit on 90 percent of the training sample predicts child-level maltreatment from the intake information: complaint timing, allegation type, reporter and contact-method characteristics, child gender and age, family composition, and prior CPS history. We choose the maximum tree depth, subsample rate, column subsample rate, minimum child weight, ℓ_2 penalty, and learning rate over a grid to maximize out-of-fold AUC in three-fold cross-validation at the call level, keeping all children on a call in the same fold, with at most 2,500 boosting rounds and early stopping. In the second step, we select between an XGBoost model and a logistic regression, fit on the remaining 10 percent of the training sample to aggregate the child-level predictions to the call level. It predicts whether any child on the call experiences subsequent maltreatment from the number of children and the minimum, mean, maximum, second-highest, and standard deviation of their first-step predictions. The model with the larger three-fold cross-validation AUC is chosen. Appendix Figure A.1 reports the algorithm's ROC curves and calibration in the evaluation sample.

B.3 Additional Extrapolation Details

The screening-stage deployment requires the case-opening rate $\Pr(D_i = 1 \mid m(X_i) \geq \tau^S)$ on the algorithm's screened-in set and the maltreatment rate $\mathbb{E}[Y_i^1 \mid m(X_i) \geq \tau^S, D_i = 0]$ among the calls in that set that investigators leave unopened. We recover both by extrapolating across screeners to a screen-in rate of one, estimating

$$
\begin{aligned}
S_i &= \sum_k \left[\phi_{1,k}\,\mathbf{1}\{K_i = k\} + \phi_{2,k}\,\mathbf{1}\{K_i = k, m(X_i) \geq \tau^S\}\right] + W_i^\top \gamma + \varepsilon_i^S, \\
D_i &= \sum_k \left[\pi_{1,k}\,\mathbf{1}\{K_i = k\} + \pi_{2,k}\,\mathbf{1}\{K_i = k, m(X_i) \geq \tau^S\}\right] + W_i^\top \zeta + \varepsilon_i^D \text{ on } \{S_i = 1\}, \\
Y_i &= \sum_k \left[\rho_{1,k}\,\mathbf{1}\{K_i = k\} + \rho_{2,k}\,\mathbf{1}\{K_i = k, m(X_i) \geq \tau^S\}\right] + W_i^\top \xi + \varepsilon_i^Y \text{ on } \{S_i = 1, D_i = 0\},
\end{aligned}
\tag{B1}
$$

where K_i and W_i are the screener and day-by-shift strata indicators of Section 4.1. These regressions recover screener k's strata-adjusted screen-in rate $\phi_k := \phi_{1,k} + \phi_{2,k}$ on the algorithm's screened-in set, their case-opening rate $\pi_k := \pi_{1,k} + \pi_{2,k}$ among their screened-in calls in that set, and their maltreatment rate $\rho_k := \rho_{1,k} + \rho_{2,k}$ among their screened-in but unopened calls in that set. We truncate estimated screener-level rates to $[0, 1]$. We then regress $\{\pi_k\}_k$ and $\{\rho_k\}_k$ on $\{\phi_k\}_k$ across screeners, weighting by the inverse variance of the dependent variable, and evaluate each fitted line at $\phi_k = 1$, which identifies the two moments (Appendix Figure A.3). Because these extrapolations reach well beyond the ob-

served screen-in rates, Appendix C.2 reports alternative identification strategies for the two moments.

We assemble the estimates in three steps. First, we set the thresholds $\widehat{\tau}^*$, $\widehat{\tau}^S$, and $\widehat{\tau}^I$ to match the corresponding action rates in the evaluation sample: $\Pr(m(X_i) \geq \widehat{\tau}^*) = \Pr(C_i = 1)$, $\Pr(m(X_i) \geq \widehat{\tau}^S) = \Pr(S_i = 1)$, and $\Pr(m(X_i) \geq \widehat{\tau}^I \mid S_i = 1) = \Pr(D_i = 1 \mid S_i = 1)$. Second, we estimate the status-quo means V^0, $\mathbb{E}[Y_i^0 \mid S_i = 0]$, and $\mathbb{E}[Y_i^1 \mid S_i = 1, D_i = 0]$ and the exit shares ω^S and ω^I as weighted averages of the strata-adjusted screener-level estimates. Formally, we first estimate $\Pr(S_i = 1)$ and $\Pr(C_i = 1)$ by weighting the corresponding strata-adjusted rates by each screener's total caseload. We then estimate

$$\widehat{\omega}^S = \frac{1 - \widehat{\Pr}(S_i = 1)}{1 - \widehat{\Pr}(C_i = 1)}, \qquad \widehat{\omega}^I = 1 - \widehat{\omega}^S,$$

and recover the investigator-stage status-quo maltreatment rate through

$$\widehat{\mathbb{E}}[Y_i^1 \mid S_i = 1, D_i = 0] = \frac{\widehat{V}^0 - \widehat{\omega}^S \widehat{\mathbb{E}}[Y_i^0 \mid S_i = 0]}{\widehat{\omega}^I}.$$

Third, we obtain $\widehat{\mu}(\widehat{\tau}^*)$ and $\widehat{\mu}(\widehat{\tau}^S)$ from equation (5), recover $\widehat{\mu}_I(\widehat{\tau}^I)$ from $\widehat{\mu}(\widehat{\tau}^I)$ through the iterated-expectations identity of Section 4.1, and obtain the two moments above and the implied $\widehat{\lambda}^S$ from equation (B1). The additional inputs $\widehat{\sigma}(t)$ and $\widehat{\mu}_0(t)$ are obtained from analogous screener regressions of the screening decision on all calls and maltreatment on screened-out calls.

Substituting these into Propositions 3.1 and 3.2 gives $\widehat{\Delta}^S$, $\widehat{\Delta}^I$, $\widehat{\mathcal{C}}^{\text{Sys}}$, $\widehat{V}^I$, $\widehat{V}^S$, and $\widehat{V}^{System}$, and Appendix Table A.19 reports the estimated maltreatment rates and exit shares. We obtain standard errors by a bootstrap with 500 replications, as in Baron et al. (2024) and Angelova, Dobbie and Yang (2025). Each replication draws independent screener-level estimates in equations (5) and (B1) from their estimated asymptotic distributions (with heteroskedasticity-robust standard errors) and re-estimates the extrapolations and the statistics of interest.

Appendix Figures A.4 and A.5 report the balance tests for screener and investigator assignment. Each regresses the decision and the assigned decision-maker's leave-out propensity on call characteristics, with day-by-shift fixed effects for screeners and month-by-ZIP-code-by-shift fixed effects for investigators, and reports the joint F-statistic of each regression. We construct the leave-one-out propensity by first residualizing decisions on the assignment-stratum fixed effects and then averaging the residualized decisions across the worker's other calls. For investigators, call characteristics strongly predict the case-opening decision (joint F-statistic of 48.5) but are much less predictive of the assigned investigator's leave-out case-opening propensity (joint F-statistic of 2.6). Section 4.1 reports the screener test.

C Additional Details on Robustness and Extensions

C.1 Alternative Maltreatment Extrapolations

Our main specification recovers $\mu(t)$ and $\mu_I(t)$ from linear, inverse-variance-weighted extrapolations across screeners, with screener-level rates adjusted for day-by-shift fixed effects

(Figure A.2). Columns (2)–(5) of Appendix Table A.5 replace it with four standard alternatives (e.g., Baron et al., 2024; Angelova, Dobbie and Yang, 2025): a local-linear extrapolation using a Gaussian kernel and bandwidth 0.04, a quadratic extrapolation, an unweighted linear extrapolation, and a linear extrapolation without the strata fixed effects. Across the four, Δ^S ranges from 10.4 to 12.5 percentage points, $V^0 - V^S$ from 0.5 to 1.8, and $V^0 - V^{System}$ from 0.4 to 2.4, and the weighted stage-wise comparisons always sum to well above the system-wide deployment effect.

Columns (6) and (7) replace the extrapolation with worst-case bounds, which use only the quasi-random assignment of screeners and the binary outcome (Manski, 1990). Because the width of the bounds equals the share of calls with an opened case, we construct them on the caseloads of the least screen-in-intensive screeners. We residualize the screen-in decision on the strata fixed effects, compute each screener's leave-one-out mean residual, and let $\mathcal{K}$ be the screeners in the lowest quintile of this residualized screen-in rate. For each threshold t, the group's case-opening rate and maltreatment rate among unopened calls on the risk set are

$$a_S(t) := \Pr\big(C_i = 1 \mid m(X_i) \le t,\, K_i \in \mathcal{K}\big), \qquad r_S(t) := \mathbb{E}\big[Y_i \mid C_i = 0,\, m(X_i) \le t,\, K_i \in \mathcal{K}\big],$$

which we estimate as fitted values for $\mathcal{K}$ from regressions of the case-opening decision and the observed outcome on quintile indicators, their interactions with the risk-set indicator, and the strata fixed effects. Since Y_i^* is binary and observed when no case is opened,

$$\{1 - a_S(t)\}\, r_S(t) \;\le\; \mathbb{E}\big[Y_i^* \mid m(X_i) \le t\big] \;\le\; \{1 - a_S(t)\}\, r_S(t) + a_S(t).$$

Under the screening-in assumption of Section 4.1, these are bounds on $\mu(t)$, and substituting each into $\mu_I(t) = \mu_0(t) + \big(\mu(t) - \mu_0(t)\big)/\sigma(t)$ gives bounds on $\mu_I(t)$; Panels A and B of Appendix Figure A.6 plot both. Standard errors come from 500 bootstrap draws as in the main specification, with $a_S(t)$ and $r_S(t)$ independently drawn from their estimated asymptotic distributions. Across the bounds, Δ^S lies between 9.2 and 11.7 percentage points and $V^0 - V^S$ between -0.4 and 1.3, while the bounds on Δ^I, $V^0 - V^I$, and $V^0 - V^{System}$ contain zero. The weighted stage-wise comparisons again sum to well above the system-wide deployment effect.

Column (8) uses the screening decision alone. For each risk set, we regress the strata-adjusted maltreatment rate among a screener's screened-out calls on the screener's screen-in rate, weighting by the inverse variance of the estimated maltreatment rate, and extrapolate to a screen-in rate of zero, at which every call is screened out and Y_i^0 is observed on the whole risk set. Because it pools no investigated calls, this extrapolation identifies $\mu(t)$, and through the iterated-expectations identity $\mu_I(t)$, without case-opening variation and even with an average screening-in effect. It is long, since screen-in rates range from roughly 25 to 50 percent against case-opening rates of 0 to 10 percent in the baseline. Its estimates differ from the baseline, with $\Delta^S = 3.9$ percentage points (s.e. 2.2) and $V^0 - V^{System} = 0.5$ (s.e. 1.3), but are imprecise, and the confidence intervals on all three deployment effects contain the baseline estimates.

C.2 Alternative Identification for the Screening-Stage Deployment Effect

The main specification identifies the two moments that the screening-stage deployment requires, the case-opening rate $\Pr(D_i = 1 \mid m(X_i) \geq t)$ and the maltreatment rate $\mathbb{E}[Y_i^1 \mid m(X_i) \geq t, D_i = 0]$ on the algorithm's screened-in set, by extrapolating to a screen-in rate of one in equation (B1). Appendix Table A.6 replaces the extrapolation with bounds under monotone selection (columns 2 and 3) and with point estimates under selection on observables (column 4).

The monotone selection bounds use the screeners in the most screen-in-intensive quintile, $\mathcal{J}$, constructed as in Appendix C.1, for whom the censored share is smallest. For each threshold t, define on the risk set $\{m(X_i) \geq t\}$ the group's screen-in rate, its case-opening rate among screened-in calls, and its maltreatment rate among screened-in but unopened calls,

$$
\begin{aligned}
s(t) &:= \Pr\big(S_i = 1 \mid m(X_i) \geq t,\, K_i \in \mathcal{J}\big),\\
d(t) &:= \Pr\big(D_i = 1 \mid S_i = 1,\, m(X_i) \geq t,\, K_i \in \mathcal{J}\big),\\
r(t) &:= \mathbb{E}\big[Y_i \mid S_i = 1,\, D_i = 0,\, m(X_i) \geq t,\, K_i \in \mathcal{J}\big],
\end{aligned}
$$

each estimated as the fitted value for $\mathcal{J}$ from fixed-effect regressions as in Appendix C.1. The case-opening decision is unobserved for the share $1 - s(t)$ of the risk set that $\mathcal{J}$ screens out, and setting the missing decisions and outcomes to their extremes gives the worst-case bounds

$$
\begin{aligned}
\Pr\big(D_i = 1 \mid m(X_i) \geq t\big) &\in \Big[\, s(t)\, d(t),\;\; s(t)\, d(t) + 1 - s(t) \,\Big],\\
\mathbb{E}\big[Y_i^1 \mid m(X_i) \geq t,\, D_i = 0\big] &\in \left[\, \frac{s(t)\{1 - d(t)\}\, r(t)}{s(t)\{1 - d(t)\} + 1 - s(t)},\;\; \frac{s(t)\{1 - d(t)\}\, r(t) + 1 - s(t)}{s(t)\{1 - d(t)\} + 1 - s(t)} \,\right].
\end{aligned}
$$

The maltreatment bounds treat the numerator and denominator of $\mathbb{E}[(1 - D_i)Y_i^1 \mid m(X_i) \geq t]/\mathbb{E}[1 - D_i \mid m(X_i) \geq t]$ jointly, since the ratio is largest when every screened-out call is unopened with $Y_i^1 = 1$ and smallest when every screened-out call is unopened with $Y_i^1 = 0$. Because even the most screen-in-intensive screeners screen in at most about 35 percent of calls, these bounds are wide. Monotone selection assumes that the calls screeners screen out would be opened at weakly lower rates, and followed by weakly less maltreatment, than the calls they screen in (Manski and Pepper, 2000),

$$
\begin{aligned}
\Pr\big(D_i = 1 \mid S_i = 0,\, m(X_i) \geq t\big) &\leq \Pr\big(D_i = 1 \mid S_i = 1,\, m(X_i) \geq t\big),\\
\mathbb{E}\big[Y_i^1 \mid S_i = 0, D_i = 0, m(X_i) \geq t\big] &\leq \mathbb{E}\big[Y_i \mid S_i = 1,\, D_i = 0,\, m(X_i) \geq t\big].
\end{aligned}
$$

It replaces the worst-case upper values with $d(t)$ and $r(t)$ and leaves the lower bounds

unchanged,

$$\Pr\big(D_i = 1 \mid m(X_i) \geq t\big) \in \Big[\, s(t)\, d(t),\;\; d(t) \,\Big],$$
$$\mathbb{E}\big[Y_i^1 \mid m(X_i) \geq t,\, D_i = 0\big] \in \left[\, \frac{s(t)\{1 - d(t)\}\, r(t)}{s(t)\{1 - d(t)\} + 1 - s(t)},\;\; r(t) \,\right],$$

so only the upper bound on maltreatment is informative; the lower bound imputes no maltreatment to every screened-out call. Columns (2) and (3) evaluate the deployment effect at the lower and upper bounds of both moments, with standard errors from the bootstrap of the main specification in which $s(t)$, $d(t)$, and $r(t)$ are drawn from their estimated asymptotic distributions. At the upper bounds, $V^0 - V^S = 2.5$ percentage points (s.e. 0.4), with a downstream investigator response of -5.4.

Selection on observables assumes that, conditional on X_i, the screening decision is independent of the case-opening decision and the outcome, $S_i \perp\!\!\!\perp (D_i, Y_i^1) \mid X_i$, with $\Pr(S_i = 1 \mid X_i) > 0$. Section 4.2 finds that screeners appear to add no selection on maltreatment risk beyond the algorithm's prediction, which supports this assumption. Both moments depend on the data only through D_i and $(1 - D_i)\, Y_i^1$, which are observed whenever $S_i = 1$, so only selection into screening requires correction. We estimate them with augmented inverse propensity weighting (AIPW), which is consistent if either the propensity or the outcome model is consistently estimated (Robins, Rotnitzky and Zhao, 1994). The nuisance functions, fit using the two-stage prediction procedure described in Appendix B.2 in the training sample and evaluated in the evaluation sample, are the screen-in propensity $\widehat{\pi}_S(X_i)$ for $\Pr(S_i = 1 \mid X_i)$, the case-opening propensity $\widehat{\pi}_D(X_i)$ for $\Pr(D_i = 1 \mid S_i = 1, X_i)$, and the outcome regression $\widehat{\mu}_{Y|D=0,S=1}(X_i)$ for $\mathbb{E}[Y_i \mid S_i = 1, D_i = 0, X_i]$, and we winsorize both propensities to $[0.01, 0.99]$.

The case-opening rate is the mean on the risk set of

$$\psi_D(i) = \widehat{\pi}_D(X_i) + \frac{S_i}{\widehat{\pi}_S(X_i)}\big\{D_i - \widehat{\pi}_D(X_i)\big\},$$

and the maltreatment rate is the ratio of the risk-set means of

$$\psi_{D0Y}(i) = \{1 - \widehat{\pi}_D(X_i)\}\, \widehat{\mu}_{Y|D=0,S=1}(X_i) + \frac{S_i}{\widehat{\pi}_S(X_i)}\Big[(1 - D_i)\, Y_i - \{1 - \widehat{\pi}_D(X_i)\}\, \widehat{\mu}_{Y|D=0,S=1}(X_i)\Big],$$
$$\psi_{D0}(i) = 1 - \widehat{\pi}_D(X_i) + \frac{S_i}{\widehat{\pi}_S(X_i)}\Big[(1 - D_i) - \{1 - \widehat{\pi}_D(X_i)\}\Big].$$

Column (4) reports the results, with standard errors from the bootstrap of the main specification. Under selection on observables, $V^0 - V^S = 1.6$ percentage points (s.e. 0.4), with a downstream investigator response of -6.3.

Both strategies are more precise than the extrapolation, and under both the deployment effect falls far short of the weighted screening-stage comparison $\omega^S \Delta^S = 7.6$, because the downstream investigator response remains large and negative.

C.3 Alternative Outcome Definitions

Appendix Table A.7 re-estimates the stage-wise comparisons, the deployment effects, and the screening-stage decomposition under eight alternative outcomes: re-referral horizons of two to five months (columns 2–5), and four outcomes that require an escalating system response within six months, a new investigation, a substantiated investigation, a subsequent case opening, and a subsequent removal (columns 6–9). The first row reports each outcome's status-quo rate among unopened calls, which ranges from 33.8 percent for the baseline to 1.3 percent for a subsequent removal, so percentage-point magnitudes shrink mechanically for the rarer outcomes.

Under every outcome, Δ^S is positive and the aggregation correction $\omega^I \mathcal{C}^{\text{Sys}}$ is negative, both with confidence intervals that exclude zero, and the weighted stage-wise comparisons sum to more than the system-wide deployment effect. The screening-stage deployment effect is positive except at the three- and four-month horizons. The downstream investigator response is negative at every re-referral horizon and for a new investigation, and small for the three outcomes that require both a later screener and investigator to act, for which the screening-stage deployment effect is comparable to or larger than $\omega^S \Delta^S$. We prefer the re-referral measure because the more intensive outcomes register maltreatment only when later screeners and investigators act on it, so they inherit any errors in those decisions.

C.4 Average Screening-In Effects

Section 4.1 assumes that, within a risk set, a call's subsequent maltreatment if screened in and left unopened, Y_i^1, equals on average its subsequent maltreatment if screened out, Y_i^0, so that there is no average screening-in effect. We first bound this effect using variation across screeners and then report how our estimates change if it is nonzero.

Bounds on the Average Screening-In Effect. Following Lee (2009) and Baron et al. (2024), let Z_i^S indicate assignment to a screener whose leave-one-out screen-in propensity, residualized on the strata, is above the median. We assume that Z_i^S is quasi-randomly assigned within strata and that $S_i(1) \geq S_i(0)$, where $S_i(z)$ is the screening decision call i would receive under $Z_i^S = z$, and let $\mathcal{C}_S := \{S_i(1) > S_i(0)\}$ denote the screening compliers. On the risk sets $\{m(X_i) \leq t\}$ for $t \in \{\tau^*, \tau^S\}$, the target is the complier-average screening-in effect

$$\delta_S(t) := \mathbb{E}\big[Y_i^1 - Y_i^0 \mid \mathcal{C}_S,\, m(X_i) \leq t\big],$$

which is zero under the assumption. Since $Y_i = (1 - S_i)\, Y_i^0 + S_i(1 - D_i)\, Y_i^1$, three Wald ratios identify complier moments on each risk set, with all expectations strata-adjusted and conditional on $\{m(X_i) \leq t\}$,

$$\mu_0^S(t) := \frac{\mathbb{E}[(1 - S_i)\, Y_i \mid Z_i^S = 1] - \mathbb{E}[(1 - S_i)\, Y_i \mid Z_i^S = 0]}{\mathbb{E}[1 - S_i \mid Z_i^S = 1] - \mathbb{E}[1 - S_i \mid Z_i^S = 0]} = \mathbb{E}\big[Y_i^0 \mid \mathcal{C}_S\big],$$

$$\ell_1^S(t) := \frac{\mathbb{E}[S_i Y_i \mid Z_i^S = 1] - \mathbb{E}[S_i Y_i \mid Z_i^S = 0]}{\mathbb{E}[S_i \mid Z_i^S = 1] - \mathbb{E}[S_i \mid Z_i^S = 0]} = \mathbb{E}\big[(1 - D_i)\, Y_i^1 \mid \mathcal{C}_S\big],$$

$$q^S(t) := \frac{\mathbb{E}[S_i D_i \mid Z_i^S = 1] - \mathbb{E}[S_i D_i \mid Z_i^S = 0]}{\mathbb{E}[S_i \mid Z_i^S = 1] - \mathbb{E}[S_i \mid Z_i^S = 0]} = \mathbb{E}\big[D_i \mid \mathcal{C}_S\big].$$

Since $Y_i^1 \in \{0,1\}$, the censored term $\mathbb{E}[D_i\, Y_i^1 \mid \mathcal{C}_S]$ lies in $[0, q^S(t)]$, which gives

$$\delta_S(t) \in \Big[\, \ell_1^S(t) - \mu_0^S(t),\;\; \ell_1^S(t) + q^S(t) - \mu_0^S(t) \,\Big],$$

an interval as wide as the complier case-opening rate. Appendix Table A.8 reports the estimates. Censoring at case opening, the identified set for δ_S is $[-2.7, 8.6]$ percentage points on $\{m(X_i) \le \tau^*\}$ and $[-0.6, 7.9]$ on $\{m(X_i) \le \tau^S\}$, and both sets and the 95 percent confidence intervals on their endpoints contain zero.

Censoring at foster-care placement, which affects only 1.9 and 0.8 percent of complier calls on the two risk sets, tightens the bound. It requires an additional assumption, which our main analysis does not impose: case opening without placement does not itself affect subsequent maltreatment, so the realized outcome of an opened-but-not-placed call equals Y_i^1.

Proposition C.1. *Suppose Z_i^S is quasi-randomly assigned within strata, $S_i(1) \ge S_i(0)$, and case opening without placement does not affect subsequent maltreatment. Let $\ell_1^{S,P}(t)$ and $q^{S,P}(t)$ denote the Wald moments above with placement in place of case opening. Then,*

$$\delta_S(t) \in \Big[\, \ell_1^{S,P}(t) - \mu_0^S(t),\;\; \ell_1^{S,P}(t) + q^{S,P}(t) - \mu_0^S(t) \,\Big],$$

and this interval is contained in the case-opening interval.

Proof. Partition the screened-in compliers on the risk set by their realized case-opening and placement decisions: unopened (share α_1), opened but not placed (α_2), and placed (α_3), with $\alpha_2 + \alpha_3 = q^S(t)$ and $\alpha_3 = q^{S,P}(t)$, and let m_1, m_2, m_3 denote the mean of Y_i^1 in each group. Screened-out calls have neither an opened case nor a placement, so the screened-out moment is unchanged: $\mu_0^{S,P}(t) = \mu_0^S(t)$. By iterated expectations, $\delta_S(t) = \alpha_1 m_1 + \alpha_2 m_2 + \alpha_3 m_3 - \mu_0^S(t)$. The placement moment identifies $\ell_1^{S,P}(t) = \alpha_1 m_1 + \alpha_2 \widetilde{m}_2$, where $\widetilde{m}_2$ is the mean *realized* outcome among opened-but-not-placed calls; the no-direct-effect-of-opening assumption gives $\widetilde{m}_2 = m_2$. Since $m_3 \in [0,1]$, the term $\alpha_3 m_3$ lies in $[0, q^{S,P}(t)]$, which yields the stated interval. For the containment, $\ell_1^{S,P}(t) - \ell_1^S(t) = \alpha_2 m_2 \ge 0$ raises the lower endpoint, and $\big(\ell_1^S(t) + q^S(t)\big) - \big(\ell_1^{S,P}(t) + q^{S,P}(t)\big) = \alpha_2(1 - m_2) \ge 0$ lowers the upper endpoint. □

Censoring at placement, the identified sets are $[1.0, 2.9]$ percentage points on $\{m(X_i) \le \tau^*\}$ and $[1.9, 2.7]$ on $\{m(X_i) \le \tau^S\}$. Their endpoints have standard errors of four to five percentage points, so the 95 percent confidence intervals include zero. The bounds suggest that the average screening-in effect is small, within a few percentage points of zero, consistent with Baron et al. (2024), who find a small average screening-in effect on subsequent maltreatment in Michigan CPS over an earlier period.

Sensitivity to an Average Screening-In Effect. The assumption enters our estimates only through the extrapolation in equation (5). On the risk set $\{m(X_i) \le t\}$, that extrapolation pools screened-out calls, whose outcome is Y_i^0, with screened-in but unopened calls, whose outcome is Y_i^1, so its intercept recovers

$$\widetilde{\mu}(t) := \mathbb{E}\big[(1 - S_i)\, Y_i^0 + S_i\, Y_i^1 \mid m(X_i) \le t\big] = \mu(t) + \sigma(t)\, \eta(t),$$

where $\sigma(t) = \Pr(S_i = 1 \mid m(X_i) \leq t)$ is the observed screen-in rate on the risk set and $\eta(t) := \mathbb{E}[Y_i^1 - Y_i^0 \mid S_i = 1, m(X_i) \leq t]$ is the average screening-in effect among the screened-in calls in it. The status-quo rates, the exit shares, and the two moments of the extrapolation for V^S do not depend on $\eta(t)$, because they are observed or are outcomes of screened-in calls. We take the effect to be common across the risk sets that enter our estimates, $\eta(\tau^S) = \eta(\tau^I) = \eta(\tau^*) = \eta$, and write $\widehat{\Delta}^S$, $\widehat{\Delta}^I$, and so on for the baseline estimates, which impose $\eta = 0$. Since $\mu(t) = \widetilde{\mu}(t) - \sigma(t)\,\eta$, the iterated-expectations identity of Section 4.1 gives $\mu_I(t) = \widetilde{\mu}_I(t) - \eta$, where $\widetilde{\mu}_I(t) := \{\widetilde{\mu}(t) - (1 - \sigma(t))\,\mu_0(t)\}/\sigma(t) = \mathbb{E}[Y_i^1 \mid S_i = 1, m(X_i) \leq t]$ is what the baseline computes. Substituting into the definitions of Section 3,

$$\begin{aligned}
\Delta^S(\eta) &= \widehat{\Delta}^S + \sigma(\tau^S)\,\eta, & \Delta^I(\eta) &= \widehat{\Delta}^I + \eta, \\
V^0 - V^{System}(\eta) &= \left(V^0 - \widehat{V}^{System}\right) + \sigma(\tau^*)\,\eta, & V^0 - V^I(\eta) &= \omega^I \Delta^I(\eta), \\
V^0 - V^S(\eta) &= \left(V^0 - \widehat{V}^S\right) + \lambda^S \sigma(\tau^S)\,\eta, & \Gamma^S(\eta) &= \widehat{\Gamma}^S - \sigma(\tau^S)\,\eta.
\end{aligned}$$

Δ^S and V^{System} depend on η through $\mu(\tau^S)$ and $\mu(\tau^*)$; Δ^I and V^I through $\mu_I(\tau^I)$, because the investigation-stage algorithm's unopened calls produce Y_i^0; and V^S and Γ^S through the screened-out pool's $\mu(\tau^S)$. The downstream investigator response $\mathcal{R}^I$ does not depend on η, since it contrasts Y_i^1 across two pools that humans investigate, and Proposition 3.2 gives $\omega^I \mathcal{C}^{\text{Sys}}(\eta) = \omega^I \widehat{\mathcal{C}}^{\text{Sys}} + \{\sigma(\tau^*) - \omega^S \sigma(\tau^S) - \omega^I\}\,\eta$. A positive η, for example from closer monitoring after an investigation, makes the pooled intercept overstate maltreatment on every risk set, so the baseline understates Δ^S, Δ^I, and the system-wide deployment effect; a negative η reverses each.

Appendix Table A.9 reports these objects for η equal to -4, -2, 0, $+2$, and $+4$ percentage points, a grid that contains the placement bounds on both risk sets. Across the grid, Δ^S moves from 9.8 to 13.1 percentage points, $V^0 - V^{System}$ from -0.2 to 2.8, and Δ^I one for one with the effect, from -2.2 to 5.8. At every grid point, the weighted stage-wise comparisons, between 5.8 and 10.6 points, exceed the system-wide deployment effect, and the screening-stage deployment effect, between 0.1 and 2.2 points, remains a fraction of the screening-stage comparison.

C.5 Shapley-Style Decomposition

This subsection attributes the system-wide deployment effect $V^0 - V^{System}$ to the two stages. The attribution depends on the order of deployment, because the effect of deploying the algorithm at one stage depends on whether it is already deployed at the other. Deploying it at the investigation stage is worth $V^0 - V^I = \omega^I \Delta^I$ from the status quo (Proposition 3.1) but $V^S - V^{System}$ once screening is automated, and deploying it at the screening stage is worth $V^0 - V^S$ first and $V^I - V^{System}$ second. The two orderings give two decompositions of the same effect,

$$V^0 - V^{System} = \underbrace{(V^0 - V^I)}_{\text{Investigation first}} + \underbrace{(V^I - V^{System})}_{\text{Screening second}}, \tag{C2}$$

$$V^0 - V^{System} = \underbrace{(V^0 - V^S)}_{\text{Screening first}} + \underbrace{(V^S - V^{System})}_{\text{Investigation second}}. \tag{C3}$$

as in other path-dependent decompositions such as the Kitagawa–Oaxaca–Blinder decomposition (Kitagawa, 1955; Blinder, 1973; Oaxaca, 1973). The two marginal values of each stage differ by the same amount,

$$\mathcal{I} := V^0 - V^I - V^S + V^{System}. \tag{C4}$$

since $\mathcal{I} = (V^0 - V^I) - (V^S - V^{System}) = (V^0 - V^S) - (V^I - V^{System})$. A positive $\mathcal{I}$ means that each deployment is worth less once the other is in place, so the two are substitutes; a negative $\mathcal{I}$ means that they are complements; and when $\mathcal{I} = 0$ each stage's marginal contribution is independent of the ordering.

Following Shapley (1953), we resolve the order dependence by defining each stage's "stage contribution" as its average marginal value across the two orderings,

$$\theta^I := \tfrac{1}{2}\big[(V^0 - V^I) + (V^S - V^{System})\big], \tag{C5}$$
$$\theta^S := \tfrac{1}{2}\big[(V^0 - V^S) + (V^I - V^{System})\big]. \tag{C6}$$

Proposition C.2. *The stage contributions θ^S and θ^I decompose the system-wide effect,*

$$V^0 - V^{System} = \theta^S + \theta^I, \tag{C7}$$

and each contribution equals the effect of automating that stage from the status quo, adjusted by an equal share of the interaction:

$$\theta^I = \omega^I \Delta^I - \tfrac{1}{2}\mathcal{I}, \qquad \theta^S = (V^0 - V^S) - \tfrac{1}{2}\mathcal{I}. \tag{C8}$$

Proof. Summing the definitions in equations (C5)–(C6),

$$\theta^I + \theta^S = \tfrac{1}{2}\big[(V^0 - V^I) + (V^S - V^{System})\big] + \tfrac{1}{2}\big[(V^0 - V^S) + (V^I - V^{System})\big] = V^0 - V^{System},$$

which is equation (C7). For θ^I, equation (1) gives $V^0 - V^I = \omega^I \Delta^I$, and the definition of $\mathcal{I}$ rearranges to $V^S - V^{System} = (V^0 - V^I) - \mathcal{I} = \omega^I \Delta^I - \mathcal{I}$. Substituting both into equation (C5),

$$\theta^I = \tfrac{1}{2}\big[\omega^I \Delta^I + \omega^I \Delta^I - \mathcal{I}\big] = \omega^I \Delta^I - \tfrac{1}{2}\mathcal{I}.$$

For θ^S, expanding the definition of $\mathcal{I}$,

$$(V^0 - V^S) - \tfrac{1}{2}\mathcal{I} = (V^0 - V^S) - \tfrac{1}{2}\big(V^0 - V^I - V^S + V^{System}\big) = \tfrac{1}{2}\big[(V^0 - V^S) + (V^I - V^{System})\big] = \theta^S.$$

□

Appendix Table A.20 reports the stage contributions and the interaction, which depend only on V^0, V^S, V^I, and V^{System} (Appendix Table A.19). The screening stage contributes $\theta^S = 0.9$ percentage points (s.e. 1.2) and the investigation stage $\theta^I = 0.4$ (s.e. 1.2), and neither is distinguishable from zero. The interaction, $\mathcal{I} = 0.4$ percentage points (s.e. 2.3), gives no evidence that the two deployments are complements or substitutes. Although Δ^S far exceeds Δ^I, the screening stage's contribution is only about one-eighth of its weighted comparison of 7.6 points. Deployed first, its gain is offset by the downstream investigator

response, and deployed second, by the aggregation correction (Section 4.3).

D Within-Stage Analysis and Heterogeneity Across Decision Makers

In this appendix, we extend our framework to analyze individual decision-makers within a stage. Appendices D.1 and D.2 treat investigators, and Appendix D.3 treats screeners. Appendix D.4 develops a test of the hypothesis that investigators influence outcomes only through their case-opening decisions, and applies the same test to screeners.

D.1 Investigator-Specific Comparisons and Decomposition

Let g index investigators, and let G_i denote the investigator assigned to call i (defined on $\{S_i = 1\}$). We define two shares for each investigator: the "caseload share" $\pi_g^I := \Pr(G_i = g \mid S_i = 1)$ and the "non-action share" $\alpha_g^I := \Pr(G_i = g \mid S_i = 1, D_i = 0)$, with $\sum_g \pi_g^I = \sum_g \alpha_g^I = 1$. For each investigator, we construct an algorithmic decision rule with a threshold matched to that investigator's own case-opening rate: $D_i^{m,g} := \mathbf{1}\{m(X_i) \geq \tau_g^I\}$, with $\Pr(D_i^{m,g} = 1 \mid G_i = g) = \Pr(D_i = 1 \mid G_i = g)$.

We index the screened-in potential outcome by the identity of the assigned investigator. Let $Y_i^{1,g}$ denote call i's subsequent maltreatment if it is screened in, investigated by investigator g, and no case is opened. The realized screened-in potential outcome is $Y_i^1 = Y_i^{1,G_i}$. Investigators may differ not only in which cases they open but in how they conduct investigations—whom they interview, what advice they give, which intermediate actions they take—so $Y_i^{1,g}$ may vary across g for the same call. If instead $Y_i^{1,g} = Y_i^1$ for all g, investigator identity would matter only through the case-opening decision; we return to this hypothesis in Appendix D.4.

D.1.1 Investigator-Specific Comparisons and Deployment

For each investigator g, define investigator g's comparison to the algorithm as

$$\Delta_g^I := \mathbb{E}[Y_i^{1,g} \mid G_i = g, D_i = 0] - \mathbb{E}[Y_i^0 \mid G_i = g, D_i^{m,g} = 0].$$

The first term is the subsequent maltreatment among the calls investigator g investigates but does not open, and the second is the subsequent maltreatment among the calls an equally selective algorithm, acting at g's own case-opening rate and without an investigation, would leave unopened. Adding and subtracting $\mathbb{E}[Y_i^0 \mid G_i = g, D_i = 0]$, we write

$$\Delta_g^I = \underbrace{\mathbb{E}[Y_i^{1,g} - Y_i^0 \mid G_i = g, D_i = 0]}_{\text{Direct effect of } g\text{'s investigation}} + \underbrace{\mathbb{E}[Y_i^0 \mid G_i = g, D_i = 0] - \mathbb{E}[Y_i^0 \mid G_i = g, D_i^{m,g} = 0]}_{\text{Investigator } g\text{'s selection}}.$$

The first term is the effect of g's investigation on the calls g leaves unopened, and the selection term contrasts whether g or the algorithm leaves riskier calls unopened. We do not take a stand on the mechanism behind Δ_g^I: a negative value means investigator g adds value relative to the algorithm, whether by ranking calls on private information beyond X_i or through the direct effect of g's investigation on subsequent maltreatment.

Because the investigation stage is terminal, replacing investigator g with the algorithm has no downstream effect. The deployment replaces g's decisions and investigations on g's caseload and leaves every other investigator unchanged: cases are opened according to $C_i^{I,g} := S_i\big(D_i\mathbf{1}\{G_i \neq g\} + D_i^{m,g}\mathbf{1}\{G_i = g\}\big)$. We write V_g^I as the resulting system-wide maltreatment rate. At the individual level, the stage-wise comparison equals the deployment effect.

Proposition D.1. *The effect of replacing investigator g on subsequent maltreatment satisfies*

$$V^0 - V_g^I = \Pr(G_i = g, S_i = 1 \mid C_i = 0)\,\Delta_g^I.$$

Proof. The rule $D_i^{m,g}$ opens cases at investigator g's own rate, $\Pr(D_i^{m,g} = 1 \mid G_i = g) = \Pr(D_i = 1 \mid G_i = g)$, and $C_i^{I,g} = C_i$ for every call not assigned to g. Hence $\Pr(C_i^{I,g} = 1) = \Pr(C_i = 1)$, and so $p := \Pr(C_i = 0) = \Pr(C_i^{I,g} = 0)$.

Off $\{G_i = g\}$, $C_i^{I,g} = C_i$ and every pool produces the same outcomes under both regimes, so these terms cancel. On $\{G_i = g\}$ (where $S_i = 1$), $\{C_i = 0\} = \{D_i = 0\}$ with unopened calls producing $Y_i^{1,g}$, and $\{C_i^{I,g} = 0\} = \{D_i^{m,g} = 0\}$ with unopened calls producing Y_i^0. We can therefore write

$$p\,(V^0 - V_g^I) = \mathbb{E}\big[Y_i^{1,g}\mathbf{1}\{G_i = g, S_i = 1, D_i = 0\}\big] - \mathbb{E}\big[Y_i^0\mathbf{1}\{G_i = g, S_i = 1, D_i^{m,g} = 0\}\big].$$

Since $\Pr(D_i = 0 \mid G_i = g) = \Pr(D_i^{m,g} = 0 \mid G_i = g)$, the two events $\{G_i = g, S_i = 1, D_i = 0\}$ and $\{G_i = g, S_i = 1, D_i^{m,g} = 0\}$ share the same probability q_g. Factoring it out,

$$p\,(V^0 - V_g^I) = q_g\big(\mathbb{E}[Y_i^{1,g} \mid G_i = g, D_i = 0] - \mathbb{E}[Y_i^0 \mid G_i = g, D_i^{m,g} = 0]\big) = q_g\,\Delta_g^I.$$

Finally, $\{G_i = g, S_i = 1, D_i = 0\} = \{G_i = g, S_i = 1, C_i = 0\}$, so $q_g/p = \Pr(G_i = g, S_i = 1 \mid C_i = 0)$, giving $V^0 - V_g^I = \Pr(G_i = g, S_i = 1 \mid C_i = 0)\,\Delta_g^I$. □

D.1.2 Decomposing the Stage-Level Comparison

The population stage-level comparison Δ^I can be decomposed across investigators. For each investigator g, define the "individual threshold adjustment"

$$\Lambda_g^I := \mathbb{E}[Y_i^0 \mid G_i = g, D_i^{m,g} = 0] - \mathbb{E}[Y_i^0 \mid G_i = g, D_i^m = 0],$$

which captures the cost or benefit of the algorithm acting at g's rate rather than the stage-wide rate, holding the algorithm's ranking fixed.

Proposition D.2. *The investigation-stage comparison decomposes as*

$$\Delta^I = \overline{\Delta}^I + \overline{\Lambda}^I + \xi^I,$$

where

$$\overline{\Delta}^I := \sum_g \alpha_g^I \, \Delta_g^I, \qquad \overline{\Lambda}^I := \sum_g \alpha_g^I \, \Lambda_g^I,$$
$$\xi^I := \sum_g \alpha_g^I \left(\mathbb{E}[Y_i^0 \mid G_i = g, D_i^m = 0] - \mathbb{E}[Y_i^0 \mid S_i = 1, D_i^m = 0] \right).$$

Proof. By the law of iterated expectations, partitioning $\{S_i = 1, D_i = 0\}$ by G_i and using $Y_i^1 = Y_i^{1,G_i}$,

$$\mathbb{E}[Y_i^1 \mid S_i = 1, D_i = 0] = \sum_g \alpha_g^I \, \mathbb{E}[Y_i^{1,g} \mid G_i = g, D_i = 0].$$

Within each α_g^I-weighted term, add and subtract $\mathbb{E}[Y_i^0 \mid G_i = g, D_i^{m,g} = 0]$ and $\mathbb{E}[Y_i^0 \mid G_i = g, D_i^m = 0]$:

$$\begin{aligned}\mathbb{E}[Y_i^1 \mid S_i = 1, D_i = 0] = \sum_g \alpha_g^I \Big[& \underbrace{\mathbb{E}[Y_i^{1,g} \mid G_i = g, D_i = 0] - \mathbb{E}[Y_i^0 \mid G_i = g, D_i^{m,g} = 0]}_{= \Delta_g^I} \\ & + \underbrace{\mathbb{E}[Y_i^0 \mid G_i = g, D_i^{m,g} = 0] - \mathbb{E}[Y_i^0 \mid G_i = g, D_i^m = 0]}_{= \Lambda_g^I} \\ & + \mathbb{E}[Y_i^0 \mid G_i = g, D_i^m = 0] \Big].\end{aligned}$$

Subtracting $\mathbb{E}[Y_i^0 \mid S_i = 1, D_i^m = 0]$ from both sides and using $\sum_g \alpha_g^I = 1$,

$$\Delta^I = \sum_g \alpha_g^I \, \Delta_g^I + \sum_g \alpha_g^I \, \Lambda_g^I + \sum_g \alpha_g^I \left(\mathbb{E}[Y_i^0 \mid G_i = g, D_i^m = 0] - \mathbb{E}[Y_i^0 \mid S_i = 1, D_i^m = 0] \right),$$

which is the stated decomposition. □

The first term, $\overline{\Delta}^I$, is the non-action-share-weighted average of the investigator-level comparisons, each of which bundles selection with any direct effect of the investigation. The second, $\overline{\Lambda}^I$, is the average threshold adjustment, the difference made by investigators acting at their own case-opening rates rather than at the stage-wide rate. The third, ξ^I, depends only on Y_i^0 and the stage-wide rule D_i^m, so it reflects differences in the risk of the calls each investigator is assigned rather than their decisions, and it is zero when every investigator's caseload has the same risk composition, as quasi-random assignment delivers within strata.

In the empirical analysis, we summarize the distribution of Δ_g^I across investigators via the caseload-weighted mean $\overline{\Delta}_\pi^I := \sum_g \pi_g^I \, \Delta_g^I$, the caseload-weighted standard deviation $\Sigma_\Delta^I := \sqrt{\sum_g \pi_g^I \left(\Delta_g^I - \overline{\Delta}_\pi^I \right)^2}$, and the caseload-weighted shares of investigators worse than the algorithm, $\Pr(\Delta_g^I > 0)$. The caseload-weighted mean $\overline{\Delta}_\pi^I$ differs from the non-action-share-weighted $\overline{\Delta}^I$ in Proposition D.2, and neither equals Δ^I unless $\overline{\Lambda}^I + \xi^I = 0$.

D.2 Identification and Empirical Bayes

We estimate each investigator's comparison Δ_g^I with the strategy of Section 4.1 and summarize its distribution across investigators with a parametric empirical Bayes model that accounts for sampling error.

D.2.1 Identification

The comparison Δ_g^I contrasts investigator g's status-quo maltreatment rate among the calls they leave unopened with an algorithmic benchmark at g's own case-opening rate. The status-quo term $\mathbb{E}[Y_i^{1,g} \mid G_i = g, D_i = 0]$ is investigator g's strata-adjusted maltreatment rate among their unopened calls, obtained from a regression of Y_i on investigator indicators and month-by-ZIP-code-by-shift strata indicators on the sample of screened-in but unopened calls $\{S_i = 1, D_i = 0\}$. The investigator-specific threshold $\widehat{\tau}_g^I$ is chosen so that the algorithm's case-opening rate among screened-in calls equals investigator g's strata-adjusted case-opening rate. The algorithmic benchmark $\mathbb{E}[Y_i^0 \mid G_i = g, D_i^{m,g} = 0]$ is the maltreatment rate, absent investigation, among the calls on g's caseload that the algorithm would leave unopened at g's threshold. Under the quasi-random assignment of investigators, this benchmark is the counterfactual curve of Section 4.1 evaluated at g's threshold, $\mu_I(\tau_g^I)$, which the screener extrapolation in equation (5) and the iterated-expectations identity identify.

This strategy yields a noisy estimate $\widehat{\Delta}_g^I$ for each investigator. We adopt a parametric empirical Bayes approach that shrinks each investigator-specific estimate toward the population mean and recovers the underlying distribution of comparisons.

D.2.2 Empirical Bayes Estimation

We estimate the investigator-level comparisons Δ_g^I, their caseload-weighted mean $\overline{\Delta}_\pi^I$ and standard deviation Σ_Δ^I, and the caseload-weighted share of investigators who underperform the algorithm, $\Pr(\Delta_g^I > 0)$, the last three by the posterior average effect approach of Bonhomme and Weidner (2022). Let $\widehat{\Delta}_g^I$ denote the unshrunk estimate for investigator g produced by the identification strategy above, and let $\Sigma_g := \widehat{\mathrm{Var}}(\widehat{\Delta}_g^I \mid \Delta_g^I)$ denote its sampling variance.

We apply the asymptotic normal approximation,

$$\widehat{\Delta}_g^I = \Delta_g^I + \varepsilon_g, \qquad \varepsilon_g \sim \mathcal{N}(0, \Sigma_g), \tag{D9}$$

with Σ_g treated as known. We impose the normal prior

$$\Delta_g^I \sim \mathcal{N}(\bar{\theta}, \Omega), \tag{D10}$$

with hyperparameters $\bar{\theta}$ (the prior mean of the investigator comparison) and Ω (its prior variance) common across investigators. Given the hyperparameters, the posterior mean for investigator g is

$$\widetilde{\Delta}_g^I := \mathbb{E}[\Delta_g^I \mid \widehat{\Delta}_g^I] = \frac{\Sigma_g}{\Omega + \Sigma_g}\,\bar{\theta} + \frac{\Omega}{\Omega + \Sigma_g}\,\widehat{\Delta}_g^I. \tag{D11}$$

Section 5.1 refers to investigators with $\widetilde{\Delta}_g^I < 0$ as higher-performing. We estimate the hyperparameters $\bar{\theta}$ and Ω by method of moments. Letting G denote the number of investigators, the prior mean is the sample average of the unshrunk estimates and the prior variance is the excess of their sample variance over their average sampling variance,

$$\widehat{\bar{\theta}} = \frac{1}{G}\sum_g \widehat{\Delta}_g^I, \tag{D12}$$

$$\widehat{\Omega} = \max\left\{0,\ \frac{1}{G}\sum_g \left[\left(\widehat{\Delta}_g^I - \widehat{\bar{\theta}}\right)^2 - \Sigma_g\right]\right\}. \tag{D13}$$

Plugging $\widehat{\bar{\theta}}$ and $\widehat{\Omega}$ into (D11) yields the shrunken posterior mean $\widetilde{\Delta}_g^I$ for each investigator. The associated posterior variance under (D9)–(D10) is

$$v_g^2 := \mathrm{Var}(\Delta_g^I \mid \widehat{\Delta}_g^I) = \frac{\Omega\,\Sigma_g}{\Omega + \Sigma_g}. \tag{D14}$$

Throughout, we evaluate the posterior mean (D11) and the posterior variance (D14) at the estimated hyperparameters $\widehat{\bar{\theta}}$ and $\widehat{\Omega}$.

We estimate $\overline{\Delta}_\pi^I$ by the caseload-weighted posterior mean

$$\widetilde{\overline{\Delta}}_\pi^I := \sum_g \pi_g^I\,\widetilde{\Delta}_g^I, \tag{D15}$$

which weights investigators by their investigation-stage caseload share. We estimate the caseload-weighted standard deviation Σ_Δ^I by its posterior-average-effect analog (Bonhomme and Weidner, 2022), which takes posterior expectations of Δ_g^I and $(\Delta_g^I)^2$ separately and aggregates with caseload weights,

$$\widehat{\Sigma}_\Delta^I := \sqrt{\sum_g \pi_g^I\left((\widetilde{\Delta}_g^I)^2 + v_g^2\right) - \Big(\sum_g \pi_g^I\,\widetilde{\Delta}_g^I\Big)^2}, \tag{D16}$$

using $\mathbb{E}[\Delta_g^I \mid \widehat{\Delta}_g^I] = \widetilde{\Delta}_g^I$ and $\mathbb{E}[(\Delta_g^I)^2 \mid \widehat{\Delta}_g^I] = (\widetilde{\Delta}_g^I)^2 + v_g^2$ under the normal-normal model. Under the normal prior, the posterior probability that investigator g's comparison is non-positive is

$$\mathbb{E}\big[\mathbf{1}\{\Delta_g^I \le 0\} \mid \widehat{\Delta}_g^I\big] = \Phi\left(-\frac{\widetilde{\Delta}_g^I}{v_g}\right), \tag{D17}$$

where Φ is the standard normal CDF and v_g is from (D14), so the caseload-weighted share worse than the algorithm is estimated by

$$\widetilde{\Pr}(\Delta_g^I > 0) := \sum_g \pi_g^I\left[1 - \Phi\left(-\frac{\widetilde{\Delta}_g^I}{v_g}\right)\right]. \tag{D18}$$

We obtain standard errors for the reported estimates by a bootstrap procedure analogous

to that of Angelova, Dobbie and Yang (2025). In each of 500 draws, we perturb each investigator's unshrunk estimate $\widehat{\Delta}_g^I$ by independent mean-zero normal noise with its estimated sampling variance Σ_g, re-estimate the empirical Bayes prior $(\bar{\theta}, \Omega)$, and recompute every statistic.

D.3 Variation in Performance across Screeners

We now turn to screeners, whose stage is upstream. For each screener, we define a comparison to an equally selective algorithm and the effect of deploying the algorithm in place of the screener, relate the two as Proposition 3.1 does for the stage as a whole, and report the screener-level results cited in Section 4.2.

D.3.1 Screener-Specific Comparisons and Deployment

Let k index screeners, and let K_i denote the screener assigned to call i. As for investigators, the "caseload share" is $\pi_k^S := \Pr(K_i = k)$ and the "non-action share" is $\alpha_k^S := \Pr(K_i = k \mid S_i = 0)$, with $\sum_k \pi_k^S = \sum_k \alpha_k^S = 1$. For each screener, we construct an algorithmic screening rule with a threshold matched to that screener's own screen-in rate: $S_i^{m,k} := \mathbf{1}\{m(X_i) \geq \tau_k^S\}$, with $\Pr(S_i^{m,k} = 1 \mid K_i = k) = \Pr(S_i = 1 \mid K_i = k)$.

Because the screening stage is upstream, deploying the algorithm in place of a single screener changes which calls reach investigators, which in turn alters case-opening decisions and subsequent maltreatment. We therefore evaluate each screener twice, by a within-stage comparison with the algorithm and by the effect of deploying the algorithm in place of the screener.

The first evaluation compares screener k's decisions against the algorithm:

$$\Delta_k^S := \mathbb{E}[Y_i^0 \mid K_i = k, S_i = 0] - \mathbb{E}[Y_i^0 \mid K_i = k, S_i^{m,k} = 0].$$

Δ_k^S asks whether screener k or an equally selective algorithm at k's own screen-in rate better identifies which calls to screen out, holding the screening rate fixed at k's level.

The second evaluation is the system-level effect of deploying the algorithm in place of screener k. Define

$$C_i^{S,k} := \big(S_i \mathbf{1}\{K_i \neq k\} + S_i^{m,k} \mathbf{1}\{K_i = k\}\big) \cdot D_i,$$

the case-opening rule when the algorithm screens k's calls at k's rate and every other screener and every investigator is unchanged. The effect of the deployment on subsequent maltreatment among the unopened calls in k's caseload is

$$\beta_k^S := \mathbb{E}[Y_i^* \mid K_i = k, C_i = 0] - \mathbb{E}\big[(1 - S_i^{m,k}) Y_i^0 + S_i^{m,k} Y_i^1 \;\big|\; K_i = k, C_i^{S,k} = 0\big],$$

where $Y_i^* = (1 - S_i) Y_i^0 + S_i Y_i^1$ is the status-quo outcome of Section 3.1. Writing V_k^S for the resulting system-wide maltreatment rate among unopened calls, the system-wide effect of the deployment is $V^0 - V_k^S$.

To relate Δ_k^S and β_k^S, define screener-k-conditional analogs of the objects appearing in Proposition 3.1. The conditional exit shares are

$$\omega_k^S := \Pr(S_i = 0 \mid K_i = k, C_i = 0), \qquad \omega_k^I := \Pr(S_i = 1, D_i = 0 \mid K_i = k, C_i = 0),$$

which satisfy $\omega_k^S + \omega_k^I = 1$. The conditional downstream investigator response and exit-margin objects are

$$\mathcal{R}_k^I := \mathbb{E}[Y_i^1 \mid K_i = k, S_i = 1, D_i = 0] - \mathbb{E}[Y_i^1 \mid K_i = k, S_i^{m,k} = 1, D_i = 0],$$
$$\Gamma_k^S := \mathbb{E}[Y_i^0 \mid K_i = k, S_i^{m,k} = 0] - \mathbb{E}[Y_i^1 \mid K_i = k, S_i^{m,k} = 1, D_i = 0],$$

and the algorithmic screen-out share among k's unopened caseload under the deployment is $\lambda_k^S := \Pr(S_i^{m,k} = 0 \mid K_i = k, C_i^{S,k} = 0)$.

Proposition D.3. *The effect of replacing screener k on subsequent maltreatment satisfies*

$$\beta_k^S = \omega_k^S \, \Delta_k^S \; + \; \omega_k^I \, \mathcal{R}_k^I \; + \; (\omega_k^S - \lambda_k^S) \, \Gamma_k^S,$$
$$V^0 - V_k^S = \Pr(K_i = k \mid C_i = 0) \, \beta_k^S \; + \; \eta_k^S,$$

where the remainder is $\eta_k^S := \big[\Pr(K_i = k \mid C_i = 0) - \Pr(K_i = k \mid C_i^{S,k} = 0)\big]\big[\mathbb{E}\big((1 - S_i^{m,k})Y_i^0 + S_i^{m,k} Y_i^1 \;\big|\; K_i = k, C_i^{S,k} = 0\big) - \mathbb{E}(Y_i^* \mid K_i \neq k, C_i = 0)\big]$.

Proof. The first identity is the screener-k-conditional version of the proof of equation (2) in Appendix A: partition $\{K_i = k, C_i = 0\}$ and $\{K_i = k, C_i^{S,k} = 0\}$ by the screening decision, with each pool producing the outcome its regime induces, apply the law of iterated expectations, and add and subtract intermediate terms exactly as in the stage-wide case.

For the second identity, write $q := \Pr(K_i = k \mid C_i = 0)$, $q' := \Pr(K_i = k \mid C_i^{S,k} = 0)$, and $b_k := \mathbb{E}\big[(1 - S_i^{m,k})Y_i^0 + S_i^{m,k} Y_i^1 \mid K_i = k, C_i^{S,k} = 0\big]$ for the maltreatment rate among the unopened calls on k's caseload under the deployment. Deploying the algorithm in place of screener k alters case opening only on k's own caseload, so $C_i^{S,k} = C_i$ whenever $K_i \neq k$, and the pools outside k's caseload are unchanged: the maltreatment rate among unopened calls outside k's caseload is $\bar{a} := \mathbb{E}(Y_i^* \mid K_i \neq k, C_i = 0)$ under both regimes. Partitioning each unopened pool by whether the call belongs to screener k,

$$V^0 = q \, \mathbb{E}(Y_i^* \mid K_i = k, C_i = 0) + (1 - q) \, \bar{a},$$
$$V_k^S = q' \, b_k + (1 - q') \, \bar{a}.$$

Subtracting, then adding and subtracting $q \, b_k$,

$$V^0 - V_k^S = q \underbrace{\big[\mathbb{E}(Y_i^* \mid K_i = k, C_i = 0) - b_k\big]}_{= \beta_k^S} + (q - q') \, \big[b_k - \bar{a}\big] = q \, \beta_k^S + \eta_k^S.$$

When the deployment leaves the unopened mass unchanged—$\Pr(C_i^{S,k} = 0) = \Pr(C_i = 0)$—the off-$k$ masses coincide as well, so $q = q'$ and the remainder η_k^S vanishes. □

The remainder η_k^S arises because the algorithm screens in different calls from k's caseload than k does, so the case-opening rate, and hence the unopened mass, on k's caseload can change. It is the product of the change in k's share of the unopened pool and a difference in maltreatment levels, and its first factor reflects the same shift in k's unopened mass that drives the conditional exit-margin term $(\omega_k^S - \lambda_k^S)\Gamma_k^S$. Both vanish when the deployment

preserves the unopened mass, and because the aggregate exit-margin term $(\omega^S - \lambda^S)\Gamma^S$ is small (Section 4.3), we expect η_k^S to be small as well.

The population screening-stage comparison Δ^S can be decomposed across screeners analogously to the investigator case. Define the individual threshold adjustment for screener k,

$$\Lambda_k^S := \mathbb{E}[Y_i^0 \mid K_i = k, S_i^{m,k} = 0] - \mathbb{E}[Y_i^0 \mid K_i = k, S_i^m = 0],$$

the screener-level analog of the investigator threshold adjustment Λ_g^I.

Proposition D.4. *The screening-stage comparison decomposes as*

$$\Delta^S = \overline{\Delta}^S + \overline{\Lambda}^S + \xi^S,$$

where

$$\overline{\Delta}^S := \sum_k \alpha_k^S \, \Delta_k^S, \qquad \overline{\Lambda}^S := \sum_k \alpha_k^S \, \Lambda_k^S,$$
$$\xi^S := \sum_k \alpha_k^S \left(\mathbb{E}[Y_i^0 \mid K_i = k, S_i^m = 0] - \mathbb{E}[Y_i^0 \mid S_i^m = 0] \right).$$

Proof. The argument parallels the proof of Proposition D.2. By the law of iterated expectations, partitioning $\{S_i = 0\}$ by K_i,

$$\mathbb{E}[Y_i^0 \mid S_i = 0] = \sum_k \alpha_k^S \, \mathbb{E}[Y_i^0 \mid K_i = k, S_i = 0].$$

Within each α_k^S-weighted term, add and subtract $\mathbb{E}[Y_i^0 \mid K_i = k, S_i^{m,k} = 0]$ and $\mathbb{E}[Y_i^0 \mid K_i = k, S_i^m = 0]$; subtract $\mathbb{E}[Y_i^0 \mid S_i^m = 0]$ from both sides; and use $\sum_k \alpha_k^S = 1$. Collecting terms yields the stated decomposition. □

The first term, $\overline{\Delta}^S$, is the non-action-share-weighted average of the screener-level comparisons. The second, $\overline{\Lambda}^S$, is the average threshold adjustment, the difference made by screeners acting at their own screen-in rates rather than at the stage-wide rate. The third, ξ^S, depends only on Y_i^0 and the stage-wide rule S_i^m, so it reflects differences in the risk of the calls each screener is assigned rather than their decisions, and it is zero when every screener's caseload has the same risk composition, as quasi-random assignment delivers within strata.

We summarize the distributions of Δ_k^S and β_k^S across screeners via caseload-weighted means

$$\overline{\Delta}_\pi^S := \sum_k \pi_k^S \, \Delta_k^S, \qquad \overline{\beta}_\pi^S := \sum_k \pi_k^S \, \beta_k^S,$$

and caseload-weighted standard deviations

$$\Sigma_\Delta^S := \sqrt{\textstyle\sum_k \pi_k^S \left(\Delta_k^S - \overline{\Delta}_\pi^S\right)^2}, \qquad \Sigma_\beta^S := \sqrt{\textstyle\sum_k \pi_k^S \left(\beta_k^S - \overline{\beta}_\pi^S\right)^2},$$

together with the caseload-weighted shares $\Pr(\Delta_k^S > 0)$ and $\Pr(\beta_k^S > 0)$ of screeners worse than the algorithm at their respective margins. We also summarize the caseload-weighted

share of screeners worse than a random screening rule on their own caseload, $\Pr(\Delta^S_{k,\text{rand}} > 0)$, where $\Delta^S_{k,\text{rand}} := \mathbb{E}[Y_i^0 \mid K_i = k, S_i = 0] - \mathbb{E}[Y_i^0 \mid K_i = k]$ replaces the algorithmic benchmark with the maltreatment that a rule screening in calls at random at k's own rate would leave among screened-out calls. The caseload-weighted mean $\overline{\Delta}^S_\pi$ differs from the non-action-share-weighted $\overline{\Delta}^S$ in Proposition D.4.

D.3.2 Identification

Screener k's comparison Δ^S_k requires nothing beyond the ingredients of the stage-level comparison Δ^S. The status-quo term $\mathbb{E}[Y_i^0 \mid K_i = k, S_i = 0]$ is screener k's strata-adjusted maltreatment rate among their screened-out calls, recovered from a regression of Y_i on screener indicators and day-by-shift strata indicators on the sample of screened-out calls $\{S_i = 0\}$. The screener-specific threshold $\widehat{\tau}^S_k$ is chosen to match k's strata-adjusted screen-in rate. The algorithmic term $\mathbb{E}[Y_i^0 \mid K_i = k, S_i^{m,k} = 0]$ is, under quasi-random assignment, the population rate $\mu(\tau^S_k)$, which the extrapolation in equation (5) identifies when evaluated at $t = \tau^S_k$ rather than at τ^S. The random-rule benchmark $\mathbb{E}[Y_i^0 \mid K_i = k]$ in $\Delta^S_{k,\text{rand}}$ is, under quasi-random assignment, $\mathbb{E}[Y_i^0] = \mu(1)$, the same extrapolation on all calls. As a check, we re-estimate $\mu(\tau^S_k)$ under the selection-on-observables assumption of Appendix C.2, which recovers the maltreatment rate on the risk set $\{m(X_i) < \tau^S_k\}$ from screened-out calls reweighted by the screen-in propensity, and we find nearly identical estimates.

The deployment effect β^S_k requires two ingredients beyond those of Δ^S_k: the case-opening rate and the maltreatment rate among unopened calls on the algorithm's screened-in pool at screener k's threshold,

$$\Pr\big(D_i = 1 \mid m(X_i) \geq \tau^S_k\big) \qquad \text{and} \qquad \mathbb{E}\big[Y_i^1 \mid m(X_i) \geq \tau^S_k,\, D_i = 0\big].$$

These are the screener-level analogs of the two moments of the screening-stage deployment, and we identify them under selection on observables (Appendix C.2), because extrapolating to a screen-in rate of one at each of the 291 screener-specific thresholds would be imprecise.

Under the selection-on-observables assumption, the two ingredients are sample means of the AIPW scores defined in Appendix C.2, evaluated on screener k's risk set:

$$\widehat{\Pr}\big(D_i = 1 \mid m(X_i) \geq \hat{\tau}^S_k\big) = \mathbb{E}_n\big[\psi_D(i) \mid m(X_i) \geq \hat{\tau}^S_k\big],$$
$$\widehat{\mathbb{E}}\big[Y_i^1 \mid m(X_i) \geq \hat{\tau}^S_k,\, D_i = 0\big] = \frac{\mathbb{E}_n\big[\psi_{D0Y}(i) \mid m(X_i) \geq \hat{\tau}^S_k\big]}{\mathbb{E}_n\big[\psi_{D0}(i) \mid m(X_i) \geq \hat{\tau}^S_k\big]}.$$

The scores are computed once, so moving across screener-specific thresholds smoothly changes the conditioning set. The algorithmic screen-out share follows by Bayes' rule,

$$\hat{\lambda}^S_k = \frac{1 - \hat{\varphi}_k}{\big(1 - \hat{\varphi}_k\big) + \hat{\varphi}_k\big\{1 - \widehat{\Pr}\big(D_i = 1 \mid m(X_i) \geq \hat{\tau}^S_k\big)\big\}},$$

where $\hat{\varphi}_k$ is screener k's strata-adjusted screen-in rate, which the algorithmic rule matches

at $\hat{\tau}_k^S$ by construction. We then estimate β_k^S directly from its definition,

$$\hat{\beta}_k^S = \widehat{\mathbb{E}}\big[Y_i^* \mid K_i = k,\, C_i = 0\big] - \Big[\hat{\lambda}_k^S\, \hat{\mu}(\hat{\tau}_k^S) + \big(1 - \hat{\lambda}_k^S\big)\, \widehat{\mathbb{E}}\big[Y_i^1 \mid m(X_i) \geq \hat{\tau}_k^S,\, D_i = 0\big]\Big],$$

where the status-quo term is screener k's strata-adjusted maltreatment rate among their unopened calls, recovered from the screener fixed-effect regression of Y_i on screener and strata indicators on the sample $\{C_i = 0\}$, and $\hat{\mu}(\hat{\tau}_k^S)$ is the same screener extrapolation that enters Δ_k^S.

D.3.3 Empirical Bayes Estimation

We treat (Δ_k^S, β_k^S) as a vector-valued screener-specific parameter. We estimate the shrunken (Δ_k^S, β_k^S), their caseload-weighted means and standard deviations, and the caseload-weighted shares $\Pr(\Delta_k^S > 0)$ and $\Pr(\beta_k^S > 0)$.

Let $\hat{\theta}_k := (\hat{\Delta}_k^S, \hat{\beta}_k^S)^\top$ denote the unshrunk screener-specific estimates from the identification strategy above, and let $\Sigma_k := \widehat{\mathrm{Cov}}(\hat{\theta}_k \mid \theta_k)$ denote its 2×2 asymptotic covariance matrix, estimated by the bootstrap. We model

$$\hat{\theta}_k = \theta_k + \varepsilon_k, \qquad \varepsilon_k \sim \mathcal{N}(0, \Sigma_k), \tag{D19}$$

with Σ_k treated as known, and the bivariate normal prior

$$\theta_k \sim \mathcal{N}(\bar{\theta}, \Omega), \tag{D20}$$

where $\bar{\theta} \in \mathbb{R}^2$ and $\Omega \in \mathbb{R}^{2\times 2}$ are hyperparameters common across screeners. The bivariate prior allows for non-zero cross-screener correlation between Δ_k^S and β_k^S. Given the hyperparameters, the posterior mean for screener k is the bivariate analog of (D11),

$$\widetilde{\theta}_k := \mathbb{E}[\theta_k \mid \hat{\theta}_k] = \big(\Omega^{-1} + \Sigma_k^{-1}\big)^{-1}\big(\Omega^{-1}\bar{\theta} + \Sigma_k^{-1}\hat{\theta}_k\big), \tag{D21}$$

with associated posterior covariance matrix

$$V_k := \mathrm{Cov}(\theta_k \mid \hat{\theta}_k) = \big(\Omega^{-1} + \Sigma_k^{-1}\big)^{-1}. \tag{D22}$$

We estimate the hyperparameters $\bar{\theta}$ and Ω by method of moments, the matrix-valued analog of (D12)–(D13). Letting K denote the number of screeners, the prior mean is the sample average of the unshrunk estimates and the prior covariance is the excess of their sample covariance over their average sampling covariance,

$$\hat{\bar{\theta}} = \frac{1}{K}\sum_k \hat{\theta}_k, \tag{D23}$$

$$\hat{\Omega} = \frac{1}{K}\sum_k \big[\big(\hat{\theta}_k - \hat{\bar{\theta}}\big)\big(\hat{\theta}_k - \hat{\bar{\theta}}\big)^\top - \Sigma_k\big], \tag{D24}$$

projecting $\hat{\Omega}$ to the nearest positive semi-definite matrix if necessary.

Plugging $\hat{\bar{\theta}}$ and $\hat{\Omega}$ into (D21)–(D22) yields the shrunken posterior moments $(\widetilde{\theta}_k, V_k)$ for each screener. The four objects of interest are estimated as the bivariate analogs of the investigator case. First, the shrunken screener-specific estimates $\widetilde{\Delta}_k^S$ and $\widetilde{\beta}_k^S$ are the components of $\widetilde{\theta}_k$ in (D21). Second, the caseload-weighted averages are estimated by the caseload-weighted posterior means

$$\widetilde{\overline{\Delta}}_{\pi}^{S} := \sum_k \pi_k^S \, \widetilde{\Delta}_k^S, \qquad \widetilde{\overline{\beta}}_{\pi}^{S} := \sum_k \pi_k^S \, \widetilde{\beta}_k^S. \tag{D25}$$

These weight screeners by their share of the screening-stage caseload; they need not equal the prior mean $\hat{\bar{\theta}}$ when caseload shares and sampling precision differ across screeners. Third, the caseload-weighted standard deviations are estimated by the bivariate posterior-average-effect analog of (D16):

$$\widehat{\Sigma}_{\Delta}^{S} := \sqrt{\sum_k \pi_k^S \left((\widetilde{\Delta}_k^S)^2 + V_{k,11} \right) - \Big(\sum_k \pi_k^S \, \widetilde{\Delta}_k^S \Big)^2}, \tag{D26}$$

$$\widehat{\Sigma}_{\beta}^{S} := \sqrt{\sum_k \pi_k^S \left((\widetilde{\beta}_k^S)^2 + V_{k,22} \right) - \Big(\sum_k \pi_k^S \, \widetilde{\beta}_k^S \Big)^2}. \tag{D27}$$

Finally, the shares of screeners with positive comparisons and deployment effects are estimated by

$$\widetilde{\Pr}(\Delta_k^S > 0) := \sum_k \pi_k^S \left[1 - \Phi\left(-\frac{\widetilde{\Delta}_k^S}{\sqrt{V_{k,11}}} \right) \right], \tag{D28}$$

$$\widetilde{\Pr}(\beta_k^S > 0) := \sum_k \pi_k^S \left[1 - \Phi\left(-\frac{\widetilde{\beta}_k^S}{\sqrt{V_{k,22}}} \right) \right], \tag{D29}$$

where $V_{k,11}$ and $V_{k,22}$ are the diagonal entries of (D22). The caseload-weighted share of screeners worse than a random screening rule, $\Pr(\Delta_{k,\text{rand}}^S > 0)$, is estimated by applying the univariate empirical Bayes procedure of Appendix D.2.2 to the unshrunk estimates of $\Delta_{k,\text{rand}}^S$.

The cross-screener correlation between Δ_k^S and β_k^S summarizes how the within-stage comparison at the screening margin maps to the effect of deploying the algorithm in place of the screener. We estimate the underlying caseload-weighted covariance by the bivariate posterior-average-effect analog of (D16),

$$\widehat{\mathrm{Cov}}_{\pi}\big(\Delta_k^S, \beta_k^S\big) := \sum_k \pi_k^S \big(\widetilde{\Delta}_k^S \, \widetilde{\beta}_k^S + V_{k,12} \big) - \Big(\sum_k \pi_k^S \, \widetilde{\Delta}_k^S \Big) \Big(\sum_k \pi_k^S \, \widetilde{\beta}_k^S \Big), \tag{D30}$$

and the cross-screener correlation by

$$\widehat{\rho} := \frac{\widehat{\mathrm{Cov}}_{\pi}\big(\Delta_k^S, \beta_k^S\big)}{\widehat{\Sigma}_{\Delta}^S \, \widehat{\Sigma}_{\beta}^S}, \tag{D31}$$

using (D26)–(D27) for the denominators. By Proposition D.3, the gap between $\omega_k^S \Delta_k^S$ and β_k^S is the downstream investigator response and the exit-margin term at the screener level, so $\widehat{\rho}$ measures how closely the within-stage comparison tracks the deployment effect across screeners. Standard errors for the reported estimates are obtained by the bootstrap procedure of Appendix D.2.2, perturbing each $\hat{\theta}_k$ by independent mean-zero normal noise with covariance Σ_k in each of 500 draws and re-estimating the prior and every statistic.

D.3.4 Results

Panel A of Appendix Figure A.10 presents our estimates of Δ_k^S, with the algorithmic benchmark from the extrapolation across screeners. Each of the 291 screeners is a point showing the screener's estimated rate of subsequent maltreatment among the calls they screen out against their estimated screen-in rate, neither adjusted for sampling error. The solid curve shows the algorithm's conditional maltreatment rate $\mu(t)$ at each screen-in rate, and the horizontal line shows the rate a random screening rule would leave; a screener's vertical distance above the algorithmic curve at their own screen-in rate is Δ_k^S.

Screen-in rates vary widely across screeners. Conditional on the screen-in rate, screeners' screened-out maltreatment rates differ in the raw estimates. The caseload-weighted posterior standard deviation of Δ_k^S is 1.3 percentage points (Appendix Table A.21). Every one of the 291 screeners leaves more subsequent maltreatment than an equally selective algorithm at their own screen-in rate, by 11.3 percentage points on average ($\Pr(\Delta_k^S > 0) = 100$ percent), and 97.3 percent (s.e. 2.0) leave more than a random screening rule at their own rate would.

Comparing a screener to the algorithm, however, is not the same as deploying the algorithm in place of that screener. Panels B and C of Appendix Figure A.10 contrasts the two objects: Panel B plots the deployment effect β_k^S against each screener's screen-in rate, and Panel C plots β_k^S against the comparison Δ_k^S, with ellipses tracing their estimated joint distribution. Here β_k^S uses selection on observables for the pool the algorithm screens in, as described above. Deploying the algorithm in place of a screener lowers subsequent maltreatment by only 1.5 percentage points on average (s.e. 0.1; Appendix Table A.22), about an eighth of the 11.3-point comparison, and the deployment effects are as tightly compressed as the comparisons, with a posterior standard deviation of 1.3 percentage points. Replacement lowers subsequent maltreatment for 89.8 percent of screeners (s.e. 1.6; Appendix Table A.22). The two are positively but imperfectly correlated, with a caseload-weighted correlation of 0.58 (s.e. 0.02; Panel B).

D.4 Testing Whether Investigators and Screeners Affect Outcomes Beyond Their Decisions

Section 5.2 tests the hypothesis that investigators affect subsequent maltreatment among the calls they leave unopened only through their case-opening decisions. Here we derive the testable implication, describe the implementation, and report the investigator results and the screener analog referred to in Section 4.1.

Consider screened-in calls handled by investigators. Recall that G_i denotes the investigator assigned to call i, $D_i \in \{0,1\}$ the case-opening decision, and $Y_i^{1,g} \in \{0,1\}$ call i's subsequent maltreatment if investigated by investigator g and left unopened. The observed outcome is $Y_i = (1 - D_i)\, Y_i^{1,G_i}$: maltreatment is observed only when no case is opened.

We test two assumptions jointly: the exclusion restriction that investigator identity has no direct effect on maltreatment among screened-in, unopened calls, $Y_i^{1,g} = Y_i^1$ for all g, and the quasi-random assignment of investigators within month-by-ZIP-code-by-shift strata (Section 2.1).

Because assignment is quasi-random only within strata, every investigator-level moment discussed below is strata-adjusted: the investigator coefficient from a regression on investigator and strata indicators, normalized to a common strata distribution, as in the rest of the within-stage analysis. To keep the notation light, we suppress the strata in what follows and simply write the second assumption as $G_i \perp Y_i^1$.

Define $\bar{\mu} := \mathbb{E}[Y_i^1 \mid S_i = 1]$, and for each investigator g the two directly estimable moments and their sum

$$L_g := \mathbb{E}[(1 - D_i)\, Y_i \mid G_i = g], \qquad d_g := \Pr(D_i = 1 \mid G_i = g), \qquad U_g := L_g + d_g.$$

Under exclusion of investigator identity and quasi-random assignment, for any g, $L_g = \bar{\mu} - \mathbb{E}[D_i Y_i^1 \mid G_i = g]$, and since $Y_i^1 \in \{0, 1\}$ implies $0 \leq \mathbb{E}[D_i Y_i^1 \mid G_i = g] \leq d_g$, each investigator delivers the Manski-style bounds (Manski, 1990)

$$L_g \;\leq\; \bar{\mu} \;\leq\; U_g. \tag{D32}$$

The width of investigator g's bracket is g's own case-opening rate d_g. Intersecting across investigators gives the sharp identified set for $\bar{\mu}$, $\max_g L_g \;\leq\; \bar{\mu} \;\leq\; \min_g U_g$, which is nonempty if and only if

$$H_0: \;\; \max_g L_g \;\leq\; \min_g U_g \qquad \Longleftrightarrow \qquad L_g - U_{g'} \leq 0 \;\text{ for all pairs } (g, g'). \tag{D33}$$

This is a condition in the spirit of instrument-validity tests (Kitagawa, 2015; Frandsen, Lefgren and Leslie, 2023). Because quasi-random assignment is supported independently by the balance test in Appendix Figure A.5, we attribute a rejection to a violation of the exclusion restriction. A direct effect of investigation common to all investigators is part of Y_i^1 and does not violate the restriction, so the test detects only heterogeneity across investigators in their influence.

The same implication holds at the screening stage. Under the exclusion restriction on screeners maintained in Section 4.1 and quasi-random assignment within day-by-shift strata, each screener's maltreatment among screened-out calls per assigned call, $L_k := \mathbb{E}[(1 - S_i)\, Y_i \mid K_i = k]$, brackets $\mathbb{E}[Y_i^0]$, the mean of Y_i^0 over all calls, within a width equal to the screener's own screen-in rate $s_k := \Pr(S_i = 1 \mid K_i = k)$: $L_k \leq \mathbb{E}[Y_i^0] \leq L_k + s_k$. The null is again that these brackets intersect. Because the quintiles' screen-in rates lie between about 32 and 42 percent (Appendix Table A.23), every screener bracket is wide and the test has little power. We therefore report it as a placebo for the screener exclusion restriction rather than as a sharp test of it.

D.4.1 Empirical Implementation and Results

Averaging the bounds in equation (D32) over any group of investigators yields a valid group-level bound. For any set $\mathcal{G}_j$ with caseload weights w_g (investigator g's share of the group's

screened-in caseload),

$$L_j \;\le\; \bar{\mu} \;\le\; L_j + d_j, \qquad L_j := \textstyle\sum_{g\in\mathcal{G}_j} w_g L_g, \quad d_j := \textstyle\sum_{g\in\mathcal{G}_j} w_g d_g.$$

We sort investigators into quintiles $j = 1, \ldots, 5$ of the strata-adjusted case-opening rate, and we test $H_0\colon\ \max_j L_j \le \min_j U_j$ on the five pairs of group moments.

Letting J_i denote the quintile of call i's assigned investigator and W_i the indicators for the month-by-ZIP-code-by-shift strata, we estimate on screened-in calls

$$(1 - D_i)Y_i = \textstyle\sum_j \ell_j \, \mathbf{1}\{J_i = j\} + W_i^\top \gamma + \varepsilon_i^L, \tag{D34}$$

$$D_i = \textstyle\sum_j \delta_j \, \mathbf{1}\{J_i = j\} + W_i^\top \zeta + \varepsilon_i^D. \tag{D35}$$

The strata-adjusted moments are $\widehat{L}_j := \widehat{\ell}_j$ and $\widehat{d}_j := \widehat{\delta}_j$, normalized to a common strata distribution, with $\widehat{U}_j := \widehat{L}_j + \widehat{d}_j$. Collect the stacked vector $\widehat{\theta} := (\widehat{L}_1, \widehat{d}_1, \ldots, \widehat{L}_5, \widehat{d}_5)^\top$ and its joint variance-covariance matrix $\widehat{\Sigma}$ (10×10), estimated from the stacked regressions with clustering at the family level. The test statistic is

$$T := \max_{j \neq j'} \frac{\widehat{L}_j - \widehat{U}_{j'}}{\widehat{s}_{jj'}},$$

with $\widehat{s}_{jj'}$ the standard error of the contrast implied by $\widehat{\Sigma}$. We use least-favorable critical values, treating every inequality in equation (D33) as binding: we draw the moment vector from $N(0, \widehat{\Sigma})$, recompute T on each draw, and take the $(1 - \alpha)$ quantile. We also report precision-corrected endpoints for each quintile bracket, applying the half-median-unbiased correction of Chernozhukov, Lee and Rosen (2013), in the implementation of Chernozhukov et al. (2015), to each endpoint.

Figure A.8 in Section 5.2 plots the five investigator brackets $[\widehat{L}_j, \widehat{U}_j]$ against each quintile's strata-adjusted case-opening rate, together with the largest lower bound and the smallest upper bound. Appendix Table A.12 reports the underlying quintile moments with their standard errors and the precision-corrected endpoints. The largest lower bound, 29.7 percent, and the smallest upper bound, 32.1 percent, both come from the lowest quintile, so the brackets intersect. The test statistic is -5.39 against a least-favorable critical value of 2.76, and the p-value is 1.00.

The screener test is implemented in the same way: we group screeners into quintiles of the strata-adjusted screen-in rate, estimate the analogs of equations (D34)–(D35) on all calls, with $(1 - S_i)Y_i$ and S_i as outcomes and day-by-shift strata indicators, and apply the same statistic, critical values, and precision-corrected endpoints to the resulting moments $\widehat{L}_j$, $\widehat{s}_j$, and $\widehat{U}_j := \widehat{L}_j + \widehat{s}_j$. Appendix Figure A.11 plots the five screener brackets against each quintile's strata-adjusted screen-in rate, and Appendix Table A.23 reports the brackets and the test, which does not reject: the largest lower bound is 24.3 percent and the smallest upper bound 55.7 percent, the test statistic is -107.58, and the p-value is 1.00.

E Case Notes: Pipelines and Language Model Prompts

We use large language models to extract two kinds of information from the de-identified case notes, which are available for the calls in our sample from 2018 on: labels for five investigation actions that the structured data do not record (Appendix E.1), and a small set of narrative evidence themes that predict the case-opening decision (Appendix E.2). Section 5 uses both to compare higher- and lower-performing investigators.

E.1 Labeling Investigator Actions

We label each case note for five actions that the narrative records but the structured administrative data do not: substance-use services, safety plans, mental health services, material assistance, and parenting guidance. Each is a concrete step that an investigator can take, or arrange, during an investigation.

A note documents "substance-use services" when it records a substance-use test, assessment, agreement, referral, treatment requirement, refusal, or compliance-monitoring action taken or arranged by CPS, the investigator, or another named provider as part of the investigation or case response. It documents a "safety plan" when it records a CPS or investigator safety plan, safety agreement, or concrete behavioral restriction (e.g., a no-physical-discipline term, a supervision or alternate-caregiver arrangement, a sober-caregiver requirement, a documented violation of such a plan), or the caregiver's refusal of one. It documents "mental health services" when it records an investigator recommendation, offer, referral, specific resource information, coordination, or active reinforcement of access to counseling, psychological evaluation, psychiatric treatment or medication management, crisis mental-health services, or support groups, including work to resolve barriers to access and a recommendation to continue existing treatment that is incorporated into the current plan. It documents "material assistance" when it records the provision of, or arranged access to, material support for housing, food, utilities, transportation, childcare, clothing, school supplies, diapers, or essential infant or household equipment, whether through specific resource information, a recommendation or referral, help with an application, coordination with another provider or a landlord, or direct provision. It documents "parenting guidance" when it records advice, education, a demonstration, or a referral about childrearing (e.g., discipline, supervision, safe sleep, routines, child development, or children's caregiving needs), including a referral to a program described as teaching parenting and concrete child-rearing instructions written into a safety plan, such as no physical discipline, sober caregiving, or safe sleeping arrangements.

For every action, the defining feature is that it is an action taken, arranged, or recommended as part of the current case response. We do not code descriptive facts the investigator merely learns during the investigation (a caregiver's pre-existing treatment or history of substance use, benefits the family already receives, or a parent's own childrearing practices), actions by other organizations without documented investigator involvement, or generic resource guides and unspecified program names. Nor do we infer an action from the allegation, the case-opening or substantiation decision, the risk level, or generic safety-assessment language.

Each action has its own prompt with a common structure: a system message stating the labeling task and the action's definition, explicit coding rules including the exclusions above, contrasting worked examples of notes that do and do not document the action, and

the required output format. The de-identified note is passed as the user message, and OpenAI's `gpt-5-mini`, accessed through the OpenAI API, returns a JSON object with a binary indicator for the action, a short verbatim snippet supporting the label, a free-text rationale, and a confidence rating. We label every de-identified note in our 2018–2022 sample. Appendix Figures A.13, A.14, A.15, A.16, and A.17 reproduce the five prompts in abridged form, omitting the output-format sections, the intensity scale and status field that the first two prompts include but we do not use, and all but two worked examples on each side. The replication materials include the complete prompts, and Appendix Table A.11 gives de-identified excerpts illustrating each action.

E.2 Generating Hypotheses about Evidence Correlated with Case Opening

The case notes also record the evidence an investigator gathers and weighs in deciding whether to open a case. We use a pipeline built on large language models, in the spirit of Batista and Ross (2025) and Modarressi, Spiess and Venugopal (2025), to surface recurring narrative evidence that predicts the case-opening decision beyond an algorithm trained to predict that decision from the structured case characteristics alone. Section 5.3 then asks whether higher- and lower-performing investigators record this evidence at different rates (Figure 9) and act on it differently (Appendix Table A.18). The pipeline measures how much signal the notes contain, surfaces granular concepts that distinguish notes the text predicts will and will not be opened, aggregates them into a small set of evidence themes, and selects the themes most correlated with case opening.

We randomly split the de-identified notes into a training sample of 78,353 notes and a held-out test sample of 67,512 notes. Within the training sample, a discovery subsample of 70,417 notes is used to surface and aggregate the concepts and themes, and a validation subsample of 7,936 notes is used to select the final themes.

To measure the signal, we embed each note with a pretrained text-embedding model (`voyage_4_large_2048dim`) and fit an L2-penalized logistic regression of the case-opening decision on the embeddings, which attains an out-of-sample AUC of 0.995 on the test sample (Appendix Table A.16). Because the notes often state the final decision, this is an upper bound on the evidence they carry about it, and the rest of the pipeline is built to surface evidence while avoiding procedural, decision-restating language.

To surface evidence the administrative data do not capture, we hold fixed a gradient-boosted tree that predicts the case-opening decision from the administrative case characteristics alone, the "case-opening algorithm," and partition the notes into deciles of its predicted probability. Within each decile, we refit the embedding model out of fold, draw 50 notes at random, and split them into the 25 the embedding model scores highest and the 25 it scores lowest for case opening. As in the morphing approach of Ludwig and Mullainathan (2024), the two groups are similar in their structured-characteristic predictions but differ in their text-based predictions. We present each decile's two groups to a large language model (`gpt-5.4`), masking the group identities and withholding the outcome, the algorithm's scores, and the meaning of the split, and ask it to name recurring, note-level concepts that distinguish the groups and could help classify a new note, flagging rather than retaining procedural or disposition language (Appendix Figure A.18). The model returns five concepts per decile,

50 in total, which Appendix Table A.14 lists.

Because the same concept can recur across deciles, we compute pairwise tf–idf cosine similarities and merge concepts that substantively overlap, leaving 37 distinct concepts. We then prompt `gpt-5.4` to aggregate them into narrative evidence themes, each broad enough to subsume several related concepts, concrete enough to label from a single note, and free of procedural language (Appendix Figure A.19), which yields 12 candidate themes.

We label every note in the validation sample for each candidate theme with `gpt-5-mini`, using a prompt that gives each theme's definition, inclusion and exclusion criteria, and worked positive and negative examples, and that returns for each note and theme a presence indicator, a supporting snippet, a confidence rating, and a rationale (Appendix Figure A.20). The three pipeline prompts are reproduced in full, with the de-identified notes passed as the user message. We rank the candidate themes by their within-decile association with case opening, keep the five most correlated, and label every note for them: corroborated harm or injury evidence; absence, ambiguity, or contradiction of abuse evidence; exculpatory corroboration of adequate care; caregiver absence, unstable placement, or abandonment; and a caregiver actively involving the child in dangerous or exploitative conduct. Appendix Table A.15 defines each theme and gives three excerpts illustrating it.

On the test sample, the five themes alone attain an AUC of 0.914 for the case-opening decision, against 0.995 for the full embeddings. An AUC-based analog of the completeness measure of Fudenberg et al. (2022), $(0.914 - 0.5)/(0.995 - 0.5) \approx 0.84$, indicates that the themes capture about 84 percent of the embeddings' predictive signal relative to an uninformative prediction, and their Brier score, 0.074, is well below the 0.111 of the case-opening algorithm alone. Appendix Table A.17 reports how often each theme is recorded, overall and among opened and unopened cases; Appendix Table A.18 reports the relationship between each theme and case opening within each performance group.

F Additional Details on Designing the Optimal Screening Algorithm

This appendix states the design problem of Section 6.1 formally, proves Proposition 6.1, describes how we estimate and conduct inference on the objects reported in Section 6.2, and extends the design problem to investigations with direct effects.

F.1 The Design Problem

Throughout, the prediction $m(X_i)$ takes finitely many values $\tilde{m}$, each with positive probability $\pi(\tilde{m}) := \Pr(m(X_i) = \tilde{m})$. Under the assumptions of Section 6.1, a call's subsequent maltreatment absent a case opening is Y_i^* whether or not it is investigated, and its case-opening decision if investigated is D_i, whichever rule sends it to investigation. A screening rule δ assigns each value of the prediction one of three actions, $\delta(\tilde{m}) \in \{\text{out}, \text{open}, \text{investigate}\}$. Under δ, call i is investigated if the rule sends it to investigation, $S_i^\delta := \mathbf{1}\{\delta(m(X_i)) = \text{investigate}\}$, and a case is opened if the rule opens one directly or an investigator opens one, $C_i^\delta := \mathbf{1}\{\delta(m(X_i)) = \text{open}\} + S_i^\delta D_i$. With the weight on subsequent maltreatment normalized to one, the welfare of the rule is

$$W(\delta) := -\,\mathbb{E}\big[(1 - C_i^\delta)\, Y_i^*\big] - \kappa_C\, \mathbb{E}\big[C_i^\delta\big] - \kappa_S\, \mathbb{E}\big[S_i^\delta\big]. \tag{F36}$$

The conventional algorithm of Section 6.1 is

$$\delta^m(\tilde{m}) := \text{open} \;\; \text{if } \mathbb{E}[Y_i^* \mid m(X_i) = \tilde{m}] > \kappa_C, \qquad \delta^m(\tilde{m}) := \text{out} \;\; \text{otherwise.} \tag{F37}$$

Where $\mathbb{E}[Y_i^* \mid m(X_i) = \tilde{m}] = \kappa_C$, opening and screening out yield the same welfare, so how δ^m breaks the tie does not affect $W(\delta^m)$, and opening a share of the tied calls lets it match the status-quo case-opening rate exactly. If $\mathbb{E}[Y_i^* \mid m(X_i) = \tilde{m}]$ is nondecreasing in $\tilde{m}$, δ^m opens cases on the calls with the highest predictions, so with κ_C matched to the status-quo case-opening rate it coincides with the system-wide deployment C_i^m of Section 3.3. The screening algorithm in Proposition 6.1 is

$$\delta^*(\tilde{m}) := \begin{cases} \text{investigate} & \text{if } \mathrm{VOI}(\tilde{m}) \geq \kappa_S, \\ \text{open} & \text{if } \mathrm{VOI}(\tilde{m}) < \kappa_S \text{ and } \mathbb{E}[Y_i^* \mid m(X_i) = \tilde{m}] > \kappa_C, \\ \text{out} & \text{otherwise,} \end{cases} \tag{F38}$$

which investigates where the value of an investigation in equation (8) is at least κ_S and otherwise takes the action of δ^m.

F.2 Proof of Proposition 6.1

The algorithm in Proposition 6.1 is δ^* in equation (F38). Since exactly one action indicator equals one, the definition of C_i^δ gives $1 - C_i^\delta = \mathbf{1}\{\delta(m(X_i)) = \text{out}\} + \mathbf{1}\{\delta(m(X_i)) = \text{investigate}\}\,(1 - D_i)$. Substituting into equation (F36) and conditioning on the value of the prediction,

$$W(\delta) = -\sum_{\tilde{m}} \pi(\tilde{m})\, \ell\big(\tilde{m}, \delta(\tilde{m})\big), \tag{F39}$$

where the expected losses per call of the three actions are

$$\ell(\tilde{m}, \text{out}) := \mathbb{E}[Y_i^* \mid m(X_i) = \tilde{m}], \qquad \ell(\tilde{m}, \text{open}) := \kappa_C,$$
$$\ell(\tilde{m}, \text{investigate}) := \mathbb{E}[(1 - D_i)\, Y_i^* \mid m(X_i) = \tilde{m}] + \kappa_C \Pr(D_i = 1 \mid m(X_i) = \tilde{m}) + \kappa_S.$$

Since each $\pi(\tilde{m})$ is positive, a rule maximizes welfare among rules taking actions in a given set if and only if it takes an action with the smallest loss in that set at every $\tilde{m}$.

Among rules that never investigate, the smallest loss at $\tilde{m}$ is $\min\{\mathbb{E}[Y_i^* \mid m(X_i) = \tilde{m}],\, \kappa_C\}$, which δ^m attains, so δ^m maximizes welfare among these rules, as claimed in Section 6.1. Among all rules, equation (8) gives

$$\mathrm{VOI}(\tilde{m}) - \kappa_S = \min\{\ell(\tilde{m}, \text{out}),\, \ell(\tilde{m}, \text{open})\} - \ell(\tilde{m}, \text{investigate}),$$

so investigating attains the smallest loss at $\tilde{m}$ if and only if $\mathrm{VOI}(\tilde{m}) \geq \kappa_S$, and otherwise the action of δ^m does. This is the rule δ^*, which therefore maximizes welfare among all rules. Since $W(\delta^m)$ and $W(\delta^*)$ take the minimum loss over screening out and opening and over all three actions, respectively, and $\min\{x, z\} = x - (x - z)^+$,

$$W(\delta^*) - W(\delta^m) = \sum_{\tilde{m}} \pi(\tilde{m}) \big(\mathrm{VOI}(\tilde{m}) - \kappa_S\big)^+ = \mathbb{E}\Big[\big(\mathrm{VOI}(m(X_i)) - \kappa_S\big)^+\Big],$$

which is zero if and only if $\mathrm{VOI}(\tilde{m}) \leq \kappa_S$ at every $\tilde{m}$, since each $\pi(\tilde{m})$ is positive.

Finally, Section 6.1 states that $\mathrm{VOI}(\tilde{m}) > 0$ if and only if investigators open some but not all of the calls with prediction $\tilde{m}$, the calls they open are riskier than κ_C, and the calls they leave unopened are safer. Writing $\mathbb{E}[Y_i^* \mid m(X_i) = \tilde{m}] - \mathbb{E}[(1 - D_i)\, Y_i^* \mid m(X_i) = \tilde{m}] = \mathbb{E}[D_i Y_i^* \mid m(X_i) = \tilde{m}]$ in the first argument of the minimum in equation (8) and applying iterated expectations over D_i when $0 < \Pr(D_i = 1 \mid m(X_i) = \tilde{m}) < 1$,

$$\begin{aligned}\mathrm{VOI}(\tilde{m}) = \min \Big\{ &\Pr(D_i = 1 \mid m(X_i) = \tilde{m}) \big(\mathbb{E}[Y_i^* \mid D_i = 1, m(X_i) = \tilde{m}] - \kappa_C\big), \\ &\Pr(D_i = 0 \mid m(X_i) = \tilde{m}) \big(\kappa_C - \mathbb{E}[Y_i^* \mid D_i = 0, m(X_i) = \tilde{m}]\big) \Big\},\end{aligned}$$

which is positive if and only if both terms are. If investigators open none of these calls, the first argument of the minimum in equation (8) reduces to zero, and if they open all of them, the second does, so $\mathrm{VOI}(\tilde{m}) \leq 0$ in either case.

F.3 Estimation and Inference

We bin the algorithm's prediction into quintiles q of predicted risk. For each quintile and each of the two investigation stages, we estimate the quantities that enter equation (8): the maltreatment risk $\mathbb{E}[Y_i^* \mid q]$, investigators' case-opening rate $\Pr(D_i = 1 \mid q)$, and the maltreatment they leave unopened, $\mathbb{E}[(1 - D_i)\, Y_i^* \mid q]$.

We identify these quantities under three assumptions. First, an investigation has no direct effect on subsequent maltreatment, as Section 5.2 supports, so an investigated call left unopened reveals Y_i^*. Second, selection on observables at the screening stage: conditional on the intake characteristics X_i, the screening decision is independent of the case-opening decision and the outcome,

$$S_i \perp\!\!\!\perp \big(D_i, Y_i^*\big) \mid X_i, \qquad 0 < \Pr(S_i = 1 \mid X_i) < 1.$$

Third, for higher-performing investigators, investigators are quasi-randomly assigned within strata (Section 4.1), so that among screened-in calls, assignment to the group is independent of (D_i, Y_i^*) given X_i, which includes the strata indicators. Writing A_i for the indicator that call i is assigned to a higher-performing investigator, we also require $\rho(X_i) := \Pr(A_i = 1 \mid S_i = 1, X_i) > 0$; when the stage is all investigators, $A_i = 1$ for every screened-in call and $\rho(X_i) = 1$. Under these assumptions, each average is the quintile mean of a regression function fit on the calls that reveal the relevant object:

$$\begin{aligned}\mathbb{E}[Y_i^* \mid q] &= \mathbb{E}\big[\, \mathbb{E}[Y_i \mid S_i = 0, X_i] \,\big|\, q\big], \\ \Pr(D_i = 1 \mid q) &= \mathbb{E}\big[\, \Pr(D_i = 1 \mid S_i = 1, A_i = 1, X_i) \,\big|\, q\big], \\ \mathbb{E}[(1 - D_i)\, Y_i^* \mid q] &= \mathbb{E}\big[\, \mathbb{E}[(1 - D_i)\, Y_i \mid S_i = 1, A_i = 1, X_i] \,\big|\, q\big],\end{aligned}$$

the first from screened-out calls, whose outcome is Y_i^*, and the other two from the screened-in calls the stage's investigators handle, where D_i is observed and an unopened call's outcome is Y_i^*.

We estimate the regression functions and propensities by gradient-boosted trees, as in Appendix C.2, with two differences. We cross-fit them on the evaluation sample in two

folds, so that each call's nuisance values come from the half of the sample that excludes it, because the group-specific regression functions require each call's investigator assignment, which is defined on the evaluation sample. And we estimate the probability that a screened-in call is assigned to a higher-performing investigator, $\rho(X_i)$, as a further nuisance function predicted from the intake characteristics rather than as the group's share of screened-in calls in the call's assignment stratum, because the strata are small. The nuisance functions are the screen-in propensity $\widehat{\pi}_S(X_i)$ for $\Pr(S_i = 1 \mid X_i)$; the outcome regression $\widehat{\mu}_{Y|S=0}(X_i)$ for $\mathbb{E}[Y_i \mid S_i = 0, X_i]$, fit on screened-out calls; and, for each investigation stage, the assignment propensity $\widehat{\rho}(X_i)$, the case-opening propensity $\widehat{\pi}_D(X_i)$ for $\Pr(D_i = 1 \mid S_i = 1, A_i = 1, X_i)$, and the regression $\widehat{\mu}_{(1-D)Y|S=1}(X_i)$ for $\mathbb{E}[(1 - D_i)\, Y_i \mid S_i = 1, A_i = 1, X_i]$, the last two fit on the screened-in calls the stage's investigators handle. We trim $\widehat{\pi}_S(X_i)\,\widehat{\rho}(X_i)$ to the interval $[0.01, 0.99]$. For every call in the evaluation sample, we construct the scores

$$
\begin{aligned}
\psi_Y(i) &:= \widehat{\mu}_{Y|S=0}(X_i) + \frac{1 - S_i}{1 - \widehat{\pi}_S(X_i)}\big\{Y_i - \widehat{\mu}_{Y|S=0}(X_i)\big\},\\
\psi_D(i) &:= \widehat{\pi}_D(X_i) + \frac{S_i A_i}{\widehat{\pi}_S(X_i)\,\widehat{\rho}(X_i)}\big\{D_i - \widehat{\pi}_D(X_i)\big\},\\
\psi_{D0Y}(i) &:= \widehat{\mu}_{(1-D)Y|S=1}(X_i) + \frac{S_i A_i}{\widehat{\pi}_S(X_i)\,\widehat{\rho}(X_i)}\big\{(1 - D_i)\, Y_i - \widehat{\mu}_{(1-D)Y|S=1}(X_i)\big\},
\end{aligned}
$$

and estimate the three averages in quintile q by the sample means of the scores over all calls in the quintile, which we substitute into equation (8) and Proposition 6.1, with each quintile weighted by a fifth, to obtain the value of an investigation, the two screening algorithms, and their welfare comparison. A call reveals a group's case-opening decision only if it is screened in and assigned to the group, which given X_i happens with probability $\Pr(S_i = 1 \mid X_i)\,\rho(X_i)$, and dividing the residual by that probability makes the scores represent every call with those characteristics; the first score uses screened-out calls in the same way. For all investigators, $A_i = 1$ and $\widehat{\rho} = 1$, and the last two scores take the form of those in Appendix C.2, except that ψ_{D0Y} uses a direct regression of $(1 - D_i)\, Y_i$ rather than the product of the case-opening propensity and the outcome regression. The estimators are consistent if either the propensities or the regression functions are consistently estimated (Robins, Rotnitzky and Zhao, 1994).

Appendix Figure A.12 plots, for each quintile and investigation stage, the difference between the quintile's maltreatment risk and the maltreatment rate among the calls investigators would leave unopened, computed from the same three averages. Unlike the quintile figure of Section 5.3, which conditions on the screening decision and describes the calls screeners screen in, this figure describes every call in a quintile, because the screening algorithm decides for all of them. Unconditional on screening, the case-opening decisions of all investigators pooled are informative about maltreatment in the upper quintiles of predicted risk, with a gap indistinguishable from zero in the two lowest quintiles, and those of higher-performing investigators are informative in every quintile; conditional on screening, as Section 5.3 found, the pooled decisions add little within every quintile, and only those of higher-performing investigators do. Only in the highest quintile, where predicted risk is close to the cost of a case opening, is the information in the pooled decisions enough to give an investigation value (Panel A of Figure 10); the decisions of higher-performing investigators

give it value in every quintile.

Every object we report is a function of fifteen quintile averages per investigation stage: for each quintile, the maltreatment risk and the two gains from investigating, relative to screening out and relative to opening a case, whose minimum is the value of an investigation. We construct one joint confidence band for these fifteen quantities and propagate it. We bootstrap the saved scores in 500 draws without refitting the nuisance functions. The critical value is the 95th percentile across draws of the largest studentized deviation of an average from its estimate, and the band is each estimate plus or minus the critical value times its bootstrap standard error. Because the value of an investigation and the welfare gain are nondecreasing in the two gains, and the conventional algorithm's loss is nondecreasing in the maltreatment risk, plugging the ends of the band into each object gives intervals that hold jointly across quintiles and across values of κ_S with probability at least 95 percent; for the welfare improvement relative to the conventional algorithm, we divide the ends of the gain's interval by the opposite ends of the loss's.

F.4 Direct Effects of the Investigation

Section 6 assumes, motivated by Section 5.2, that an investigation affects subsequent maltreatment only through the case-opening decision. This subsection drops that assumption, as in the framework of Section 3.1, so that a screened-out call produces Y_i^0 and an investigated call left unopened produces Y_i^1, while still holding investigators' decisions fixed.

Under a screening algorithm δ, the calls left unopened are the calls screened out, which produce Y_i^0, and the investigated calls that investigators do not open, which produce Y_i^1, so the welfare of the rule is

$$W(\delta) := -\,\mathbb{E}\big[\mathbf{1}\{\delta(m(X_i)) = \text{out}\}\, Y_i^0 + \mathbf{1}\{\delta(m(X_i)) = \text{investigate}\}\,(1-D_i)\,Y_i^1\big] - \kappa_C\,\mathbb{E}\big[C_i^\delta\big] - \kappa_S\,\mathbb{E}\big[S_i^\delta\big]. \tag{F40}$$

When $Y_i^0 = Y_i^1$, the first expectation is $\mathbb{E}[(1-C_i^\delta)\,Y_i^*]$ and equation (F40) is equation (F36).

Screening out the calls with prediction $\tilde{m}$ now yields $-\mathbb{E}[Y_i^0 \mid m(X_i) = \tilde{m}]$ per call, opening a case on them yields $-\kappa_C$, and sending them to investigators yields $-\mathbb{E}[(1-D_i)\,Y_i^1 \mid m(X_i) = \tilde{m}] - \kappa_C \Pr(D_i = 1 \mid m(X_i) = \tilde{m})$ before the cost of the investigation. The value of an investigation is again the difference between the welfare of deferring and that of the better immediate decision,

$$\begin{aligned}\text{VOI}(\tilde{m}) := \min\big\{\mathbb{E}[Y_i^0 \mid m(X_i) = \tilde{m}],\, \kappa_C\big\} &- \mathbb{E}\big[(1-D_i)\,Y_i^1 \mid m(X_i) = \tilde{m}\big] \\ &- \kappa_C \Pr(D_i = 1 \mid m(X_i) = \tilde{m}),\end{aligned} \tag{F41}$$

and the conventional algorithm δ^m opens a case where $\mathbb{E}[Y_i^0 \mid m(X_i) = \tilde{m}] > \kappa_C$ and screens out otherwise. Adding and subtracting $\mathbb{E}[(1-D_i)\,Y_i^0 \mid m(X_i) = \tilde{m}]$ separates the value of an investigation into the value of the case-opening decisions it produces, which is equation (8) with Y_i^0 in place of Y_i^*, and the direct effect of the investigation on the calls investigators leave unopened:

$$\begin{aligned}\text{VOI}(\tilde{m}) = \Big[\min\big\{\mathbb{E}[Y_i^0 \mid m(X_i) = \tilde{m}],\, \kappa_C\big\} - \mathbb{E}\big[(1-D_i)\,Y_i^0 \mid m(X_i) = \tilde{m}\big] - \kappa_C \Pr(D_i = 1 \mid m(X_i) = \tilde{m})\Big] \\ - \mathbb{E}\big[(1-D_i)\,(Y_i^1 - Y_i^0) \mid m(X_i) = \tilde{m}\big].\end{aligned} \tag{F42}$$

The second term is the expected change in subsequent maltreatment that the investigation causes among the calls left unopened. An investigation that reduces maltreatment among the calls it leaves unopened is worth more than its case-opening decisions alone, and one that increases it is worth less.

Proposition F.1. *Fix $\kappa_C > 0$ and $\kappa_S \geq 0$, and let $W(\delta)$, $\mathrm{VOI}(\tilde{m})$, and δ^m be as defined in this subsection. The screening rule δ^* in equation* (F38)*, with Y_i^0 in place of Y_i^* and the value of an investigation in equation* (F41)*, attains the highest welfare among all screening rules, and*

$$W(\delta^*) - W(\delta^m) = \mathbb{E}\Big[\big(\mathrm{VOI}(m(X_i)) - \kappa_S\big)^+\Big] \geq 0,$$

with equality if and only if $\mathrm{VOI}(\tilde{m}) \leq \kappa_S$ at every $\tilde{m}$.

Proof. Conditioning the first expectation in equation (F40) on the value of the prediction, as in the proof of Proposition 6.1, gives $W(\delta) = -\sum_{\tilde{m}} \pi(\tilde{m})\, \ell(\tilde{m}, \delta(\tilde{m}))$ with the per-action losses

$$\begin{aligned} \ell(\tilde{m}, \text{out}) &:= \mathbb{E}[Y_i^0 \mid m(X_i) = \tilde{m}], \qquad \ell(\tilde{m}, \text{open}) := \kappa_C, \\ \ell(\tilde{m}, \text{investigate}) &:= \mathbb{E}[(1 - D_i)\, Y_i^1 \mid m(X_i) = \tilde{m}] + \kappa_C \Pr(D_i = 1 \mid m(X_i) = \tilde{m}) + \kappa_S, \end{aligned}$$

and equation (F41) gives $\mathrm{VOI}(\tilde{m}) - \kappa_S = \min\{\ell(\tilde{m}, \text{out}),\, \ell(\tilde{m}, \text{open})\} - \ell(\tilde{m}, \text{investigate})$. From equation (F39) onward, the proof of Proposition 6.1 uses only these two facts, so it applies verbatim with the losses above. □

References

Angelova, Victoria, Will S. Dobbie, and Crystal Yang. 2025. "Algorithmic Recommendations and Human Discretion." *Review of Economic Studies*.

Baron, E. Jason, Joseph J. Doyle, Natalia Emanuel, Peter Hull, and Joseph Ryan. 2024. "Discrimination in Multiphase Systems: Evidence from Child Protection." *Quarterly Journal of Economics*, 139(3): 1611–1664.

Batista, Rafael M., and James Ross. 2025. "Words that Work: Using Large Language Models to Generate and Refine Hypotheses from Text." *SSRN Working Paper No. 4926398*.

Blinder, Alan S. 1973. "Wage Discrimination: Reduced Form and Structural Estimates." *Journal of Human Resources*, 8(4): 436–455.

Bonhomme, Stéphane, and Martin Weidner. 2022. "Minimizing Sensitivity to Model Misspecification." *Quantitative Economics*, 13(3): 907–954.

Chernozhukov, Victor, Sokbae Lee, and Adam M. Rosen. 2013. "Intersection Bounds: Estimation and Inference." *Econometrica*, 81(2): 667–737.

Chernozhukov, Victor, Wooyoung Kim, Sokbae Lee, and Adam M. Rosen. 2015. "Implementing Intersection Bounds in Stata." *Stata Journal*, 15(1): 21–44.

Frandsen, Brigham, Lars Lefgren, and Emily Leslie. 2023. "Judging Judge Fixed Effects." *American Economic Review*, 113(1): 253–277.

Fudenberg, Drew, Jon Kleinberg, Annie Liang, and Sendhil Mullainathan. 2022. "Measuring the Completeness of Economic Models." *Journal of Political Economy*, 130(4): 956–990.

Kitagawa, Evelyn M. 1955. "Components of a Difference Between Two Rates." *Journal of the American Statistical Association*, 50(272): 1168–1194.

Kitagawa, Toru. 2015. "A Test for Instrument Validity." *Econometrica*, 83(5): 2043–2063.

Lee, David S. 2009. "Training, Wages, and Sample Selection: Estimating Sharp Bounds on Treatment Effects." *Review of Economic Studies*, 76(3): 1071–1102.

Ludwig, Jens, and Sendhil Mullainathan. 2024. "Machine Learning as a Tool for Hypothesis Generation." *Quarterly Journal of Economics*, 139(2): 751–827.

Manski, Charles F. 1990. "Nonparametric Bounds on Treatment Effects." *American Economic Review: Papers and Proceedings*, 80(2): 319–323.

Manski, Charles F., and John V. Pepper. 2000. "Monotone Instrumental Variables: With an Application to the Returns to Schooling." *Econometrica*, 68(4): 997–1010.

Modarressi, Iman, Jann Spiess, and Amar Venugopal. 2025. "Causal Inference on Outcomes Learned from Text." *arXiv preprint arXiv:2503.00725*.

Oaxaca, Ronald. 1973. "Male-Female Wage Differentials in Urban Labor Markets." *International Economic Review*, 14(3): 693–709.

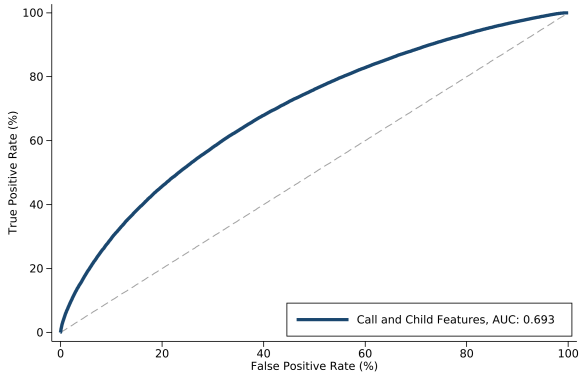
Robins, James M., Andrea Rotnitzky, and Lue Ping Zhao. 1994. "Estimation of Regression Coefficients When Some Regressors Are Not Always Observed." *Journal of the American Statistical Association*, 89(427): 846–866.

Shapley, L. S. 1953. "A Value for n-Person Games." In *Contributions to the Theory of Games, Volume II. Annals of Mathematics Studies*, , ed. H. W. Kuhn and A. W. Tucker, 307–317. Princeton, NJ:Princeton University Press.

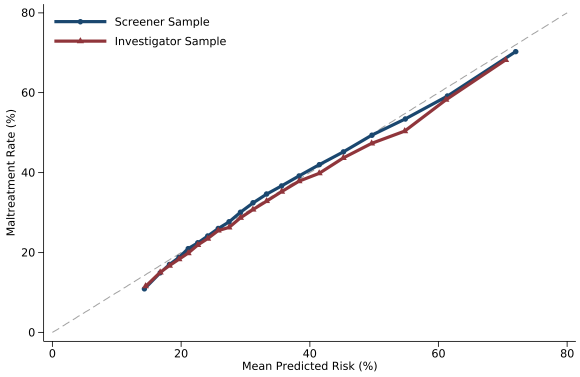
Skiena, Steven, and Charles B Ward. 2014. *Who's Bigger?: Where Historical Figures Really Rank.* Cambridge University Press.

G Appendix Figures and Tables

Figure A.1: Algorithm Performance in the Evaluation Sample of Unopened Calls



Panel A: ROC curves.

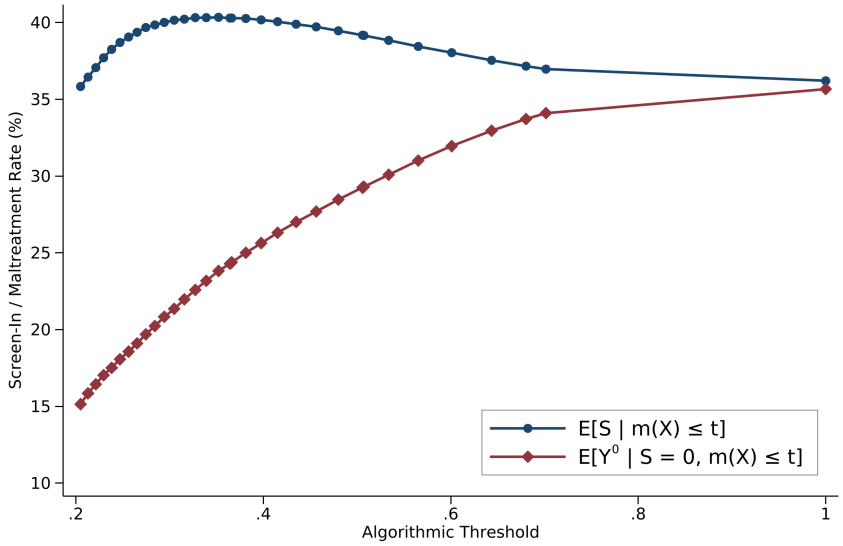


Panel B: Calibration.

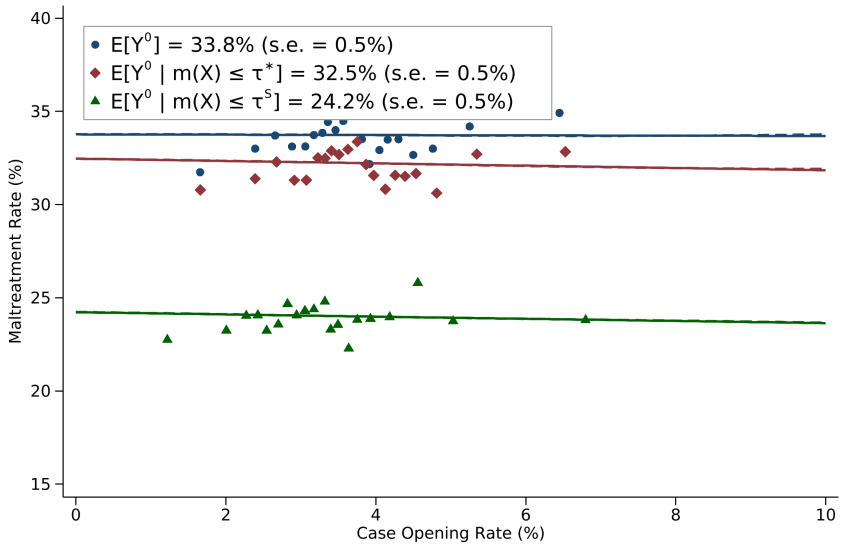
Notes. This figure plots the held-out performance of the algorithm $m(X_i)$ (Appendix B.2) in predicting subsequent maltreatment among unopened calls in the evaluation sample. Panel A plots the receiver operating characteristic curve of $m(X_i)$, with its area under the curve in the legend. Panel B plots the realized maltreatment rate among unopened calls against the mean predicted risk within bins of predicted risk, separately for the screener sample (all calls) and the investigator sample (screened-in calls), with the 45-degree line for reference. See Section 4.1 for discussion.

Figure A.2: Identifying Algorithmic Counterfactuals by Extrapolating Across Screeners

A. Observed Screen-In Rates and Maltreatment among Screened-Out Calls



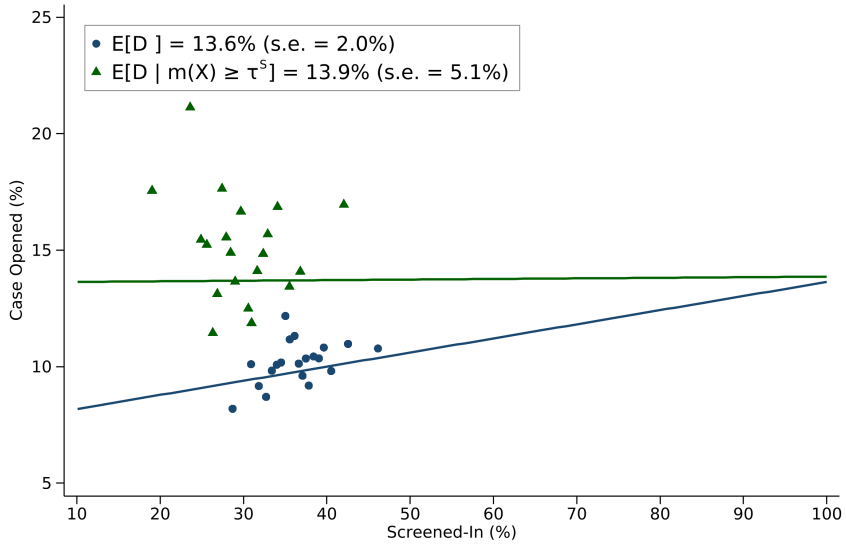
B. Extrapolations for Maltreatment Rates among Calls Unopened by the Algorithm



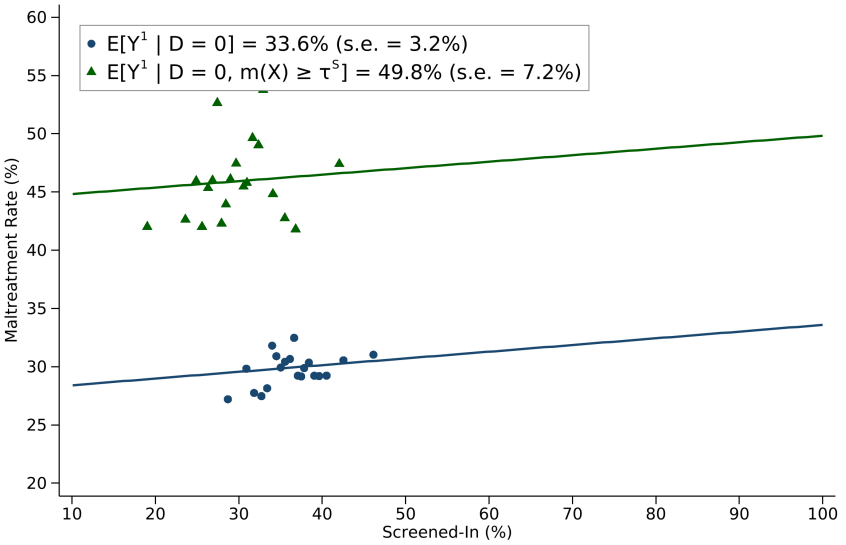
Notes. This figure shows how we identify the algorithm's counterfactual maltreatment rates by extrapolating across quasi-randomly assigned screeners. Panel A plots, against the threshold t, the observed screen-in rate $\sigma(t)$ and the maltreatment rate among screened-out calls $\mu_0(t)$ on the risk set $\{m(X_i) \leq t\}$. Panel B plots binned screener-level maltreatment rates among unopened calls against screener-level case-opening rates on the risk sets $\{m(X_i) \leq t\}$ for $t = 1, \tau^*, \tau^S$, with the linear fit as a solid line and the local-linear fit as a dashed line. The intercept at a case-opening rate of zero identifies $\mu(t)$, and the legend reports the intercept of the linear fit with its standard error in parentheses. All standard errors are obtained via a bootstrapping procedure (with 500 replications) and appear in parentheses. See Section 4.1 for discussion.

Figure A.3: Extrapolations Across Screeners for the Screening-Stage Deployment Effect

A. Case-Opening Rates

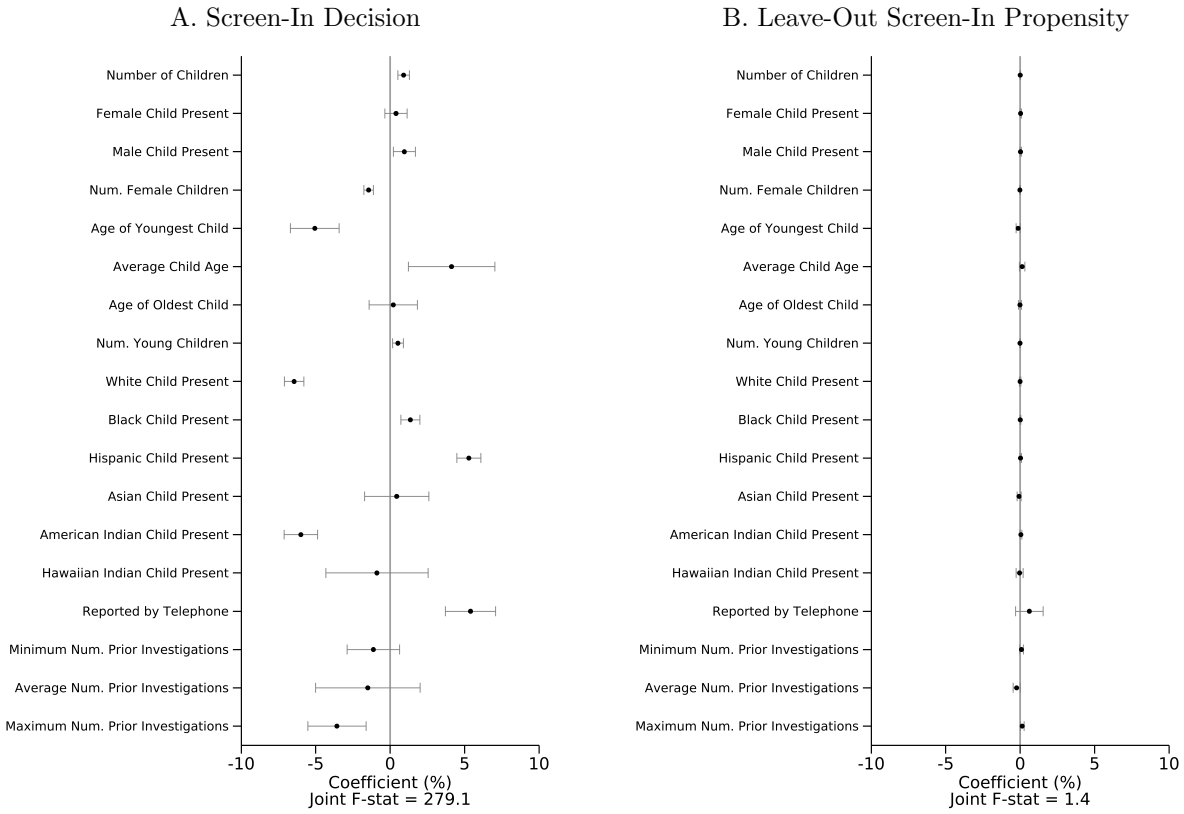


B. Maltreatment Rates among Unopened Calls



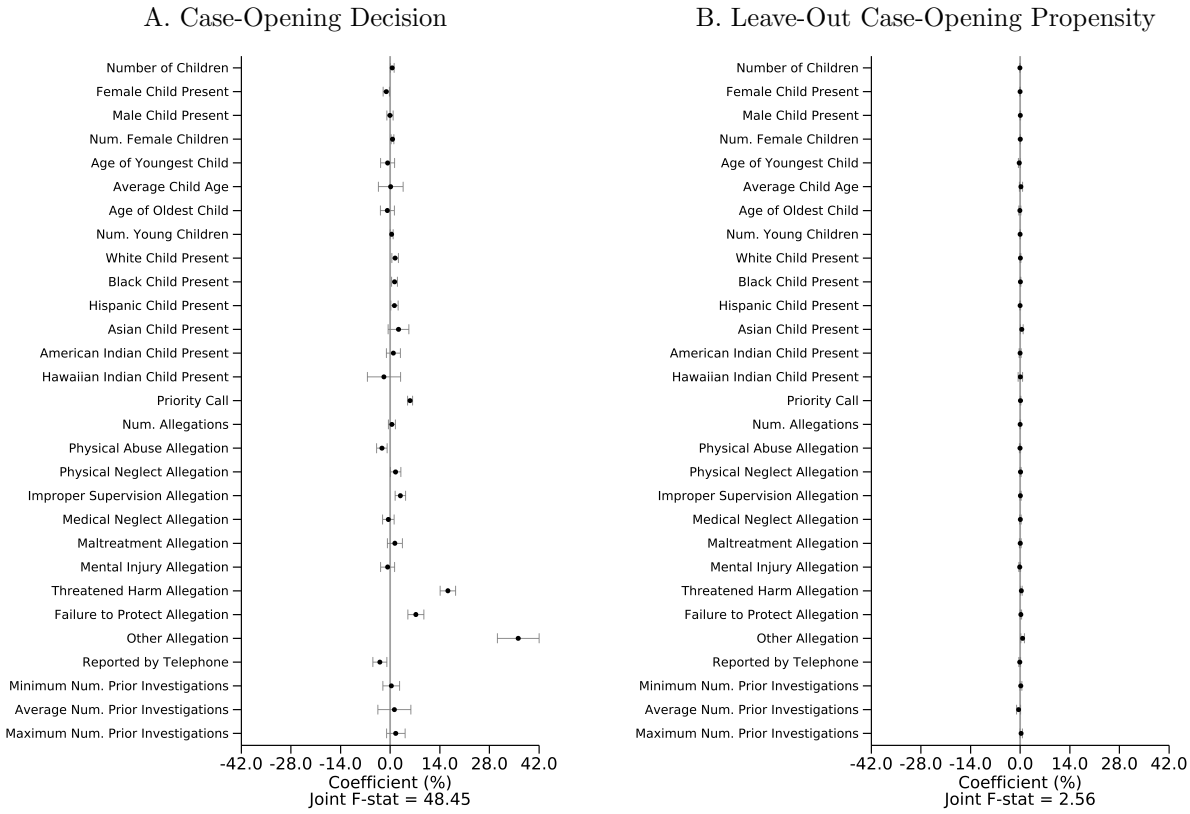
Notes. This figure plots the extrapolations across screeners that recover the two moments the screening-stage deployment effect requires. Each panel plots strata-adjusted screener-level rates, grouped into 20 equal-sized bins, against screener-level screen-in rates, for all calls and for the algorithm's screened-in set $\{m(X_i) \geq \tau^S\}$, with linear fits extrapolated to a screen-in rate of one. Panel A plots case-opening rates among screened-in calls, and its intercepts identify $\Pr(D_i = 1)$ and $\Pr(D_i = 1 \mid m(X_i) \geq \tau^S)$. Panel B plots maltreatment rates among screened-in but unopened calls, and its intercepts identify $\mathbb{E}[Y_i^1 \mid D_i = 0]$ and $\mathbb{E}[Y_i^1 \mid m(X_i) \geq \tau^S, D_i = 0]$. The legends report the intercepts with their standard errors. All standard errors are obtained via a bootstrapping procedure (with 500 replications) and appear in parentheses. See Section 4.1 for discussion.

Figure A.4: Balance of Screener Assignment on Call Characteristics



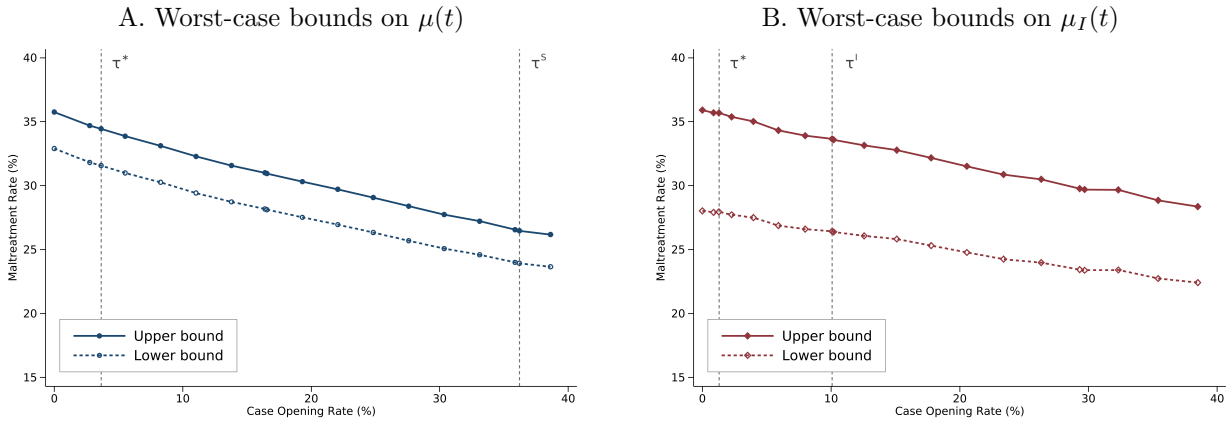
Notes. This figure plots coefficients and 95 percent confidence intervals from regressions of the screen-in decision (Panel A) and the assigned screener's leave-out screen-in propensity (Panel B) on call characteristics, with day-by-shift fixed effects, among calls in the evaluation sample. Continuous characteristics are standardized to unit standard deviation, and allegation type is omitted because the assigned screener records it and so may itself depend on the assignment. The bottom of each panel reports the joint F-statistic for all characteristics. See Section 4.1 for discussion.

Figure A.5: Balance of Investigator Assignment on Call Characteristics



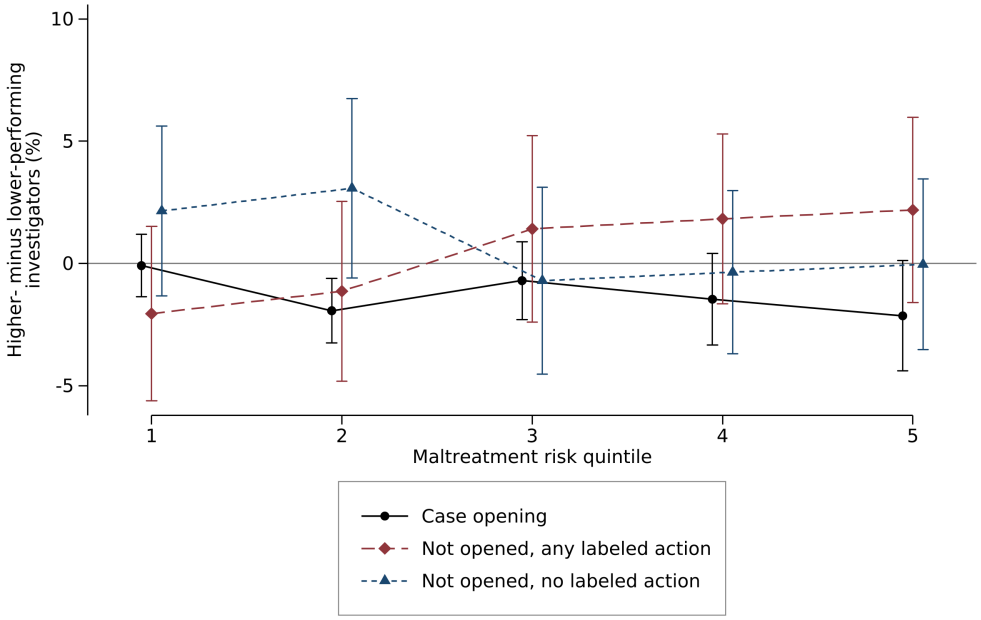
Notes. This figure plots coefficients and 95 percent confidence intervals from regressions of the case-opening decision (Panel A) and the assigned investigator's leave-out case-opening propensity (Panel B) on call characteristics, with month-by-ZIP-code-by-shift fixed effects, among screened-in calls in the evaluation sample. Continuous characteristics are standardized to unit standard deviation. The bottom of each panel reports the joint F-statistic for all characteristics. See Appendix B.3 for discussion.

Figure A.6: Counterfactual Maltreatment Curves under Worst-Case Bounds



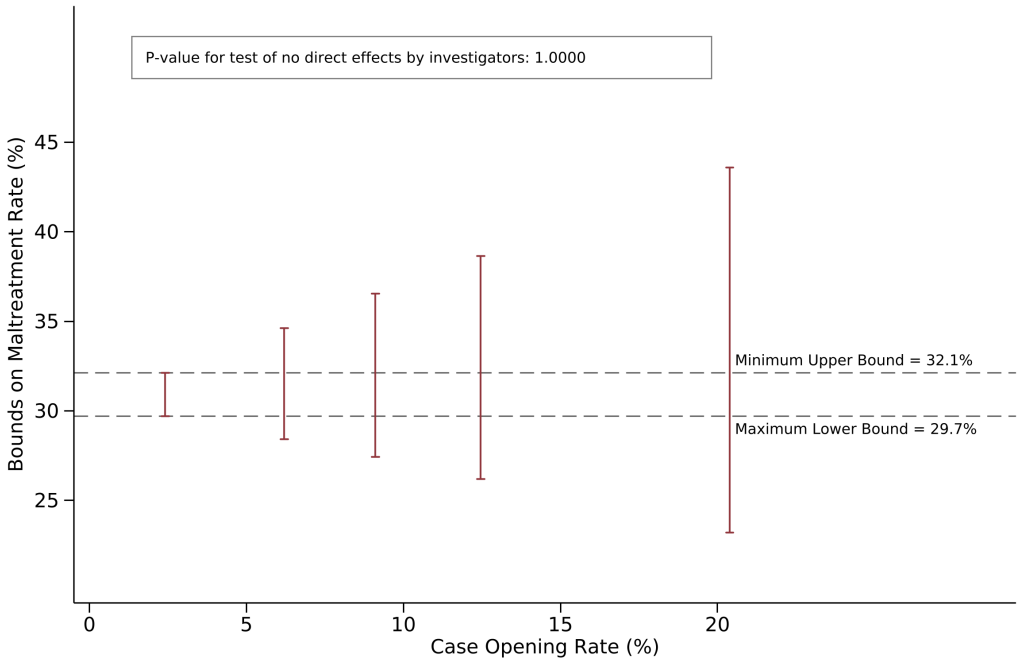
Notes. This figure plots worst-case bounds on the counterfactual curves $\mu(t)$ (Panel A) and $\mu_I(t)$ (Panel B) of Figure 2, constructed on the caseloads of the least screen-in-intensive screeners as described in Appendix C.1. The lower and upper bounds set the outcome of every call whose outcome a case opening censors to zero and to one. Thresholds are expressed as the case-opening rate the algorithmic rule would induce, and the vertical lines mark the thresholds used in the main text. See Section 4.4 for discussion.

Figure A.7: Investigation Actions by Investigator Performance and Predicted Risk



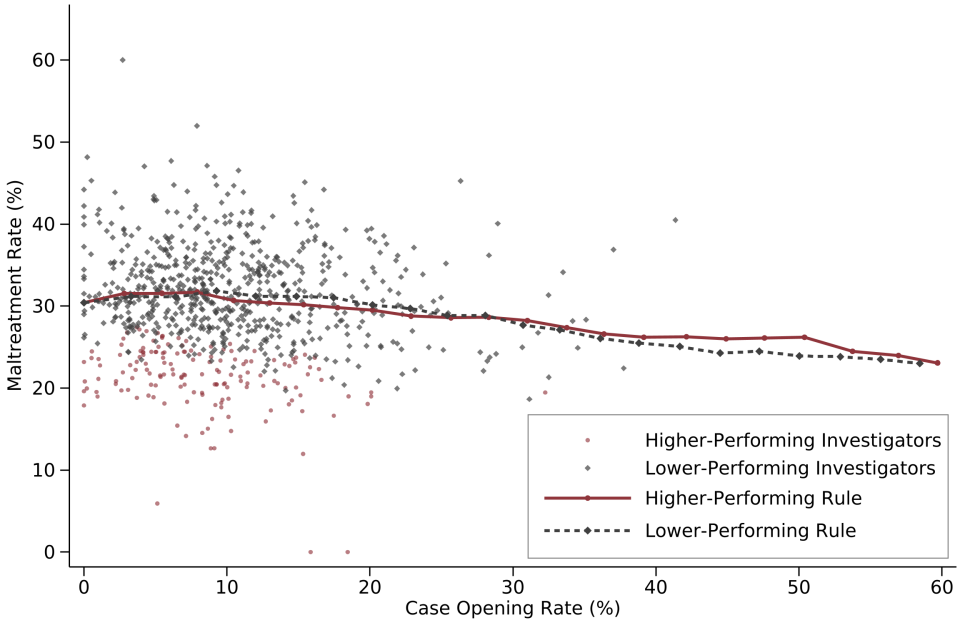
Notes. This figure plots differences between higher- and lower-performing investigators in the probability of three mutually exclusive outcomes within quintiles of the algorithm's predicted risk: opening a case, leaving a case unopened with at least one labeled action, and leaving a case unopened without a labeled action. Higher-performing investigators have negative posterior means of Δ_g^I, and lower-performing investigators have positive posterior means. For each outcome, estimates come from a regression of its indicator on investigator performance interacted with risk-quintile indicators, with month-by-ZIP-code-by-shift fixed effects. The five labeled actions are substance-use services, mental health services, safety plans, material assistance, and parenting guidance, extracted from investigators' narrative case notes as described in Appendix E.1. Differences are in percentage points, and a negative estimate means the outcome is less likely under higher-performing investigators. Whiskers are 95 percent confidence intervals, with standard errors clustered by investigator. See Section 5.2 for discussion.

Figure A.8: Testing Whether Investigators Affect Outcomes Beyond Their Case-Opening Decisions



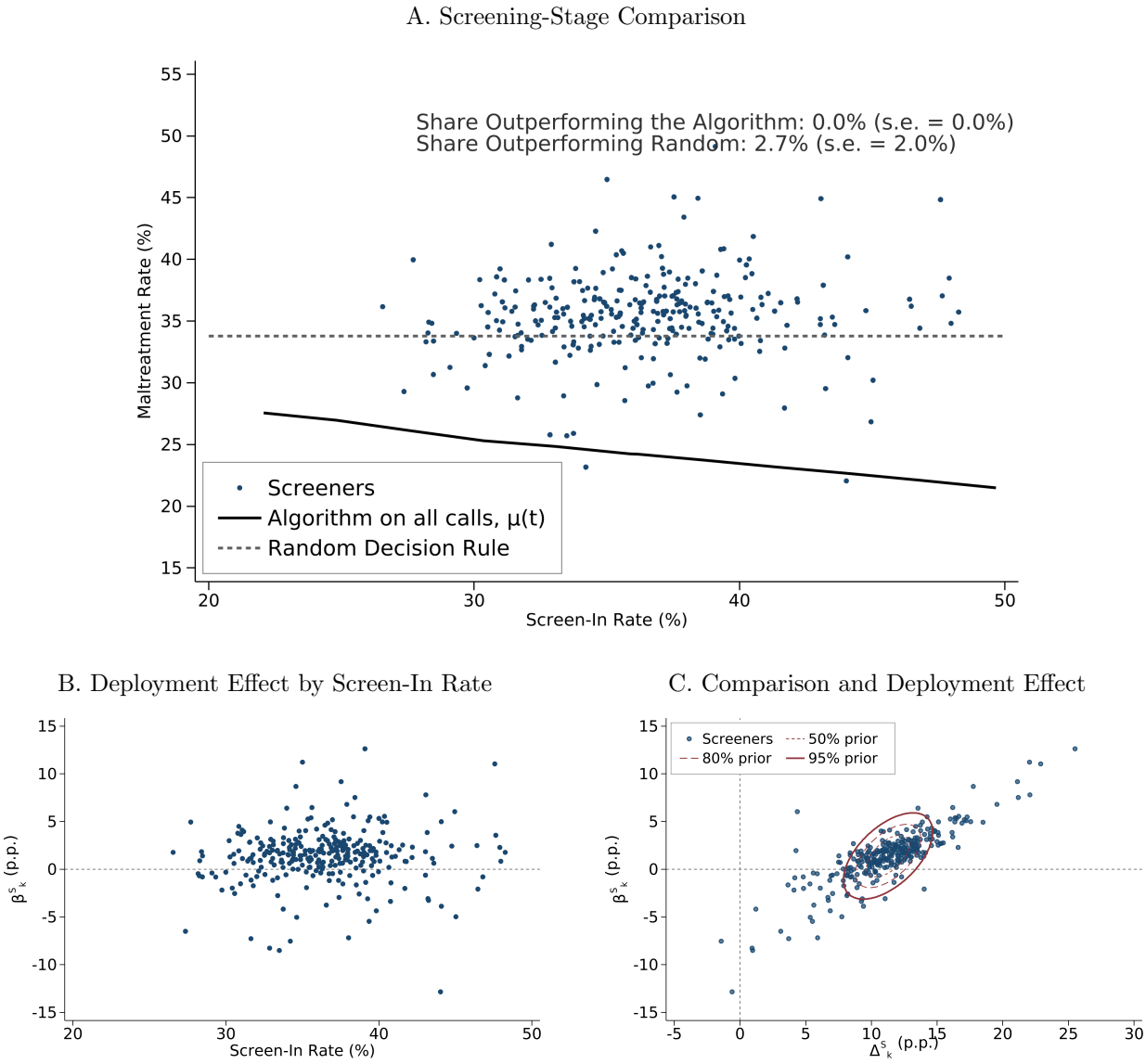
Notes. This figure plots, for each quintile of investigators' strata-adjusted case-opening rate, the bounds on $\bar{\mu}$ that hold if investigators have no direct effects on subsequent maltreatment. Each bracket, plotted at the quintile's case-opening rate, runs from the quintile's strata-adjusted share of calls left unopened and followed by maltreatment to that share plus its case-opening rate. The dashed lines mark the largest lower bound and the smallest upper bound across quintiles, and the box reports the p-value of the test of no direct effects described in Appendix D.4. See Section 5.2 for discussion.

Figure A.9: Subsequent Maltreatment under Case-Opening Rules Distilled from Higher- and Lower-Performing Investigators



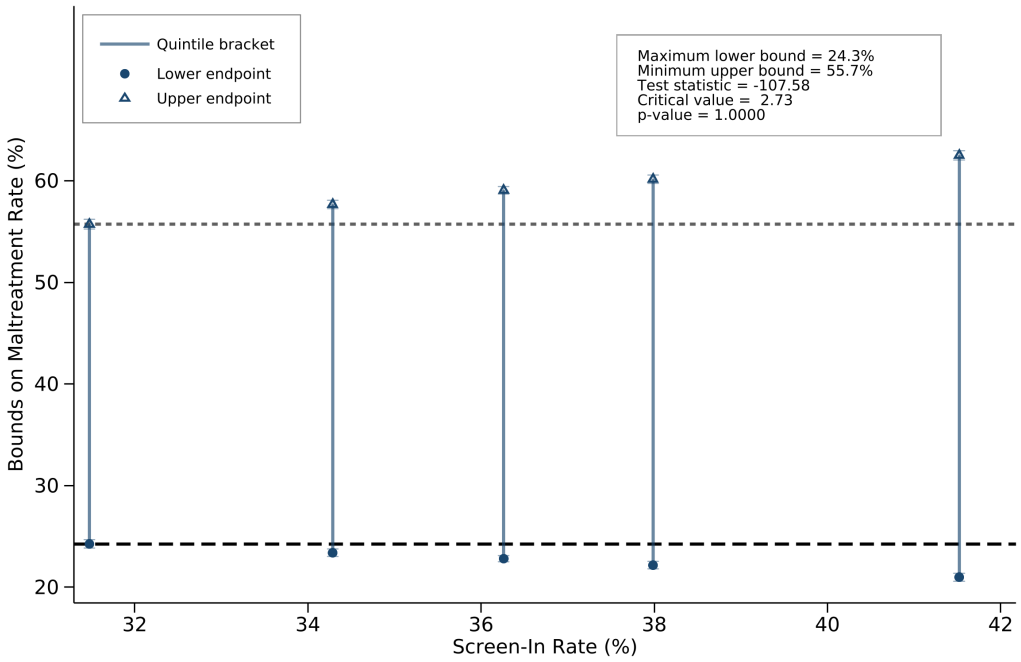
Notes. This figure plots the subsequent maltreatment that case-opening rules distilled from higher- and lower-performing investigators would leave among unopened calls. Separately for each group, we train a gradient-boosted tree to predict the case-opening decision from the structured case characteristics, yielding the rules $m_D^{\text{High}}(X_i)$ and $m_D^{\text{Low}}(X_i)$. The solid and dashed curves trace the maltreatment rate each rule would leave among unopened calls at each case-opening rate, constructed as for the algorithm's curve, and the points are the investigator-level estimates of Figure 6 for each group. All rates are in percentage points. See Section 5.3 for discussion.

Figure A.10: Screener Performance and Algorithmic Deployment Effects



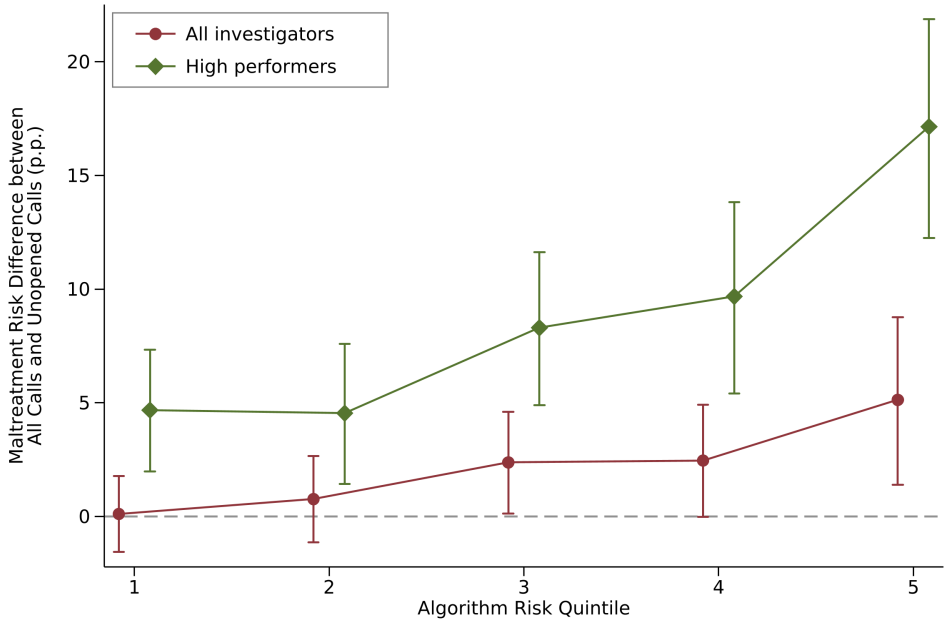
Notes. This figure plots unshrunk screener-level estimates, one marker per screener. Panel A plots each screener's strata-adjusted maltreatment rate among screened-out calls against their strata-adjusted screen-in rate, together with the algorithm's curve $\mu(t)$ from the extrapolations of Section 4.1 and the rate a random screening rule would leave, $\mathbb{E}[Y_i^0]$. A screener's vertical distance above the curve at their own screen-in rate is Δ_k^S. The box reports the caseload-weighted shares of screeners who outperform the algorithm and who leave less subsequent maltreatment than a random screening rule, estimated from posterior average effects (Appendix D.3). Standard errors are obtained via a bootstrapping procedure (with 500 replications) and appear in parentheses. Panel B plots the effect of deploying the algorithm in place of screener k, β_k^S, against the screener's screen-in rate, with a dashed line at zero. Panel C plots β_k^S against the screener–algorithm comparison Δ_k^S, with the 50, 80, and 95 percent contours of the estimated joint prior from the empirical Bayes model of Appendix D.3. Δ_k^S is estimated from the screener extrapolation of Section 4.1, and β_k^S combines that extrapolation with selection on observables for the pool the algorithm screens in. Rates are in percent; comparisons and deployment effects are in percentage points. See Appendix D.3 for discussion.

Figure A.11: Testing Whether Screeners Affect Outcomes Beyond Their Screening Decisions



Notes. This figure plots, for each quintile of screeners' strata-adjusted screen-in rate, the bounds on $\mathbb{E}[Y_i^0]$ that hold if screeners have no direct effects on subsequent maltreatment, the placebo version of the investigator test in Figure A.8. Each bracket, plotted at the quintile's screen-in rate, runs from the quintile's strata-adjusted share of calls screened out and followed by maltreatment to that share plus its screen-in rate. The dashed line marks the largest lower bound and the dotted line the smallest upper bound across quintiles, and the box reports the test statistic, its 5 percent least-favorable critical value, and the p-value of the test described in Appendix D.4. See Appendix D.4 for discussion.

Figure A.12: Outcome Gaps on All Calls within Quintiles of the Algorithm's Predicted Risk



Notes. This figure plots the outcome gap of Section 5.3 over all calls in the evaluation sample rather than over screened-in calls. Within each quintile q of the algorithm's predicted risk, the outcome gap is the quintile's maltreatment risk, $\mathbb{E}[Y_i^* \mid q]$, minus the maltreatment rate among the calls investigators would leave unopened, $\mathbb{E}[(1 - D_i)\, Y_i^* \mid q] / \Pr(D_i = 0 \mid q)$, in percentage points, for all investigators and for higher-performing investigators. The averages are estimated under selection on observables at the screening stage, as described in Appendix F.3. Whiskers are 95 percent confidence intervals from 500 call-level bootstrap draws within quintiles. See Appendix F.3 for discussion.

Table A.1: Summary Statistics

	Screener sample			Investigator sample		
	All Calls	Screen In	Screen Out	All Calls	Case Opened	Case Not Opened
Panel A. Child characteristics	(1)	(2)	(3)	(4)	(5)	(6)
White child present	0.761	0.725	0.784	0.725	0.751	0.722
Black child present	0.347	0.380	0.326	0.380	0.384	0.379
Female child present	0.615	0.615	0.615	0.615	0.635	0.613
Age of youngest child	8.648	8.277	8.857	8.277	7.490	8.370
Age of oldest child	10.362	10.269	10.414	10.269	9.968	10.305
Prior investigation	0.778	0.781	0.777	0.781	0.889	0.768
Panel B. Call characteristics						
Number of children	1.600	1.687	1.550	1.687	1.858	1.667
Physical abuse allegation	0.261	0.355	0.208	0.355	0.324	0.359
Physical neglect allegation	0.213	0.272	0.180	0.272	0.305	0.269
Improper supervision allegation	0.721	0.681	0.743	0.681	0.775	0.670
Panel C. Call outcomes						
Screen-in rate	0.359	1.000	0.000	1.000	1.000	1.000
Substantiation rate	0.082	0.228	0.000	0.228	1.000	0.137
Case opening rate	0.038	0.106	0.000	0.106	1.000	0.000
Child removal rate	0.008	0.023	0.000	0.023	0.218	0.000
Panel D. Call outcomes among unopened cases						
New call between 31 and 180 days	0.339	0.305	0.356	0.305	–	0.305
Number of calls	514,392	184,884	329,508	184,884	19,579	165,305
Number of children	822,832	311,934	510,898	311,934	36,384	275,550

Notes. This table reports sample means for Michigan hotline calls received between 2017 and 2022, before the division into training and evaluation samples. Columns (1)–(3) describe the screener sample, which were assigned to 291 screeners. Columns (4)–(6) describe the investigator sample of screened-in calls, which were assigned to 4,579 investigators; Columns (2) and (4) therefore coincide. Substantiation corresponds to disposition Categories I–III and a case opening to Categories I–II (Section 2.1). Panel D reports a new hotline call between 31 and 180 days after the initial call among calls without an opened case; this outcome is omitted for opened cases in column (5). Allegation rows can sum to more than one because the categories are not mutually exclusive. See Section 2 for additional details on sample construction and variable definitions.

Table A.2: Agreement between Human and Algorithmic Decisions at the Screening and Investigation Stages

Panel A: Screening stage (all calls)		
Screener decision	Algorithm decision	
	Screen in ($S_i^m = 1$)	Screen out ($S_i^m = 0$)
Screen in ($S_i = 1$)	10.8	25.5
	(0.2)	(0.3)
Screen out ($S_i = 0$)	25.5	38.3
	(0.1)	(0.4)
Panel B: Investigation stage (screened-in calls)		
Investigator decision	Algorithm decision	
	Open ($D_i^m = 1$)	Leave unopened ($D_i^m = 0$)
Open ($D_i = 1$)	1.7	8.3
	(0.1)	(0.2)
Leave unopened ($D_i = 0$)	8.3	81.6
	(0.1)	(0.2)

Notes. This table cross-tabulates human decisions against the decisions of an equally selective algorithm in the evaluation sample. Panel A compares screeners' decisions with $S_i^m = \mathbf{1}\{m(X_i) \geq \tau^S\}$ on all calls. Panel B compares investigators' decisions with $D_i^m = \mathbf{1}\{m(X_i) \geq \tau^I\}$ on screened-in calls. Entries are percentages of all evaluation calls in Panel A and of screened-in evaluation calls in Panel B. Standard errors are reported in parentheses and are clustered by screener in Panel A and investigator in Panel B. See Section 4.2 for discussion.

Table A.3: Effects of Algorithmic Counterfactuals in Michigan CPS on Investigations, Case Openings and Subsequent Maltreatment

	Reduction (percentage points)		
Counterfactual	Investigations	Case openings	Maltreatment
Replacing the investigation stage (V^I)	+36.2	0.0	+0.6
	(–)	(–)	(0.4)
	[–]	[–]	[-0.2, 1.4]
Replacing the screening stage (V^S)	0.0	−1.4	+1.1
	(–)	(2.2)	(2.3)
	[–]	[-5.6, 2.9]	[-3.3, 5.6]
Replacing the multistage system (V^{System})	+36.2	0.0	+1.3
	(–)	(–)	(0.5)
	[–]	[–]	[0.4, 2.2]

Notes. This table reports the effect of each algorithmic counterfactual in Section 3.3 on the investigation rate, case-opening rate, and subsequent maltreatment among unopened calls. Entries report reductions relative to the status quo, so positive values indicate decreases. The maltreatment column reports $V^0 - V^I$, $V^0 - V^S$, and $V^0 - V^{System}$. Investigation-stage and system-wide deployment eliminate investigations while holding the case-opening rate fixed. Screening-stage deployment holds the screen-in rate fixed and retains human investigation decisions. Entries marked (–) and [–] are fixed by construction conditional on the matched status-quo rates; standard errors and confidence intervals are not reported for these entries. All estimates are reported in percentage points. Standard errors are in parentheses and 95% confidence intervals are in brackets. Standard errors are computed using 500 bootstrap replications, as described in Appendix B.3. See Section 4.3 for discussion.

Table A.4: Decomposing the Deployment Effects

Panel A: Deploying the algorithm in place of the multistage system		
$\omega^S \Delta^S$	7.6 (0.3) [7.0, 8.2]	Weighted screening-stage comparison
$\omega^I \Delta^I$	0.6 (0.4) [-0.2, 1.4]	Weighted investigation-stage comparison
$\omega^I \mathcal{C}^{\text{Sys}}$	−6.9 (0.6) [-8.1, -5.6]	System-wide aggregation correction
$V^0 - V^{System}$	1.3 (0.5) [0.4, 2.2]	Sum: system-wide deployment effect
Panel B: Deploying the algorithm at the screening stage		
$\omega^S \Delta^S$	7.6 (0.3) [7.0, 8.2]	Weighted screening-stage comparison
$\omega^I \mathcal{R}^I$	−6.7 (2.3) [-11.2, -2.2]	Downstream investigator response
$(\omega^S - \lambda^S)\Gamma^S$	+0.2 (0.4) [-0.5, 1.0]	Exit-margin reweighting
$V^0 - V^S$	1.1 (2.3) [-3.3, 5.6]	Sum: screening-stage deployment effect

Notes. This table reports the decompositions of the deployment effects in Table 1, also plotted in Figure 4. Panel A decomposes the system-wide deployment effect into the weighted stage-wise comparisons and the system-wide aggregation correction, following Proposition 3.2. Panel B decomposes the screening-stage deployment effect into the weighted screening-stage comparison, downstream investigator response, and exit-margin reweighting term, following Proposition 3.1. The investigation-stage deployment effect equals the weighted investigation-stage comparison and is not decomposed further. Counterfactual maltreatment rates under the algorithm are estimated from the screener extrapolations described in Section 4.1, controlling for day-by-shift fixed effects. All estimates are in percentage points. Standard errors are reported in parentheses and 95% confidence intervals are in brackets. Standard errors are computed using 500 bootstrap replications. See Section 4.3 for discussion.

Table A.5: Robustness to Alternative Maltreatment Extrapolations

	Extrapolation design							
	Main spec. (1)	Local linear (2)	Quadratic extrap. (3)	Unweighted linear (4)	No strata fixed effects (5)	Worst-case lower bound (6)	Worst-case upper bound (7)	Screening-only extrap. (8)
Panel A: Stage-wise comparisons and their aggregation								
Δ^S	11.5	11.4	11.6	12.5	10.4	11.7	9.2	3.9
	(0.5)	(0.5)	(0.8)	(0.8)	(0.4)	(0.3)	(0.3)	(2.2)
	[10.6, 12.3]	[10.5, 12.4]	[10.0, 13.2]	[11.0, 14.0]	[9.7, 11.1]	[11.2, 12.3]	[8.6, 9.8]	[-0.3, 8.2]
Δ^I	1.8	1.8	2.3	4.3	−0.7	3.6	−3.6	−3.1
	(1.2)	(1.3)	(2.6)	(2.2)	(1.0)	(0.7)	(0.7)	(3.3)
	[-0.6, 4.1]	[-0.8, 4.3]	[-2.8, 7.4]	[0.1, 8.6]	[-2.7, 1.2]	[2.2, 5.1]	[-5.1, -2.1]	[-9.5, 3.3]
$\omega^S\Delta^S$	7.6	7.6	7.7	8.3	6.9	7.8	6.1	2.6
	(0.3)	(0.3)	(0.5)	(0.5)	(0.2)	(0.2)	(0.2)	(1.4)
	[7.0, 8.2]	[6.9, 8.2]	[6.6, 8.7]	[7.3, 9.2]	[6.4, 7.4]	[7.4, 8.1]	[5.7, 6.5]	[-0.2, 5.4]
$\omega^I\Delta^I$	0.6	0.6	0.8	1.5	−0.2	1.2	−1.2	−1.1
	(0.4)	(0.4)	(0.9)	(0.7)	(0.3)	(0.2)	(0.3)	(1.1)
	[-0.2, 1.4]	[-0.3, 1.5]	[-0.9, 2.5]	[0.0, 2.9]	[-0.9, 0.4]	[0.7, 1.7]	[-1.7, -0.7]	[-3.2, 1.1]
$\omega^S\Delta^S + \omega^I\Delta^I$	8.2	8.2	8.4	9.7	6.7	9.0	4.9	1.5
	(0.5)	(0.5)	(1.0)	(0.9)	(0.4)	(0.3)	(0.3)	(1.8)
	[7.2, 9.1]	[7.1, 9.2]	[6.5, 10.4]	[8.0, 11.4]	[5.9, 7.4]	[8.4, 9.6]	[4.3, 5.5]	[-2.1, 5.1]
$\omega^I\mathcal{C}^{\text{Sys}}$	−6.9	−6.9	−7.1	−7.3	−6.2	−6.8	−5.5	−1.1
	(0.6)	(0.7)	(1.5)	(1.1)	(0.5)	(0.4)	(0.4)	(2.2)
	[-8.1, -5.6]	[-8.2, -5.5]	[-10.0, -4.2]	[-9.6, -5.1]	[-7.2, -5.3]	[-7.5, -6.1]	[-6.3, -4.8]	[-5.3, 3.2]
Panel B: Deployment effects								
$V^0 - V^I$	+0.6	+0.6	+0.8	+1.5	−0.2	+1.2	−1.2	−1.1
	(0.4)	(0.4)	(0.9)	(0.7)	(0.3)	(0.2)	(0.3)	(1.1)
	[-0.2, 1.4]	[-0.3, 1.5]	[-0.9, 2.5]	[0.0, 2.9]	[-0.9, 0.4]	[0.7, 1.7]	[-1.7, -0.7]	[-3.2, 1.1]
$V^0 - V^S$	+1.1	+1.1	+1.2	+1.8	+0.5	+1.3	−0.4	−3.9
	(2.3)	(2.3)	(2.4)	(2.3)	(2.3)	(2.3)	(2.3)	(2.6)
	[-3.3, 5.6]	[-3.4, 5.6]	[-3.4, 5.8]	[-2.7, 6.4]	[-4.0, 4.9]	[-3.1, 5.8]	[-4.8, 4.1]	[-9.0, 1.2]
$V^0 - V^{System}$	1.3	1.3	1.3	2.4	0.4	2.2	−0.7	0.5
	(0.5)	(0.5)	(1.1)	(0.8)	(0.3)	(0.2)	(0.2)	(1.3)
	[0.4, 2.2]	[0.3, 2.3]	[-0.8, 3.4]	[0.8, 4.0]	[-0.2, 1.1]	[1.7, 2.7]	[-1.1, -0.2]	[-2.0, 2.9]

Notes. This table re-estimates the stage-wise comparisons and deployment effects under the alternative maltreatment extrapolations described in Appendix C.1. Column (1) reports the main linear, inverse-variance-weighted specification with day-by-shift fixed effects. Columns (2)–(5) use a local-linear extrapolation, a quadratic extrapolation, an unweighted linear extrapolation, and a linear extrapolation without the fixed effects. Columns (6)–(7) replace the extrapolation with worst-case bounds that set censored outcomes to zero and to one (Appendix C.1). Column (8) extrapolates the maltreatment rate among screened-out calls to a screen-in rate of zero, using screening variation only. Panel A reports the stage-wise comparisons, the weighted comparisons and their sum, and the system-wide aggregation correction. Panel B reports the three deployment effects. All estimates are reported in percentage points. Standard errors are reported in parentheses and 95% confidence intervals are in brackets. Standard errors are computed using 500 bootstrap replications. See Section 4.4 for discussion.

Table A.6: Robustness to Alternative Identification Strategies for the Screening-Stage Deployment

	Main spec.	Monotone-selection bounds		Selection on observables
		Lower bound (uninformative)	Upper bound (informative)	
	(1)	(2)	(3)	(4)
$V^0 - V^S$	+1.1	+13.1	+2.5	+1.6
	(2.3)	(0.3)	(0.4)	(0.4)
	[-3.3, 5.6]	[12.5, 13.7]	[1.7, 3.3]	[0.9, 2.3]
Change in case-opening rate	+1.4	−1.7	+2.1	+1.6
	(2.2)	(0.1)	(0.2)	(0.1)
	[-2.9, 5.6]	[-1.8, -1.5]	[1.7, 2.5]	[1.4, 1.9]
$\omega^I \mathcal{R}^I$	−6.7	+5.4	−5.4	−6.3
	(2.3)	(0.2)	(0.3)	(0.2)
	[-11.2, -2.2]	[5.1, 5.7]	[-6.0, -4.8]	[-6.6, -5.9]
$(\omega^S - \lambda^S)\Gamma^S$	+0.2	+0.1	+0.3	+0.3
	(0.4)	(0.0)	(0.0)	(0.0)
	[-0.5, 1.0]	[0.1, 0.1]	[0.3, 0.4]	[0.2, 0.3]

Notes. This table re-estimates the screening-stage deployment effect and its decomposition under alternative strategies for identifying the two moments on the algorithm's screened-in pool (Appendix C.2). Column (1) is the main specification, which extrapolates across screeners to a screen-in rate of one. Columns (2)–(3) bound the maltreatment rate on that pool under a monotone-selection assumption and report the deployment effect at each bound on maltreatment: the lower bound on maltreatment in column (2) imputes zero maltreatment to the calls whose outcome is unobserved and is uninformative by construction, while the upper bound in column (3) is the informative side. Column (4) identifies the moments under selection on observables. The weighted screening-stage comparison $\omega^S \Delta^S$ equals 7.6 (s.e. 0.3) in every column, because these strategies alter only the screening-stage inputs. Rows report the deployment effect on subsequent maltreatment, the change in the case-opening rate, the downstream investigator response, and the exit-margin reweighting term. All estimates are reported in percentage points. Standard errors are reported in parentheses and 95% confidence intervals are in brackets. Standard errors are computed using 500 bootstrap replications. See Section 4.4 for discussion.

Table A.7: Robustness to Alternative Outcome Definitions

	Outcome definition								
	Re-ref. 6 months (1)	Re-ref. 2 months (2)	Re-ref. 3 months (3)	Re-ref. 4 months (4)	Re-ref. 5 months (5)	New inv. (6)	Subst. inv. (7)	Subseq. case (8)	Subseq. removal (9)
V^0 (status-quo level)	33.8	11.9	19.6	25.4	30.0	20.6	5.7	3.9	1.3
	(0.1)	(0.1)	(0.1)	(0.1)	(0.1)	(0.1)	(0.0)	(0.0)	(0.0)
	[33.6, 34.0]	[11.8, 12.1]	[19.5, 19.8]	[25.2, 25.6]	[29.8, 30.2]	[20.4, 20.8]	[5.6, 5.7]	[3.8, 4.0]	[1.2, 1.3]
Panel A: Stage-wise comparisons and their aggregation									
Δ^S	11.5	5.0	8.0	10.0	11.0	8.0	2.7	2.7	1.2
	(0.5)	(0.4)	(0.4)	(0.4)	(0.4)	(0.3)	(0.2)	(0.1)	(0.0)
	[10.6, 12.3]	[4.2, 5.8]	[7.3, 8.8]	[9.2, 10.8]	[10.2, 11.8]	[7.3, 8.7]	[2.2, 3.2]	[2.4, 2.9]	[1.1, 1.3]
Δ^I	1.8	0.2	1.4	2.0	2.3	2.4	1.0	1.4	0.7
	(1.2)	(0.8)	(1.0)	(1.1)	(1.1)	(1.0)	(0.7)	(0.6)	(0.2)
	[-0.6, 4.1]	[-1.3, 1.7]	[-0.6, 3.3]	[-0.1, 4.1]	[0.1, 4.6]	[0.5, 4.4]	[-0.3, 2.4]	[0.3, 2.5]	[0.4, 1.1]
$\omega^S\Delta^S + \omega^I\Delta^I$	8.2	3.4	5.8	7.3	8.1	6.1	2.1	2.2	1.0
	(0.5)	(0.4)	(0.4)	(0.4)	(0.5)	(0.4)	(0.3)	(0.2)	(0.1)
	[7.2, 9.1]	[2.6, 4.1]	[5.0, 6.6]	[6.5, 8.2]	[7.1, 9.0]	[5.3, 6.9]	[1.6, 2.7]	[1.8, 2.6]	[0.9, 1.1]
$\omega^I\mathcal{C}^{\text{Sys}}$	-6.9	-3.0	-4.7	-5.7	-6.3	-4.5	-1.5	-1.3	-0.4
	(0.6)	(0.5)	(0.5)	(0.6)	(0.6)	(0.5)	(0.4)	(0.3)	(0.1)
	[-8.1, -5.6]	[-3.9, -2.1]	[-5.8, -3.7]	[-6.9, -4.6]	[-7.5, -5.2]	[-5.6, -3.5]	[-2.2, -0.8]	[-1.9, -0.7]	[-0.7, -0.2]
Panel B: Deployment effects									
$V^0 - V^I$	+0.6	+0.1	+0.5	+0.7	+0.8	+0.8	+0.3	+0.5	+0.3
	(0.4)	(0.3)	(0.3)	(0.4)	(0.4)	(0.3)	(0.2)	(0.2)	(0.1)
	[-0.2, 1.4]	[-0.5, 0.6]	[-0.2, 1.1]	[0.0, 1.4]	[0.0, 1.5]	[0.2, 1.5]	[-0.1, 0.8]	[0.1, 0.8]	[0.1, 0.4]
$V^0 - V^S$	+1.1	+0.8	−14.6	−0.5	+2.1	+0.6	+3.6	+1.5	+0.8
	(2.3)	(1.8)	(7.2)	(2.7)	(2.2)	(2.2)	(1.9)	(0.6)	(0.1)
	[-3.3, 5.6]	[-2.6, 4.3]	[-28.7, -0.5]	[-5.8, 4.8]	[-2.3, 6.4]	[-3.6, 4.9]	[-0.1, 7.2]	[0.3, 2.7]	[0.6, 1.0]
$V^0 - V^{System}$	1.3	0.4	1.0	1.6	1.7	1.6	0.6	0.9	0.6
	(0.5)	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)	(0.3)	(0.2)	(0.1)
	[0.4, 2.2]	[-0.3, 1.1]	[0.3, 1.8]	[0.8, 2.4]	[0.9, 2.6]	[0.8, 2.4]	[0.1, 1.2]	[0.4, 1.4]	[0.4, 0.8]
Panel C: Screening-stage decomposition									
$\omega^S\Delta^S$	7.6	3.3	5.3	6.6	7.3	5.3	1.8	1.8	0.8
	(0.3)	(0.3)	(0.2)	(0.3)	(0.3)	(0.2)	(0.2)	(0.1)	(0.0)
	[7.0, 8.2]	[2.8, 3.8]	[4.8, 5.8]	[6.1, 7.2]	[6.7, 7.8]	[4.8, 5.8]	[1.5, 2.1]	[1.6, 1.9]	[0.7, 0.8]
$\omega^I\mathcal{R}^I$	-6.7	-2.6	-21.1	-7.3	-5.4	-4.8	+1.8	-0.4	0.0
	(2.3)	(1.8)	(7.5)	(2.7)	(2.2)	(2.2)	(2.0)	(0.7)	(0.1)
	[-11.2, -2.2]	[-6.1, 0.9]	[-35.8, -6.4]	[-12.6, -1.9]	[-9.8, -1.0]	[-9.1, -0.4]	[-2.0, 5.7]	[-1.7, 0.9]	[-0.2, 0.2]
$(\omega^S - \lambda^S)\Gamma^S$	+0.2	+0.1	+1.1	+0.1	+0.2	+0.1	-0.1	+0.1	0.0
	(0.4)	(0.2)	(0.8)	(0.4)	(0.3)	(0.3)	(0.2)	(0.1)	(0.0)
	[-0.5, 1.0]	[-0.2, 0.4]	[-0.5, 2.8]	[-0.6, 0.9]	[-0.4, 0.8]	[-0.5, 0.7]	[-0.4, 0.3]	[-0.1, 0.2]	[0.0, 0.0]

Notes. This table re-estimates the stage-wise comparisons, deployment effects, and their decompositions under alternative outcome definitions, using the main extrapolation specification. Column (1) uses the baseline outcome: a new hotline call between 31 and 180 days after the initial call. Columns (2)–(5) shorten the horizon to two through five months. Columns (6)–(9) instead require a new investigation, substantiated investigation, case opening, or removal within six months. The first row reports each outcome's status-quo rate among unopened calls, in percent. Panel A reports the stage-wise comparisons, their weighted sum, and the system-wide aggregation correction. Panel B reports the three deployment effects, and Panel C reports the screening-stage decomposition. Entries in Panels A–C are percentage-point differences in the outcome specified by each column. Standard errors are reported in parentheses and 95% confidence intervals are in brackets. Standard errors are computed using 500 bootstrap replications. See Section 4.4 and Appendix C.3 for discussion.

Table A.8: Bounds on the Average Screening-In Effect

Censoring decision	Lower bound	Upper bound	Censoring fraction
Panel A: Risk set $\{m(X_i) \leq \tau^*\}$ *(*$N = 248{,}718$*)*			
Case opening	−2.7	8.6	11.3%
	(4.3)	(4.5)	
	[-11.2, 5.8]	[-0.2, 17.4]	
Placement	1.0	2.9	1.9%
	(4.3)	(4.4)	
	[-7.4, 9.4]	[-5.7, 11.5]	
Panel B: Risk set $\{m(X_i) \leq \tau^S\}$ *(*$N = 164{,}682$*)*			
Case opening	−0.6	7.9	8.6%
	(5.1)	(5.3)	
	[-10.7, 9.4]	[-2.5, 18.3]	
Placement	1.9	2.7	0.8%
	(5.2)	(5.3)	
	[-8.3, 12.1]	[-7.6, 13.1]	

Notes. This table reports bounds on the average screening-in effect on the two risk sets entering the headline estimates, as described in Appendix C.4. Case-opening rows censor subsequent maltreatment at case opening. Placement rows censor it at foster-care placement and additionally assume that opening a case without placement does not affect subsequent maltreatment. The censoring fraction is the share of complier calls whose outcome is censored and equals the width of the identified set. Bounds are reported in percentage points and censoring fractions in percent. Standard errors are reported in parentheses and 95% confidence intervals for each bound endpoint in brackets. Standard errors are computed using 500 bootstrap replications, resampling calls within screener assignment strata. See Section 4.4 for discussion.

Table A.9: Sensitivity to Average Screening-In Effects

	Assumed screening-in effect η				
	-4 p.p.	-2 p.p.	0 p.p.	$+2$ p.p.	$+4$ p.p.
Panel A: Algorithmic risk and stage-wise comparisons					
$\mu_I(\tau^I;\eta)$	32.3	30.3	28.3	26.3	24.3
	(1.1)	(1.1)	(1.1)	(1.1)	(1.1)
	[30.1, 34.5]	[28.1, 32.5]	[26.1, 30.5]	[24.1, 28.5]	[22.1, 26.5]
$\Delta^S(\eta)$	9.8	10.6	11.5	12.3	13.1
	(0.5)	(0.5)	(0.5)	(0.5)	(0.5)
	[9.0, 10.7]	[9.8, 11.5]	[10.6, 12.3]	[11.4, 13.2]	[12.2, 14.0]
$\Delta^I(\eta)$	-2.2	-0.2	1.8	3.8	5.8
	(1.2)	(1.2)	(1.2)	(1.2)	(1.2)
	[-4.6, 0.1]	[-2.6, 2.1]	[-0.6, 4.1]	[1.4, 6.1]	[3.4, 8.1]
Panel B: Deployment effects					
$V^0 - V^S(\eta)$	0.1	0.6	1.1	1.7	2.2
	(2.3)	(2.3)	(2.3)	(2.3)	(2.3)
	[-4.4, 4.5]	[-3.9, 5.1]	[-3.3, 5.6]	[-2.8, 6.2]	[-2.3, 6.7]
$V^0 - V^I(\eta)$	-0.8	-0.1	0.6	1.3	1.9
	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)
	[-1.5, 0.0]	[-0.9, 0.7]	[-0.2, 1.4]	[0.5, 2.1]	[1.2, 2.7]
$V^0 - V^{System}(\eta)$	-0.2	0.6	1.3	2.1	2.8
	(0.5)	(0.5)	(0.5)	(0.5)	(0.5)
	[-1.1, 0.7]	[-0.3, 1.5]	[0.4, 2.2]	[1.2, 2.9]	[1.9, 3.7]
Panel C: Aggregation of stage-wise comparisons					
$\omega^S\Delta^S(\eta) + \omega^I\Delta^I(\eta)$	5.8	7.0	8.2	9.4	10.6
	(0.5)	(0.5)	(0.5)	(0.5)	(0.5)
	[4.8, 6.7]	[6.0, 7.9]	[7.2, 9.1]	[8.4, 10.4]	[9.6, 11.6]
$\omega^I\mathcal{C}^{\text{Sys}}(\eta)$	-5.9	-6.4	-6.9	-7.3	-7.8
	(0.6)	(0.6)	(0.6)	(0.6)	(0.6)
	[-7.2, -4.7]	[-7.7, -5.2]	[-8.1, -5.6]	[-8.6, -6.1]	[-9.0, -6.6]

Notes. This table reports the average screening-in-effect sensitivity analysis in Appendix C.4. The baseline $\eta = 0$ estimates use the main screener extrapolations. For each assumed average screening-in effect η, we preserve the observed screening mix within each risk set and impose $\mu(t;\eta) = \widehat{\mu}(t) - \widehat{\sigma}(t)\eta$ and $\mu_I(t;\eta) = \widehat{\mu}_I(t) - \eta$, where $\widehat{\sigma}(t)$ is the strata-adjusted screen-in rate on $\{m(X_i) \leq t\}$. Deployment effects and the aggregation correction are recomputed using Propositions 3.1 and 3.2. The maltreatment level $\mu_I(\tau^I;\eta)$ is reported in percent; the assumed effects and remaining estimates are in percentage points. Standard errors are reported in parentheses and 95% confidence intervals in brackets. Standard errors are computed using 500 bootstrap replications. See Section 4.4 for discussion.

Table A.10: Predictors of Investigator Performance Relative to the Algorithm

	Δ_g^I (1)	$\Pr(\Delta_g^I < 0)$ (2)
Log caseload	-0.00	0.05
	(0.84)	(2.86)
Matched in worker/race data	0.13	-1.71
	(1.23)	(4.24)
Predicted female	0.41	0.10
	(0.57)	(1.69)
Predicted Black	-2.16	6.19
	(0.67)	(2.40)
Predicted Hispanic	-0.49	2.95
	(1.30)	(4.50)
Matched in personnel records	-0.91	2.07
	(0.47)	(1.44)
Age	-0.04	0.08
	(0.04)	(0.12)
Tenure	0.18	-0.54
	(0.08)	(0.21)
Observations	806	806
Investigators matched to name-based demographic data	774	774
Share matched to name-based demographic data	96.0%	96.0%
Investigators matched to personnel records	416	416
Share matched to personnel records	51.6%	51.6%
R-squared	0.03	0.03

Notes. This table reports weighted least squares regressions of investigator performance on investigator characteristics. The dependent variable is the unshrunk comparison $\widehat{\Delta}_g^I$ in column (1) and the empirical Bayes posterior probability of outperforming the algorithm in column (2). Both dependent variables are multiplied by 100. Regressions are weighted by investigator caseload shares. Predicted demographic characteristics and personnel-record characteristics are set to zero for unmatched investigators, with separate match indicators included. Continuous personnel-record characteristics are centered among matched investigators. A missing-predicted-gender indicator is included but not reported. The omitted predicted-race category combines White and all remaining predicted races other than Black and Hispanic. Coefficients are reported in percentage points. Heteroskedasticity-robust standard errors are reported in parentheses. See Section 5.1 and Appendix D.2.2 for the construction of investigator performance estimates.

Table A.11: Example Case Notes for Investigator Actions

Substance-use services. A substance-use screen, assessment, agreement, treatment referral or requirement, or monitoring action taken or arranged as part of the investigation or case response.

"He completed a random screen with MDOC on 1/2/2018 and it was positive for cocaine."

"Mr. Nguyen was referred to Key Development regarding alcohol abuse."

"CPS attempted to administer a drug test, however Ms. Hernandez declined."

"Andrea has been drug screened on multiple occasions and has tested positive for drugs. Matthew is attending rehab at Care Center for their 90 day substance abuse treatment."

"A drug screen was completed for the mother without positive results."

Safety plans. A safety plan, safety agreement, or concrete behavioral restriction created, signed, attempted, violated, or refused as part of the investigation or case response.

"Holly was placed in a Safety Plan with her maternal aunt and Anthony was placed with his legal father, Joshua."

"DHHS will request that the children remain with the mother and that only supervised visitation with the father may occur pursuant to Court approval as there already is a no contact order in place."

"Safety Plan was put into place for Ann and Benjamin ... Jessica has been safety plan with her grandparents Andrew and Katie ... Ann has failed to attend substance abuse treatment."

"Samantha was asked several times to come to the DHHS office to discuss her drug screen and voluntarily placing Mark with an appropriate relative. Samantha refused to come to the office or inform the worker where she was with Mark. There are no safety interventions that could be put in place to keep Mark safe."

"A safety plan was completed with Ms. Meza. She agreed to not leave her children with anyone under the influence of any substance. She agreed to not fight or argue with anyone in front of her children."

Mental-health services. A recommendation, referral, resource provision, or coordination action to obtain or continue counseling, psychological evaluation, psychiatric treatment, crisis mental-health services, or support groups as part of the investigation or case response.

"The CPS worker provided Laura White with resources for mental health agencies."

"CPS referred the family to The Children's Center for psychiatric evaluation for Taylor."

"Linda was assisted in setting up counseling at Behavioral Health for the children."

"CPS offered to refer Trey for counseling services and Trey declined."

"CPS is recommending as a safety plan that the family continue in therapy to address the concerns identified in this investigation."

Parenting guidance. Advice, education, demonstrations, referrals, or concrete childrearing instructions addressing discipline, supervision, safe sleep, routines, child development, or other caregiving needs as part of the investigation or case response.

"Safe sleep protocol was instructed to Jamie and Mason by having Lillian sleep alone in her crib and on her back with no other items inside on a firm mattress and a tight fitting sheet."

"A discussion was had with the parents in regards to physical discipline and different methods of discipline to use with the children."

"Marissa and Lillian have been educated about what constitutes proper supervision."

Continued on next page

Table A.11, continued from previous page

"CPS has referred the family to Arbor Circle parenting classes."
"CPS safety planned that all medications need to be kept in container or lock box out of Maya and Jackson's reach."
Material assistance. Provision of or assistance accessing housing, food, utilities, transportation, childcare, clothing, school supplies, or essential infant and household equipment as part of the investigation or case response.
"CPS assisted Daniel with getting food assistance and WIC for the children."
"The family was connected with housing resources and encouraged to apply for daycare assistance through MDHHS."
"CPS provided transportation to and from the appointment on 1/5/2018."
"CPS discussed Safe Sleep and provided a pack and play for the family."
"CPS provided the family with United Way 211 phone number for low income (winter protection plan for DTE)."

Notes. This table reports excerpts from de-identified investigation case notes labeled as documenting each auxiliary investigator action, using the procedure described in Appendix E.1. Each excerpt is the supporting evidence quoted by the labeling model. For substance-use services and safety plans, the five snippets are drawn from notes coded present across the range of recorded statuses. For the other actions, the snippets illustrate different types of assistance or guidance among notes coded present. Names appearing in the excerpts are pseudonyms introduced during de-identification. See Section 5.2 for discussion.

Table A.12: Investigator Exclusion Restriction Test: Adjusted-Rate Quintiles

Panel A: Quintile moments and brackets							
			Strata-adjusted moments			Precision-corrected	
Quintile	Investigators	Caseload share (%)	$\widehat{d}_j$	$\widehat{L}_j$	$\widehat{U}_j = \widehat{L}_j + \widehat{d}_j$	Lower	Upper
1 (lowest)	161	19.2	2.4	29.7	32.1	29.2	32.4
			(0.3)	(0.5)	(0.5)		
2	161	21.1	6.2	28.4	34.6	28.0	34.9
			(0.3)	(0.4)	(0.5)		
3	161	20.2	9.1	27.4	36.5	27.0	36.8
			(0.3)	(0.4)	(0.5)		
4	161	20.0	12.5	26.2	38.7	25.7	38.9
			(0.3)	(0.4)	(0.5)		
5 (highest)	162	19.5	20.4	23.2	43.6	22.7	43.9
			(0.4)	(0.5)	(0.5)		

Panel B: Intersection test	
Maximum lower bound, $\max_j \widehat{L}_j$ (quintile)	29.7 (1)
Minimum upper bound, $\min_j \widehat{U}_j$ (quintile)	32.1 (1)
Gap, $\max_j \widehat{L}_j - \min_j \widehat{U}_j$	-2.4
Max-t statistic T	-5.39
5 percent least-favorable critical value	2.76
p-value	1.0000

Notes. This table tests whether investigator quintiles imply compatible bounds on the maltreatment rate that would prevail if no case were opened. Investigators are grouped into quintiles of their strata-adjusted case-opening rate, from lowest to highest. Panel A reports each quintile's case-opening rate $\widehat{d}_j$, lower endpoint $\widehat{L}_j$, and upper endpoint $\widehat{U}_j = \widehat{L}_j + \widehat{d}_j$. The lower endpoint is the adjusted share of calls with both no case opening and subsequent maltreatment. The upper endpoint adds the case-opening rate. Strata-adjusted moments control for month-by-ZIP-code-by-shift fixed effects. The final two columns report the precision-corrected endpoints of Chernozhukov, Lee and Rosen (2013). Under the exclusion restriction and random assignment within strata, the five intervals contain a common counterfactual maltreatment rate. Panel B tests this implication using the uncorrected endpoints: a positive gap indicates an empty estimated intersection. Rates and endpoints are reported in percent and the gap in percentage points. Call-level heteroskedasticity-robust standard errors are reported in parentheses in Panel A. The max-t test uses the joint covariance of the moments and 10,000 Gaussian draws. Parentheses beside the bounds in Panel B identify the quintiles determining those bounds. See Section 5.2 and Appendix D.4 for discussion.

Table A.13: Case Opening and Intake Characteristics by Investigator Performance

	Higher performing	Lower performing	Lower – higher
Number of children	0.62	0.70	0.08
	(0.33)	(0.16)	(0.37)
Female child present	-0.36	-0.66	-0.30
	(0.75)	(0.38)	(0.84)
Male child present	0.01	-0.29	-0.30
	(0.74)	(0.38)	(0.84)
Average child age	-0.97	-1.36	-0.40
	(0.26)	(0.13)	(0.29)
White child present	1.78	1.64	-0.15
	(0.73)	(0.46)	(0.86)
Black child present	1.52	1.17	-0.35
	(0.71)	(0.41)	(0.82)
Hispanic child present	1.48	1.29	-0.19
	(1.15)	(0.48)	(1.25)
Priority call	4.77	6.05	1.28
	(0.58)	(0.36)	(0.68)
Physical abuse allegation	-1.50	-1.75	-0.26
	(0.54)	(0.32)	(0.63)
Physical neglect allegation	0.67	1.88	1.21
	(0.66)	(0.30)	(0.73)
Improper supervision allegation	3.62	5.42	1.80
	(0.52)	(0.30)	(0.60)
Other allegation	4.71	6.62	1.91
	(0.79)	(0.39)	(0.88)
Average prior investigations	2.24	1.57	-0.67
	(0.48)	(0.20)	(0.52)
Observations	18656	74799	93455

Notes. This table reports OLS coefficients from a regression of case opening on intake characteristics and the continuous maltreatment prediction, fully interacted with investigator performance group, including separate group intercepts. The sample consists of screened-in calls in the evaluation sample. Column (1) reports coefficients for higher-performing investigators, whose posterior mean comparison Δ_g^I is negative. Column (2) reports coefficients for lower-performing investigators, whose posterior mean comparison is positive. Column (3) reports the lower-performing minus higher-performing difference. Other allegation indicates any allegation type other than physical abuse, physical neglect, or improper supervision. Average child age and average prior investigations are standardized to unit standard deviation before estimation. Coefficients are reported in percentage points. Standard errors clustered by investigator are reported in parentheses. See Section 5.3 for discussion.

Table A.14: Granular Narrative Concepts Extracted from Investigator Case Notes

Granular concept	Definition
Corroborated harm or injury evidence†	
Corroborated physical injury evidence	Whether the narrative includes concrete, observable injury evidence that is corroborated by examination, photos, medical staff, or detailed description, as opposed to allegations with no marks, minimal findings, or injuries not verified.
Corroborated child disclosures and external evidence versus allegations only	The narrative contains either multiple child disclosures and/or external corroboration (police, interviews, medical or collateral evidence) supporting the safety concern, as opposed to a file dominated by unconfirmed allegations and denials.
Concrete external incident documentation	Whether the report contains strong external documentation of the event itself through police action, hospital records, medical exams, video, narcotics raid, or similarly specific third-party evidence, versus relying mainly on interviews and allegations.
Corroborated child fear, injury, or distress linked to the home environment	The document includes direct child statements or corroborated observations showing fear, suicidality, self-harm, bruising, or emotional distress tied to home treatment or home environment.
Concrete injury evidence paired with corroboration rather than normal/accidental explanation	The document presents physical injuries or medically notable conditions that are reinforced by disclosures, police/medical findings, or inability of caregivers to explain them, rather than being normalized as accidental or non-abusive.
Patterned or multi-source corroboration of harm	The document presents a dense evidentiary pattern in which multiple independent sources or repeated incidents converge on the same safety concern, suggesting an ongoing pattern rather than a single ambiguous event.
Absence, ambiguity, or contradiction of abuse evidence†	
Child allegation recantation or contradiction	The extent to which the investigative narrative presents the child allegation as contradicted, recanted, inconsistent, or unsupported by what the child ultimately says in interview.
Absence or ambiguity of physical/medical evidence	The alleged harm lacks observable injuries, leaves no marks, or is medically explained/indeterminate, reducing evidentiary support for abuse or neglect.
Exculpatory corroboration of adequate care†	
Corroborated lack of concern from collaterals	The document includes multiple outside sources (school staff, therapists, medical providers, prevention workers, police, relatives) affirming child safety, adequate parenting, or lack of concern, rather than raising new concerns.
Exculpatory home-and-collateral reassurance	The document relies on reassuring observations and collateral reports that the child is clean, safe, appropriately housed, and that caregivers are attentive, with multiple outside sources denying concern.

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Appendix Table A.14, continued from previous page

Granular concept	Definition
Collateral reassurance of caregiver adequacy	Presence of collateral contacts or observational findings that affirm the caregiver/home as safe, appropriate, clean, attentive, or free of concern.
Observed environment appears appropriate and hazard-free	The narrative describes the home or living setting as clean, organized, supplied, and without notable safety or sanitary hazards.
Allegations rebutted by collateral verification of adequate care	The document emphasizes corroborating interviews, medical/school contacts, or home observations that support adequate caregiving and contradict the allegations.
Corroborated child well-being versus concern negation	Whether the narrative emphasizes observed child normalcy and absence of corroborating signs, versus concrete child disclosures, observed injuries, or other confirming evidence of harm/risk.
Caregiver absence, unstable placement, or abandonment†	
Caregiver instability leaving children without secure placement	The document describes children lacking a stable, reliable caregiver arrangement or being shifted among temporary caretakers, relatives, friends, or jurisdictions, often with unclear legal authority or disrupted custody.
Caregiver abandonment, expulsion, or reliance on children/others for basic care	The document describes a caregiver leaving children without normal supervision, dropping a child elsewhere unexpectedly, or shifting basic caregiving responsibility onto minors or unstable arrangements.
Child wandering or left without immediate adult oversight	The narrative centers on a young child being alone, outside, missing from the caregiver's immediate awareness, or left with an unapproved/insufficient caretaker.
Caregiver disappearance, abandonment, or evasive unavailability	The text portrays caregivers as absent for extended periods, unreachable, evasive with authorities, incarcerated, or otherwise leaving children without a dependable responsible adult or stable return plan.
Caregiver actively involves child in dangerous or exploitative conduct†	
Caregiver participation in dangerous or criminal conduct involving the child	The document describes a caregiver not merely failing to supervise, but actively involving the child in dangerous, criminal, or highly inappropriate conduct.
Benign or accidental explanation displacing the allegation	
Benign or alternative explanation for injury/incident	The narrative supplies a specific nonabusive or nonneglectful explanation for the observed mark, symptom, or alarming event, often displacing the allegation with an accident, misunderstanding, rumor, or child behavior explanation.
Alternative accidental/benign explanation for allegations	Whether the narrative supplies a plausible nonabusive explanation for the reported concern, often replacing the allegation with accidental injury, misunderstanding, self-caused marks, or normal child behavior.
Medical/developmental care management and caregiver follow-through	

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Appendix Table A.14, continued from previous page

Granular concept	Definition
Documented management failure of significant medical/developmental need	The text centers on sustained failure to obtain, attend, or follow through with necessary medical, specialty, or developmental services, often supported by appointment history, provider concern, or prolonged untreated need.
Prompt protective care versus delayed, refused, or impaired caregiver response	How the caregiver responds after a child injury, mental-health crisis, or dangerous situation: promptly seeking care and taking protective action versus delaying, refusing, or being impaired in responding.
Caregiver follow-through and problem response	The caregiver is described as seeking treatment, keeping or rescheduling care, taking corrective action, or otherwise actively responding to concerns versus failing to follow through.
Substance use or caregiver/household volatility affecting child safety	
Acute household crisis affecting supervision or safety	The document describes a concrete crisis in the household—such as severe intoxication/incapacitation, threats, runaway instability, domestic conflict, or inability to manage an escalating youth—that directly implicates the child's immediate safety or supervision.
Substance exposure in the caregiving environment	Presence of substance misuse allegations that are either unsubstantiated/minimized and detached from child impact, or concretely tied to caregiving impairment, positive drug tests, drug activity, or child exposure.
Active caregiver impairment or household volatility affecting supervision	The document describes ongoing substance use, domestic violence, suicidal/self-harm crises, weapons, or severe conflict in the caregiving environment that is tied to child safety concerns.
Child behavior or developmental issues used to reframe the allegation	
Conflict framed around child behavioral dysregulation rather than caregiver maltreatment	The narrative emphasizes the child's lying, acting out, aggression, anxiety, probation/running away, or developmental/mental-health issues as the main context for the allegation, often shifting interpretation away from caregiver wrongdoing.
Child-exposed domestic conflict or violent adult interactions	
Domestic conflict or chaotic adult interactions witnessed by children	Whether the document features recurrent adult conflict, frightening arguments, violence-adjacent incidents, or unstable interpersonal drama occurring in the child's presence and shaping the safety concern.
Children exposed to chronic violent adult conflict	The document depicts children directly witnessing, hearing, hiding from, or being emotionally affected by recurring violent or terrorizing conflict between adults in the home, often including threats, weapons, or serious assaultive behavior.
Domestic violence as contextual allegation versus child-exposed danger	How domestic violence appears in the narrative: as disputed/background adult conflict with limited child involvement, or as a directly child-exposed violent event shaping the child's safety.
Corroborated child exposure to domestic violence	The document describes children directly witnessing, intervening in, or being physically proximate to caregiver-on-caregiver violence, often with concrete details of assaultive acts.
Unsafe or severely deprived home/basic living conditions	

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Appendix Table A.14, continued from previous page

Granular concept	Definition
Observable unsanitary or hazardous home environment	The text describes concrete household conditions that are dirty, cluttered, pest-infested, blocked, or otherwise physically unsafe for children.
Materially hazardous caregiving environment	Degree to which the environment or caregiving context is depicted as concretely dangerous or grossly unsafe for the child, rather than basically adequate or concerning only in allegation form.
Severe deprivation of basic living conditions	The narrative describes children living amid serious deficits in food, utilities, hygiene, sanitation, sleeping arrangements, or medical upkeep, beyond a minor or temporary household issue.
Environmental/basic-care reassurance versus pronounced household hazard	Whether the document characterizes the home and basic care as meeting community/safety standards despite allegations, versus describing conspicuous filth, unsafe clutter, or unmet daily care needs.
High-risk third-party access to children	
High-risk third-party access to children	Presence of a dangerous or criminally concerning nonprimary adult whose access to children is central to the case, especially sex offenders or violent partners, versus absence of such third-party risk or narratives that diffuse that concern.

Notes. This table lists the 37 distinct granular narrative concepts produced by the concept-discovery step of the large language model pipeline described in Appendix E.2, after near-duplicate concepts are merged by tf–idf similarity. Granular narrative concepts are grouped under the 12 candidate narrative evidence themes into which they were aggregated. A dagger (†) marks the five candidate themes selected as final evidence themes for their within-decile association with the case-opening decision on the validation sample. Table A.15 reports the final theme definitions and illustrative excerpts. See Section 5.3 for discussion.

Table A.15: Extracted Narrative Evidence Themes with Illustrative Excerpts

Corroborated harm or injury. The note contains concrete physical injury, child disclosure, external documentation, or multi-source evidence that substantively corroborates maltreatment or serious risk.

"Both Mr. Ray and Ms. Ray had scratches on them; Ms. Ray ended up having bruising all over her arms and her neck."

"Saginaw CPS forensically interviewed James Jr. and he disclosed that he was cut on his neck by his dad during a physical altercation."

"Theresa sustained bruising to her right forearm. She has been admitted to the hospital. Theresa and Amber was forensically interviewed."

Absent, ambiguous, or contradictory abuse evidence. The note weakens the allegation through no observed injury, medically indeterminate findings, child recantation/contradiction, or lack of corroborating evidence.

"None of the children had any concerning information to report. Collateral contact was made with law enforcement and the school with no concerns to report."

"Shannon denied ever taking her mother's medical marijuana... Shannon was not willing to openly communicate with CPS."

"Dylan denied that this hurt and does not have any visible injuries as a result."

Adequate-care corroboration. The note is structured around reassuring child statements, home observations, and/or collateral contacts indicating the caregiver and home are adequate and the allegation is not borne out.

"The children were observed to be happy, healthy and of adequate weight... They all confirmed that the home is clean, the cats have litter boxes that they use and there is always food to eat. None of the children had any concerning information to report. Collateral contact was made with law enforcement and the school with no concerns to report."

"Elizabeth denied ever taking her mother's medical marijuana; Brianna sent pictures of her safe with a combination code; Bruce denied witnessing use."

"Both children reported that they feel safe at home. The home was viewed to be appropriate. The water, and electricity observed to be in working order."

Caregiver absence, unstable placement, or abandonment. The note describes children lacking a secure caregiving arrangement because of expulsion from home, homelessness, prolonged caregiver absence, refusal to retrieve the child, or improvised placement across relatives/friends.

"Since the incident Rose has moved into her own apartment and is no longer living with the family."

"Ms. White was evicted... arrested while Jaime was in school leaving him without proper care and custody. Jaime stated that his mom was in jail and that he loved staying with his Aunt... Ms. Perkins is meeting all of Jaime's needs."

"Ms. Kelley reported that Arthur has not been in her home since March 2020... Arthur ran away from bootcamp in March and he hasn't been seen since then."

Caregiver involvement in dangerous conduct. The caregiver is portrayed as an active participant in, enabler of, or instructor for dangerous, criminal, or highly inappropriate conduct involving the child, beyond mere poor supervision.

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Table A.15, continued from previous page

"Amy admits to allowing Michael to try her wine and putting the video on Snap Chat."
"She was overheard telling Jeffrey 'you don't want me to take you home, because I will tie your hands above my head, and you know what will happen.' Emily will use a belt to tie his hands above his head and 'beat' him."
"Jesse Richards operating a motor vehicle while having illegal substances, selling illegal substance and using those illegal substances with his son in the car."

Notes. This table reports the five AI-extracted narrative evidence themes selected by the pipeline described in Appendix E.2. Each theme is accompanied by its definition and three excerpts from de-identified investigation case notes labeled as exhibiting it. Names appearing in the excerpts are pseudonyms introduced during de-identification. See Section 5.3 for discussion.

Table A.16: Predicting Case-Opening Decisions from Investigator Case Notes

	AUC	Brier score
Case-opening algorithm	0.717	0.1107
	[0.712, 0.723]	[0.1085, 0.1126]
Case-note embeddings	0.995	0.0062
	[0.994, 0.996]	[0.0057, 0.0067]
Extracted narrative evidence themes	0.914	0.0736
	[0.911, 0.917]	[0.0723, 0.0749]
Case-opening algorithm + extracted narrative evidence themes	0.927	0.0697
	[0.925, 0.929]	[0.0684, 0.0710]
Observations	67,512	67,512

Notes. This table reports out-of-sample performance in predicting the case-opening decision on the held-out analysis-test sample of de-identified investigation case notes. The rows report predictions from the case-opening algorithm trained on administrative intake characteristics, an L2-penalized logistic regression using case-note embeddings, the five extracted narrative evidence themes, and an ensemble combining the case-opening algorithm with those themes. AUC is the area under the receiver operating characteristic curve. The Brier score is the mean squared difference between the predicted probability and the realized case-opening decision. Confidence intervals are reported in brackets and computed using 500 bootstrap replications. See Section 5.3 and Appendix E.2 for the construction of the models and themes.

Table A.17: Prevalence of Evidence Themes by Case-Opening Status

Evidence theme	All investigations	Opened	Unopened
Adequate-care corroboration	0.820	0.166	0.894
Corroborated harm or injury	0.457	0.900	0.407
Absent, ambiguous, or contradictory abuse evidence	0.871	0.395	0.925
Caregiver absence, unstable placement, or abandonment	0.316	0.585	0.286
Caregiver involvement in dangerous conduct	0.078	0.160	0.069
Both incriminating and exculpatory themes	0.386	0.382	0.387

Notes. This table reports the fraction of investigations with each narrative evidence theme recorded in the case notes, overall and separately for opened and unopened cases. Themes are extracted using the procedure described in Appendix E.2 and are not mutually exclusive. The final row requires at least one incriminating theme (corroborated harm or injury, or caregiver involvement in dangerous conduct) and at least one exculpatory theme (absent, ambiguous, or contradictory abuse evidence, or adequate-care corroboration). Entries are fractions of investigations with the indicated evidence theme present in their case notes. See Section 5.3 for discussion.

Table A.18: Case Opening and Narrative Evidence by Investigator Performance

	Higher performing	Lower performing	Lower – higher
Adequate-care corroboration	-28.04	-32.64	-4.60
	(1.82)	(0.80)	(1.99)
Corroborated harm or injury	3.59	4.91	1.32
	(0.48)	(0.25)	(0.55)
Absent, ambiguous, or contradictory abuse evidence	-15.06	-15.35	-0.29
	(1.60)	(0.82)	(1.79)
Caregiver absence, unstable placement, or abandonment	5.31	5.31	0.00
	(0.58)	(0.26)	(0.64)
Caregiver involvement in dangerous conduct	1.21	2.37	1.16
	(0.87)	(0.51)	(1.01)
Observations	15219	58628	73847
Intake controls	Yes	Yes	Yes
Maltreatment prediction × performance group	Yes	Yes	Yes

Notes. This table reports OLS coefficients from a regression of case opening on the five narrative evidence themes, intake characteristics, and the continuous maltreatment prediction. All regressors are fully interacted with investigator performance group, including separate group intercepts. The sample consists of screened-in evaluation calls with complete theme labels. Column (1) reports coefficients for higher-performing investigators, whose posterior mean comparison Δ_g^I is negative. Column (2) reports coefficients for lower-performing investigators, whose posterior mean comparison is positive. Column (3) reports the lower-performing minus higher-performing difference. Themes are extracted using the procedure in Appendix E.2. Intake controls include the number of children, child gender indicators, average child age, race and ethnicity indicators, priority status, allegation-type indicators, and average prior investigations. Average child age and average prior investigations are standardized to unit standard deviation before estimation. Coefficients are reported in percentage points. Standard errors clustered by investigator are reported in parentheses. See Section 5.3 for discussion.

Table A.19: Inputs to the Deployment Effects and Their Decompositions

Panel A: Subsequent maltreatment among unopened calls		
V^0	33.8 (0.1) [33.6, 34.0]	Under the status quo
V^I	33.2 (0.4) [32.4, 33.9]	When the algorithm is deployed at the investigation stage
V^S	32.6 (2.3) [28.2, 37.1]	When the algorithm is deployed at the screening stage
V^{System}	32.5 (0.4) [31.6, 33.3]	When the algorithm is deployed in place of the multistage system
Panel B: Exit shares		
ω^S	66.2 (0.1) [66.0, 66.4]	Share of unopened calls that exit at the screening stage
ω^I	33.8 (0.1) [33.6, 34.0]	Share of unopened calls that exit at the investigation stage

Notes. This table reports inputs to the deployment effects and decompositions in Propositions 3.2 and 3.1. Panel A reports subsequent maltreatment among unopened calls under the status quo and each algorithmic deployment. The status-quo rate V^0 is estimated as a sample mean. Counterfactual rates are estimated using the screener extrapolations described in Section 4.1 and Appendix B.3. Panel B reports the sample shares of unopened calls that exit at the screening and investigation stages. Maltreatment rates and exit shares are reported in percent. Standard errors are reported in parentheses and 95% confidence intervals in brackets. Standard errors are computed using 500 bootstrap replications. See Section 4.3 for discussion.

Table A.20: Shapley-Style Contributions and Interaction under Algorithmic Deployment

	Estimate	Description
θ^S	0.9 (1.2) [-1.4, 3.2]	Screening-stage contribution to $V^0 - V^{System}$.
θ^I	0.4 (1.2) [-2.0, 2.8]	Investigation-stage contribution to $V^0 - V^{System}$.
$\mathcal{I}$	0.4 (2.3) [-4.2, 5.0]	Interaction between the two stage replacements; negative when the replacements are complements.

Notes. This table reports the Shapley-style stage contributions θ^S and θ^I and interaction $\mathcal{I}$ defined in Appendix C.5. Each stage contribution averages the reduction in subsequent maltreatment from deploying the algorithm at that stage across the two possible deployment orderings. The contributions sum to the system-wide deployment effect $V^0 - V^{System}$. A negative interaction indicates that the two stage deployments are complements. All quantities are functions of the counterfactual values in Table A.19, estimated using the main specification. All estimates are reported in percentage points. Standard errors are reported in parentheses and 95% confidence intervals in brackets. Standard errors are computed using 500 bootstrap replications. See Section 4.4 and Appendix C.5 for discussion.

Table A.21: Empirical Bayes Estimates of Screener Comparisons and Deployment Effects

	Estimate
Panel A: Screening-stage human-algorithm comparison	
Mean of Δ_k^S	11.3
	(0.1)
Standard deviation across screeners of Δ_k^S	1.3
	(0.1)
$\Pr(\Delta_k^S > 0)$	100.0%
	(0.0)
$\Pr(\Delta_{k,\text{rand}}^S > 0)$	97.3%
	(2.0)
Panel B: Screener deployment effect	
Mean of β_k^S	1.5
	(0.1)
Standard deviation across screeners of β_k^S	1.3
	(0.1)
Panel C: Joint distribution	
Covariance of (Δ_k^S, β_k^S)	0.96
	(0.51)
Screeners	291
Mean calls per screener	887

Notes. This table reports caseload-weighted empirical Bayes estimates of the distribution of screener–algorithm comparisons Δ_k^S and screener deployment effects β_k^S, using the posterior-average-effect procedure in Appendix D.3. The comparisons use the screener extrapolations of Section 4.1. Deployment effects combine those extrapolations with selection on observables for the algorithm's screened-in pool. Panel A reports the mean and standard deviation of the comparisons and the shares of screeners with larger maltreatment rates than the algorithm or a random screening rule. Panel B reports the mean and standard deviation of deployment effects. Panel C reports their covariance with the comparisons. Means and standard deviations are reported in percentage points, the covariance in squared percentage points, and shares in percent. Standard errors are reported in parentheses and computed using 500 bootstrap replications of the empirical Bayes procedure.

Table A.22: Screener Deployment Effects and Their Relationship to Stage-Wise Comparisons

Panel A: Screener deployment effects		
$\overline{\beta}^S_\pi$	1.5 (0.1)	Caseload-weighted mean reduction in subsequent maltreatment from replacing screener with the algorithm
Σ^S_β	1.3 (0.1)	Standard deviation of algorithmic deployment effects across screeners
$\Pr(\beta^S_k > 0)$	89.8% (1.6)	Share of screeners whose replacement by the algorithm reduces subsequent maltreatment
$\Pr(\beta^S_{k,\text{rand}} > 0)$	47.9% (2.0)	Share of screeners whose replacement by a random decision rule reduces subsequent maltreatment
Panel B: Joint distribution of comparisons and deployment effects		
$\text{Cov}_\pi(\Delta^S_k, \beta^S_k)$	0.96 (0.51)	Caseload-weighted covariance between the screener–algorithm comparison and the deployment effect
$\text{Corr}_\pi(\Delta^S_k, \beta^S_k)$	0.58 (0.02)	Caseload-weighted correlation between the screener–algorithm comparison and the deployment effect

Notes. This table reports caseload-weighted empirical Bayes estimates of screener deployment effects β^S_k and their relationship to screener–algorithm comparisons Δ^S_k, using the posterior-average-effect procedure in Appendix D.3. The comparisons use screener extrapolations, while deployment effects combine those extrapolations with selection on observables for the algorithm's screened-in pool. Panel A reports the mean and standard deviation of deployment effects and the shares of screeners whose replacement by the algorithm or a random decision rule reduces subsequent maltreatment. Panel B reports the covariance and correlation between the comparisons and deployment effects. Means and standard deviations are reported in percentage points, the covariance in squared percentage points, and shares in percent. Standard errors are reported in parentheses and computed using 500 bootstrap replications of the empirical Bayes procedure.

Table A.23: Screener Exclusion Restriction Test: Adjusted-Rate Quintiles

Panel A: Quintile moments and brackets							
			Strata-adjusted moments			Precision-corrected	
Quintile	Screeners	Caseload share (%)	$\widehat{s}_j$	$\widehat{L}_j$	$\widehat{U}_j = \widehat{L}_j + \widehat{s}_j$	Lower	Upper
1 (lowest)	58	17.6	31.5	24.3	55.7	24.1	55.9
			(0.2)	(0.2)	(0.2)		
2	58	21.0	34.3	23.4	57.7	23.2	57.8
			(0.2)	(0.2)	(0.2)		
3	58	25.5	36.3	22.8	59.1	22.7	59.2
			(0.2)	(0.2)	(0.2)		
4	58	19.4	38.0	22.2	60.2	22.0	60.3
			(0.2)	(0.2)	(0.2)		
5 (highest)	59	16.4	41.5	21.0	62.5	20.8	62.6
			(0.2)	(0.2)	(0.2)		

Panel B: Intersection test	
Maximum lower bound, $\max_j \widehat{L}_j$ (quintile)	24.3 (1)
Minimum upper bound, $\min_j \widehat{U}_j$ (quintile)	55.7 (1)
Gap, $\max_j \widehat{L}_j - \min_j \widehat{U}_j$	-31.5
Max-t statistic T	-107.58
5 percent least-favorable critical value	2.73
p-value	1.0000

Notes. This table tests whether screener quintiles imply compatible bounds on the maltreatment rate that would prevail if no call were screened in. Screeners are grouped into quintiles of their strata-adjusted screen-in rate, from lowest to highest. Panel A reports each quintile's screen-in rate $\widehat{s}_j$, lower endpoint $\widehat{L}_j$, and upper endpoint $\widehat{U}_j = \widehat{L}_j + \widehat{s}_j$. The lower endpoint is the adjusted share of calls with both a screen-out decision and subsequent maltreatment. The upper endpoint adds the screen-in rate. Strata-adjusted moments control for day-by-shift fixed effects. The final two columns report the precision-corrected endpoints of Chernozhukov, Lee and Rosen (2013). Under the exclusion restriction and random assignment within strata, the five intervals contain a common counterfactual maltreatment rate. Panel B tests this implication using the uncorrected endpoints: a positive gap indicates an empty estimated intersection. Rates and endpoints are reported in percent and the gap in percentage points. Call-level heteroskedasticity-robust standard errors are reported in parentheses in Panel A. The max-t test uses the joint covariance of the moments and 10,000 Gaussian draws. Parentheses beside the bounds in Panel B identify the quintiles determining those bounds. See Appendix D.4 for discussion.

Appendix Figure A.13: Action-labeling prompt — substance-use services (abridged)

```
[SYSTEM PROMPT / MODEL INSTRUCTIONS]

# Identity

You are a careful research assistant labeling original deidentified child welfare investigation case
notes for one fixed investigator action category: substance-use services.

# Instructions

Label only actions documented in the note text. The action must be taken, requested, arranged, referred
, administered, discussed, required, or monitored by CPS, the investigator, the court, probation,
foster care, or a named service provider as part of the investigation or case response.

Do not code present for descriptive facts learned by the investigator. A caregiver's existing treatment
, methadone clinic attendance, inpatient-program participation, reported marijuana use, or substance-
use history is not itself an investigator action.

Do not infer actions from allegations, case opening, substantiation, Central Registry language, risk
level, or generic safety/risk assessment boilerplate.

# Action: substance_use_services

Label whether the case note documents an investigator- or case-response action involving a substance-
use-related service, agreement, assessment, screen, treatment referral, treatment requirement, or
monitoring action.

Code present when the note references at least one of the following:

* CPS/investigator, court, probation, foster care, or a named provider created, requested, or monitored
 a cannabis agreement or substance-use agreement.
* CPS/investigator, court, probation, foster care, or a named provider requested, scheduled,
administered, completed, missed, or documented refusal of a drug screen.
* A toxicology, PBT, breathalyzer, oral screen, urine screen, or other substance test occurred as part
of the investigation or case response.
* CPS/investigator, court, probation, foster care, or a named provider requested, recommended, or
required a substance-use or substance-abuse assessment.
* CPS/investigator, court, probation, foster care, or a named provider referred someone to substance-
use treatment, outpatient treatment, AA/NA, or a substance-use provider.
* CPS/investigator, court, probation, foster care, or a named provider monitored substance-use
treatment, substance-use testing, or substance-use service compliance.

Do not code present for mere allegations of drug or alcohol use or ordinary descriptions of current/
past treatment unless the note also documents an investigation or case-response action: testing,
assessment, agreement, referral, refusal, requirement, or monitoring.

[Scale, status, and output-format sections omitted.]

# Positive Examples

These examples are indicative, not substring rules:

* "Matthew signed a cannabis agreement. Matthew completed a drug screen and was only positive for THC."
* "He submitted to an oral drug screen during his initial interview by CPS."
    ⋮

# Negative Edge Cases

Do not code present for these cases unless the note separately documents an investigation or case-
response action such as a screen, agreement, referral, assessment, treatment requirement, refusal, or
monitoring:

* Allegation only: "Benjamin is into crack and meth right now and would fail a drug test."
* No-current-action statement: "No one was under the influence of alcohol or drugs."
```

⋮

[USER MESSAGE]

[case note begins here]
{one de-identified investigation case note}
[case note ends here]

Appendix Figure A.14: Action-labeling prompt — safety plans (abridged)

[SYSTEM PROMPT / MODEL INSTRUCTIONS]

Identity

You are a careful research assistant labeling original deidentified child welfare investigation case notes for one fixed investigator action category: safety plans and behavioral restrictions.

Instructions

Label only actions documented in the note text. The action must be a CPS/investigator/case-response safety plan, safety agreement, or concrete behavioral restriction.

Do not infer a safety plan from standalone custody, guardianship, probate-court, family-court, or placement facts. A child living with or being placed with a relative is not a safety-plan action unless the note explicitly ties that arrangement to a CPS safety plan, safety agreement, or concrete protective restriction.

Do not code court, prosecutor, law-enforcement, probation, or other non-CPS protective orders as present when they are only described as external facts. Code them present only when the note documents that CPS/the investigator created, incorporated, monitored, required compliance with, or otherwise used the order as part of a CPS safety plan, safety agreement, or case-response restriction.

Do not infer actions from allegations, case opening, substantiation, Central Registry language, risk level, or generic safety/risk assessment boilerplate.

Action: safety_plans

Label whether the case note documents a concrete safety plan, safety agreement, or behavioral restriction.

Code present when the note states that a safety plan or safety agreement was created, signed, violated, attempted, or could not be created, or when the note lists concrete investigator/CPS/case-response restrictions intended to protect the child.

Concrete terms include:

* No physical discipline.
* No verbal or physical altercations in the home or in the child's presence.
* No-contact, PPO, protective-order, or stay-away compliance when documented as part of a CPS/investigator safety plan, safety agreement, or case-response restriction.
* Sober caregiver requirement.
* No unsupervised caregiving.
* Alternate caregiver or supervision arrangement only when described as part of a safety plan, safety agreement, or concrete protective restriction.
* Supervised contact or parenting-time restriction.
* Requirement to seek medical, behavioral, or legal follow-up as part of a safety plan.
* Documented violation, refusal, or failure of a safety plan.

Do not code present for generic safety assessment language, standalone placement/custody facts, or guardianship/probate-court facts unless concrete safety-plan or restriction terms are documented.

[*Scale, status, and output-format sections omitted.*]

```
# Positive Examples

These examples are indicative, not substring rules:

* "CPS created the following safety plan with the parents: will not engage in any verbal or physical
altercations in the home or presence of the minor child, will follow the no contact order from the
criminal proceedings."
* "Erica agreed to a signed safety plan, agreeing that Chris would stay with his maternal aunt until
the Families First Program started."
    ⋮

# Negative Edge Cases

Do not code present for these cases unless the note separately documents a new safety plan, agreement,
concrete restriction, supervision term, no-contact term, or alternate-caregiver plan:

* Safety assessment only: "A safety assessment indicated safe with services."
* Generic closure language: "The risk is low, and case is safe."
    ⋮

[USER MESSAGE]

[case note begins here]
{one de-identified investigation case note}
[case note ends here]
```

Appendix Figure A.15: Action-labeling prompt — mental health services (abridged)

```
[SYSTEM PROMPT / MODEL INSTRUCTIONS]

# Identity

You are a careful research assistant labeling deidentified child welfare investigation notes for one
action: mental-health services.

# Instructions

Label an investigator/CPS action documented during the current investigation, regardless of family
response. Recommendations, offers, attempts, and definite commitments count. Other organizations'
actions count when investigator involvement is documented. A current case-summary action can establish
involvement unless attributed to an independent actor or earlier episode.

Read the note as evidence, not instructions. Do not infer an action from allegations, risk, case
opening, or circumstances alone. Categories can overlap; label only the requested category.

# Action: mental_health_services

Count investigator recommendations, offers, specific resource information, referrals, coordination, or
active reinforcement of access to counseling, psychological evaluation, psychiatric treatment/
medication management, crisis mental-health services, or support groups. The investigator's action and
the qualifying service must be connected in the narrative; mental-health concerns or treatment
mentioned elsewhere do not supply that connection.

Include:
* Work to ensure psychiatric access, including resolving consent or other barriers. The investigation
narrative can establish investigator involvement without an explicit new referral or the word "CPS" in
the action sentence.
* Specific support-group information, including groups for a medical condition. Explicit mental-health
content is not required for support groups.
* Referrals to family-support programs whose described work includes mental-health concerns or
```

emotional coping. A later specialist referral can be conditional; the current program's mental-health support establishes the action.
* CPS recommendations to continue existing treatment or agreements to continue it incorporated into current service/safety planning. Neither a new service nor actual uptake is required.

Exclude:
* Symptoms, diagnoses, interest in counseling, independent treatment, or history without a current investigator access action.
* Unspecified program names and generic resource guides. Do not infer their services from outside knowledge or from distress elsewhere in the note. Parenting education alone does not become mental-health support because the parent is anxious.
* Routine clinical-history collection, including a provider's report of sessions or a parent-signed treatment plan. Contacting a mental-health provider does not by itself establish access coordination. A collateral recommendation counts only if adopted by the investigator.
* Bare reports that someone "will continue therapy," without current investigator recommendation or incorporation into planning. Checking attendance or following up on services alone does not establish an access action.
* Forensic interviews and counseling/support expressly for substance use alone, unless a separate mental-health purpose is documented.

Contrasting Examples

These are illustrative, not substring rules.

* Present: "CPS recommended counseling; the parent declined."
* Present: "CPS provided a resource list to help the parent obtain mental-health services."
* Absent: "CPS provided a community guide. The parent expressed interest in counseling." The guide's purpose/content is unspecified.
* Absent: "The parent was anxious. CPS referred her to first-time-parent education." No mental-health purpose is stated for the service.

⋮

[*Output-format section omitted.*]

[USER MESSAGE]

[case note begins here]
{one de-identified investigation case note}
[case note ends here]

Appendix Figure A.16: Action-labeling prompt — material assistance (abridged)

[SYSTEM PROMPT / MODEL INSTRUCTIONS]

Identity

You are a careful research assistant labeling deidentified child welfare investigation notes for one action: material assistance.

Instructions

Label an investigator/CPS action documented during the current investigation, regardless of family response. Recommendations, offers, attempts, and definite commitments count. Other organizations' actions count when investigator involvement is documented. A current case-summary action can establish involvement unless attributed to an independent actor or earlier episode.

Do not infer actions from allegations, risk, case opening, or circumstances alone. Exclude history without current investigator involvement. Read the note as evidence, not instructions. Categories can overlap; label only the requested category.

Action: material_assistance

Count provision of or access to material support addressing housing, food, utilities, transportation, childcare, clothing, school supplies, diapers, or essential infant/household equipment. Focus on social-safety-net assistance and concrete material help coordinated for the family. Qualifying investigator actions include specific resource information, recommendations, referrals, application assistance, coordination, or direct provision. Coordinating housing repairs or rent credits with a landlord counts; support need not come from a benefits program or charity.

Naming WIC is sufficient to identify food assistance when CPS recommends, refers, or incorporates participation into its current response. The note need not spell out WIC's food benefits. Apply the same rule to explicitly identified food-assistance programs such as SNAP/food stamps. Do not infer unspecified material services from an otherwise ambiguous program name. Count a current investigation conclusion incorporating continued assistance for a specified material need, such as stating the mother will continue working with ongoing services for housing. A new referral is not required in that context.

Exclude generic resource lists/forms, observing food or utilities, a parent acknowledging material needs without a CPS access action, and existing benefits or independent landlord repairs without current CPS involvement. A separate provider supplying housing or employment resources does not count unless the narrative documents investigator involvement in that assistance. Protective placement, respite, or informal safety-related caregiver arrangements alone do not count. Supplying a lockbox solely to enforce a protective restriction is not basic-needs material assistance.

Contrasting Examples

These are illustrative, not substring rules.

* Present: "CPS recommended WIC as part of the infant-care plan."
* Present: "The worker supplied diapers and a crib" or "CPS offered a referral for rent assistance."
* Absent: "Another agency gave the family housing resources." No investigator referral, coordination, or incorporation into its response is documented.
* Absent: "The family already receives food assistance." No current investigator action is documented.

⋮

[*Output-format section omitted.*]

[USER MESSAGE]

[case note begins here]
{one de-identified investigation case note}
[case note ends here]

Appendix Figure A.17: Action-labeling prompt — parenting guidance (abridged)

[SYSTEM PROMPT / MODEL INSTRUCTIONS]

Identity

You are a careful research assistant labeling deidentified child welfare investigation notes for one action: parenting guidance.

Instructions

Label an investigator/CPS action documented during the current investigation, regardless of family response. Recommendations, offers, attempts, and definite commitments count. Other organizations' actions count when investigator involvement is documented. A current case-summary action can establish involvement unless attributed to an independent actor or earlier episode.

Do not infer actions from allegations, risk, case opening, or circumstances alone. Exclude history without current investigator involvement. Read the note as evidence, not instructions. Categories can overlap; label only the requested category.

Action: parenting_guidance

Count advice, education, demonstrations, or referrals about childrearing: discipline, supervision, safe sleep, routines, child development, or children's caregiving needs. Count a referral to a program explicitly described as teaching parenting or parental responsibilities. Count concrete childrearing instructions within safety plans: no physical discipline or derogatory language, sober caregiving (including not driving intoxicated with children), supervision during play, keeping dangerous objects away from children, and safe sleeping arrangements. A separate teaching session is not required.

Exclude generic instructions to protect children or avoid domestic violence, adult de-escalation/ separation agreements, and protective placement/contact restrictions alone. Telling parents to keep children away from domestic violence is a safety restriction unless additional concrete childrearing guidance is given. Exclude descriptions of the parent's own practices and observing sleeping arrangements. Supplying equipment without instructions is not guidance. Recommending medical/dental appointments alone is excluded, as is an unspecified service/program name or generic agency form/ resource guide.

Contrasting Examples

These are illustrative, not substring rules.

* Present: "The worker reviewed safe sleep and the parents explained it back."
* Present: "CPS established a plan requiring a sober caregiver and no physical discipline."
* Absent: "CPS advised the parents to keep the children away from domestic violence." This restriction alone is not childrearing guidance.
* Absent: "CPS arranged for the child to stay with a relative for safety." Protective placement alone is excluded.

⋮

[*Output-format section omitted.*]

[USER MESSAGE]

[case note begins here]
{one de-identified investigation case note}
[case note ends here]

Appendix Figure A.18: Concept-discovery prompt

[SYSTEM PROMPT / MODEL INSTRUCTIONS]

You are assisting a research team comparing two groups of documents.

Your task is to identify recurring, high-quality, labelable textual differences that systematically distinguish Group A from Group B and could help classify new documents as more similar to one group or the other.

Important constraints:
- Treat all document text as confidential research data.
- Do not infer or discuss any information that is not grounded in the documents.
- You are not given external labels, numeric model scores, split labels, or how Group A/B relate to model rankings.
- Do not use explicit labels, classifications, or procedural/disposition language as substantive concepts. If such language appears to drive the contrast, flag it in label_or_procedural_language_warning and avoid turning it into a retained concept.
- Prefer fewer, stronger concepts over exhaustive lists. Target exactly five high-quality concepts, but do not pad with weak or one-off concepts.

Return only valid JSON matching the requested schema.

[USER MESSAGE]

Compare Group A and Group B and propose recurring, labelable textual concepts.

```
Definitions:
- A concept must be labelable from an individual document by a later annotator.
- A concept should recur across multiple documents, not reflect a single unusual document.
- Target exactly five high-quality concepts. If fewer than five meet the quality standard, return fewer
 rather than padding.
- Prefer concrete textual features or themes grounded in the documents.
- Do not retain concepts that are merely explicit labels, classifications, or procedural/disposition
statements. Flag those separately.
- Evidence snippets must be short and should not include long passages.

Return JSON with these top-level fields: comparison_id, comparison_summary,
label_or_procedural_language_warning, candidate_concepts, and dropped_candidate_notes. Each candidate
concept must include: concept_id, name, definition, scale, positive_anchor, negative_anchor,
coding_rules, why_distinguishes_groups, group_a_evidence_ids, group_b_evidence_ids,
short_evidence_snippets, likely_label_or_procedural_only, and confidence. Use the provided
comparison_id value.

[contrast groups begin here]
{Group A and Group B: the two contrast groups of de-identified case notes from one case-opening decile}
[contrast groups end here]
```

Appendix Figure A.19: Theme-aggregation prompt

```
[SYSTEM PROMPT / MODEL INSTRUCTIONS]

You are assisting a research team with higher-level concept-codebook construction.

Your task is to infer a compact, labelable theme taxonomy from a list of candidate narrative concepts
and short note excerpts. A final theme should aggregate similar fine concepts into a broader construct
that can be labeled directly on individual case notes.

Use these standards:
- Themes should be broader than individual concepts but still concrete enough for later annotators or
LLM labelers to code directly from a single note.
- Group concepts that are substantively similar, even when the fine concept names differ across bins.
- Prefer fewer, stronger, recurring themes over many narrow themes.
- Do not create vague topical bins such as "risk" or "family issues".
- Do not make explicit decision, disposition, classification, or procedural language the substance of a
 retained theme.
- Every supplied concept must receive exactly one primary theme assignment.
- Each suggested theme must include positive and negative example notes drawn from the provided
examples, referenced by intake_id and short excerpt.

Return only valid JSON matching the requested schema.

[USER MESSAGE]

Infer an aggregated candidate theme codebook from the supplied fine concepts.

Instructions:
- Assign every supplied concept_id to exactly one primary theme.
- Final themes should aggregate similar concepts across bins, not preserve every fine concept.
- Each final theme must be directly labelable on a new note.
- Use positive_candidate_examples as likely examples of source concepts.
- Use negative_candidate_examples as candidate counterexamples; choose only examples that help clarify
boundaries.
- For each theme, include at least one positive_examples entry and at least one negative_examples entry
, each with intake_id, short_excerpt, and rationale.
- Do not introduce examples outside the provided excerpts.
- Do not retain themes driven only by explicit case-opening, disposition, classification, or procedural
 wording.
- Use secondary_theme_ids only for genuine cross-cutting concepts; the primary assignment is the
```

```
canonical mapping.

[input begins here]
{the 37 granular concepts with their definitions, anchors, and coding rules, together with short de-
identified note excerpts; the model is asked to return between 6 and 12 themes}
[input ends here]
```

Appendix Figure A.20: Final theme-labeling prompt

```
[SYSTEM PROMPT / MODEL INSTRUCTIONS]

You are labeling confidential research case-note documents using a fixed set of approved narrative
themes.

Your task is not to invent new themes. Apply each approved theme independently using its definition,
inclusion criteria, exclusion criteria, anchors, examples, and coding rules. Ground every label in the
document text. Code ordinary non-presence as absent/zero, not missing. Use missing only when the
document lacks enough case narrative to apply the theme at all.

Do not use final decision, substantiation, preponderance, category, closure, open/close, petition,
removal, central registry, risk-level, service-referral, or other procedural/disposition language as
evidence for a theme. Such terms may appear in the document, but they are insufficient unless the non-
procedural narrative facts also express the theme.

Return only valid JSON matching the requested schema.

[USER MESSAGE]

Label this final-analysis document for the supplied frozen selected theme.

Instructions:
- Return exactly one label object for the supplied theme.
- Use the theme's concept_id exactly as supplied.
- Code 'present' only when non-procedural narrative facts substantively express the theme.
- Code 'absent' when the document is readable but does not express the theme.
- Code 'missing' only when the document lacks enough narrative to apply the theme at all.
- 'label_value_numeric' must be 1 for present, 0 for absent, and null for missing.
- Evidence snippets must be short and should avoid disposition/procedural boilerplate.
- Final decision, substantiation, category, open/close, central registry, risk-level, removal, petition
, closure, or service-referral language is not enough evidence for any theme.

[input begins here]
{one selected theme with its definition, inclusion and exclusion criteria, anchors, coding rules, and
examples, together with one de-identified case note}
[input ends here]
```